

Subject _____

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Date : / /

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AIOD, Device Note

FINAL TERM

Wolfram Language
AID, Device for MTS

#BOT Analysts
Choo G

BJT AC analysis! —

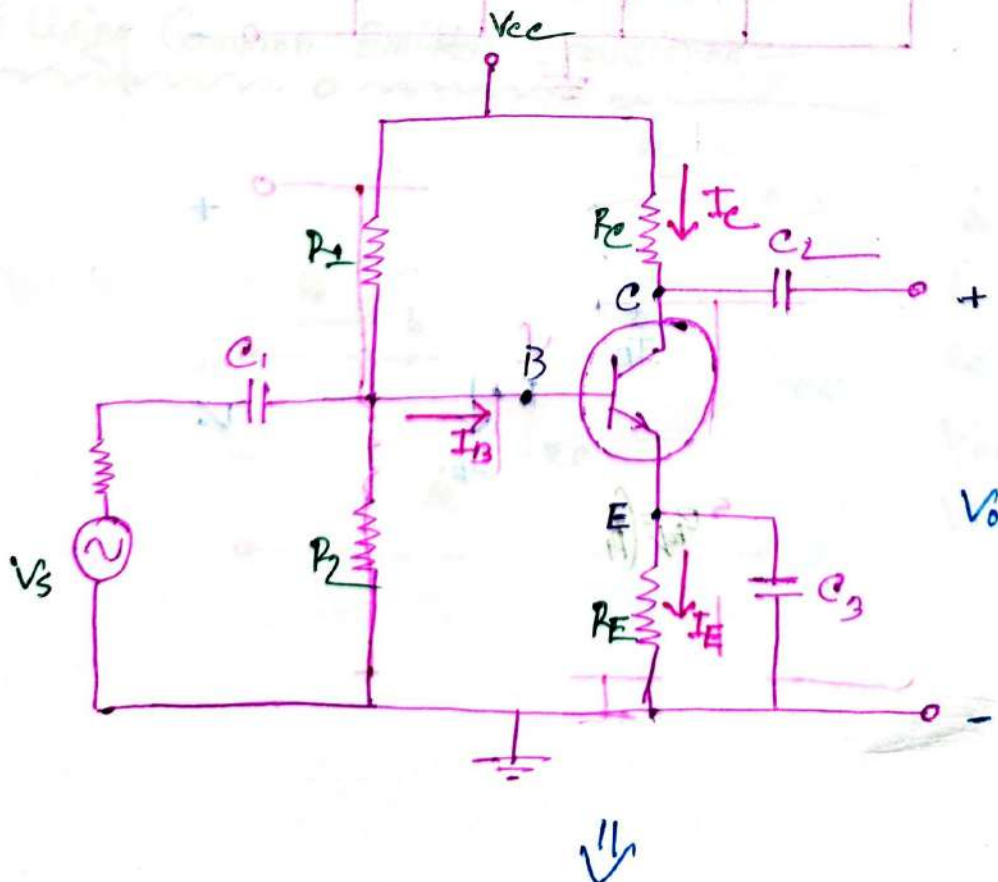
⇒ For AC response we need to do two things

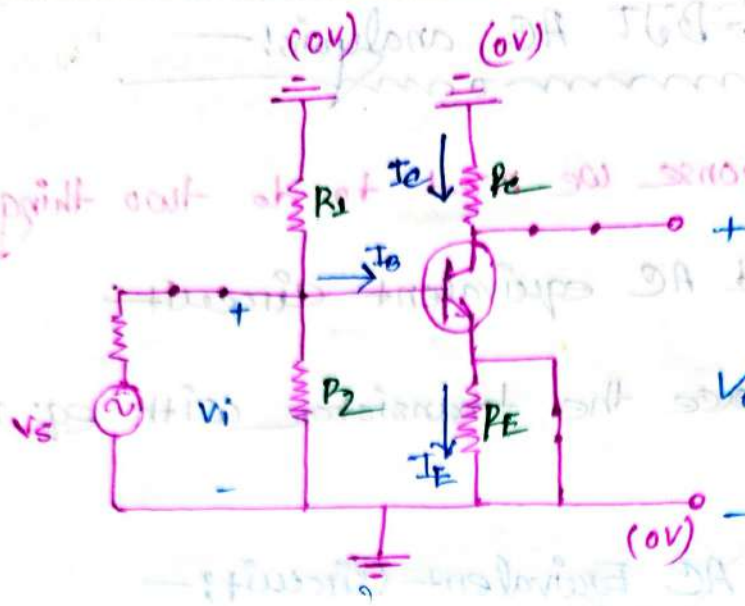
1. Find AC equivalent circuit.
2. Replace the transistor with eq. model

AC Equivalent Circuit: —

⇒ There are 3 steps: —

1. Short circuit all the DC Sources.
2. Short all the capacitor
3. Redraw the Network removing all the short circuited element





#Logic Behind
Shorting the capacitor

For AC

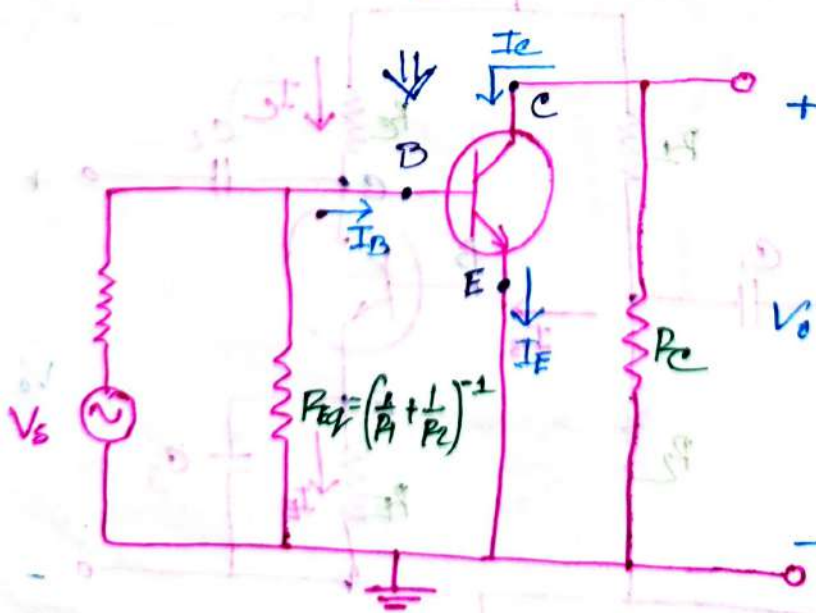
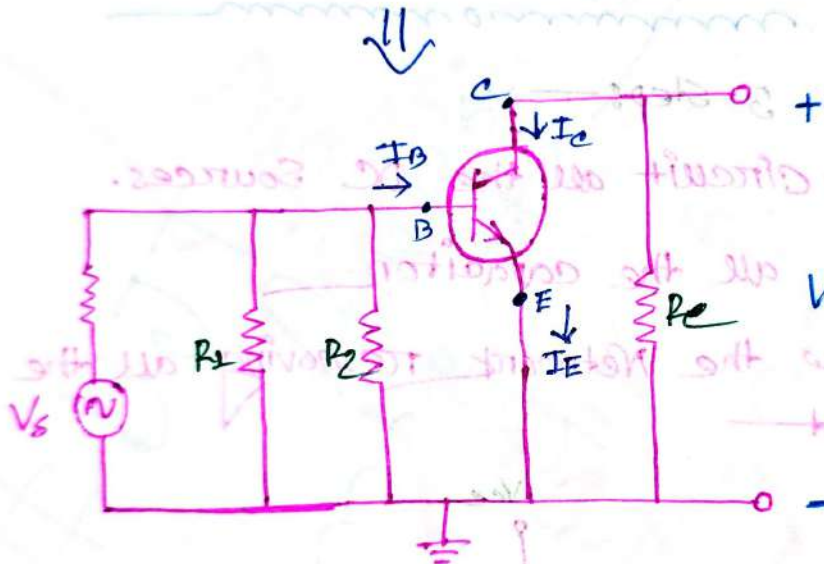
$$f = 50/60 \text{ Hz}$$

C = very large.

We know -

$$X_C = \frac{1}{2\pi f C} = \frac{1}{100 \times 10^3 \times \text{large } C}$$

$$X_C \approx 0 \Omega$$



Equivalent model of transistor:-

⇒ An equivalent model is combination of circuit elements properly chosen to best represent the actual behaviour of device under specific operating point.

There are three types:-

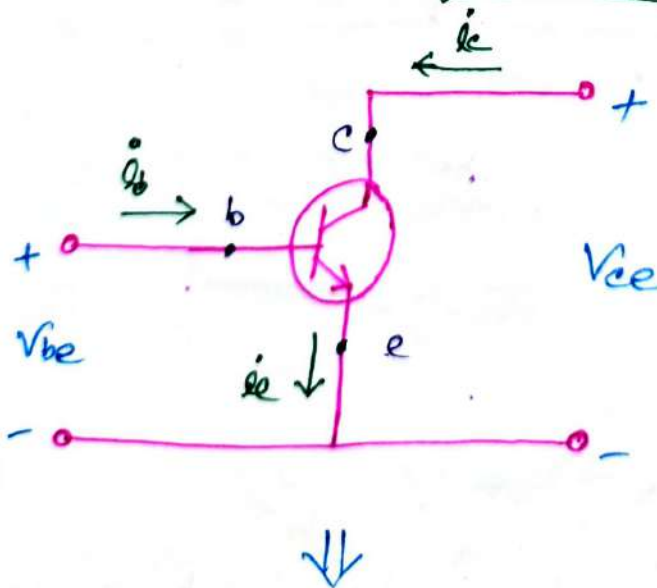
1. Hybrid model

2. Hybrid π model.

3. "T_{ce}" model.

T_{ce} Transistor Model:-

Using Common Emitter Transistor:-



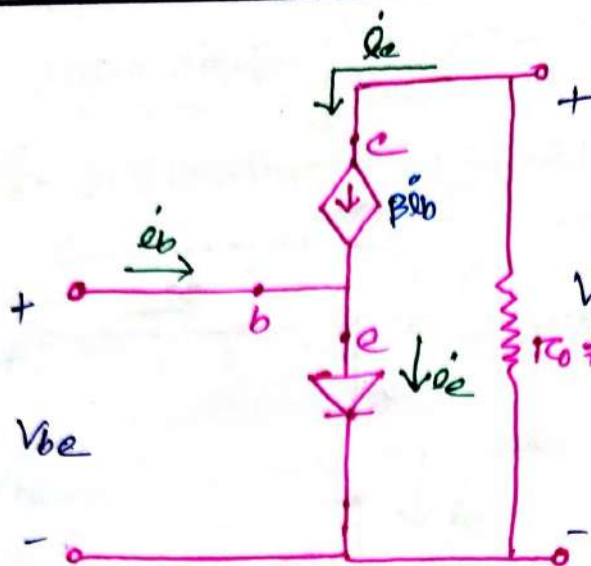
i_b = input current

i_c = output current

i_e = emitter current

V_{be} = input voltage

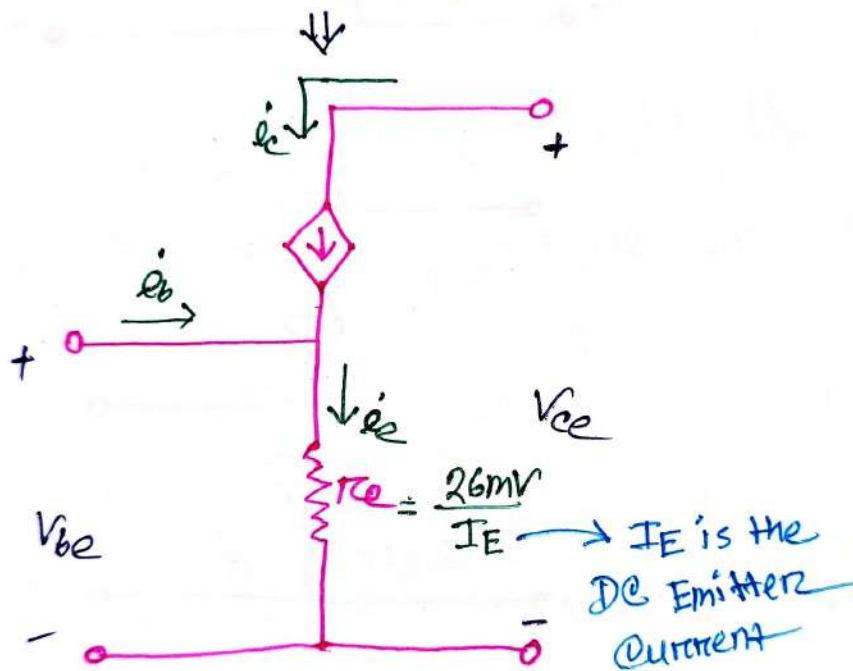
V_{ce} = output voltage.



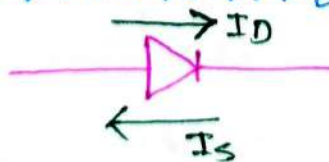
In case of F.B.D

$$r_d = \frac{\Delta V_{BE}}{\Delta I_D} = r_{ce}$$

$r_d = A_c / \text{dynamic resistance}$



Calculation of resistance " r_{ce} "



We know -

$$I_D = I_S \left(e^{\frac{V_D}{\eta V_T}} - 1 \right)$$

$$\Rightarrow \frac{dI_D}{dV_D} = I_S \frac{d}{dV_D} \left(e^{\frac{V_D}{\eta V_T}} - 1 \right)$$

$$\Rightarrow \frac{1}{r_d} = \frac{I_S \times e^{\frac{V_D}{V_T}}}{V_T}$$

Where,

$$\eta = 1$$

$$T = 300\text{K}$$

$$r_d = \frac{\Delta V_D}{\Delta I_D}$$

$$\Rightarrow \frac{1}{r_d} = \frac{I_s + I_D}{V_T}$$

$$\Rightarrow \frac{1}{r_d} = \frac{I_D}{V_T}$$

$$\Rightarrow r_d = \frac{V_T}{I_D}$$

$$\therefore r_d = r_e = \frac{26 \text{ mV}}{I_D}$$

$$\text{So, } r_e = \frac{26 \text{ mV}}{I_E}$$

$$I_s \ll I_D$$

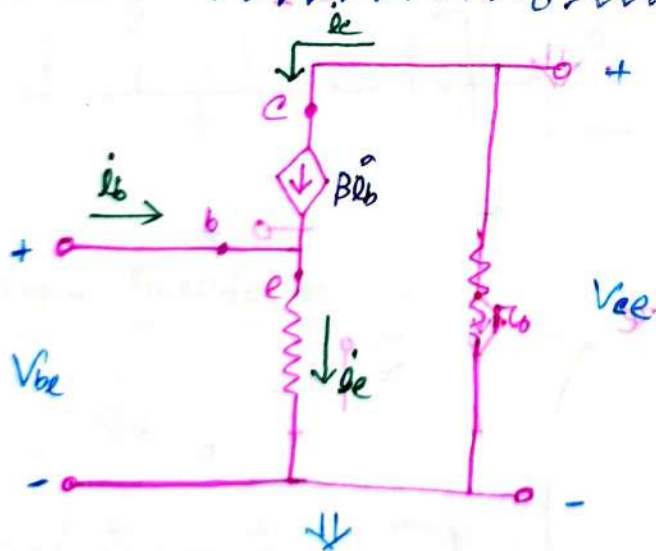
$$\therefore I_s + I_D \approx I_D$$

At room Temp $T = 300 \text{ K}$

The $V_T = 26 \text{ mV}$

$$\Rightarrow I_D = I_E$$

Modification of re model:-



$$I_e = I_b + I_c$$

$$= I_b + \beta I_b$$

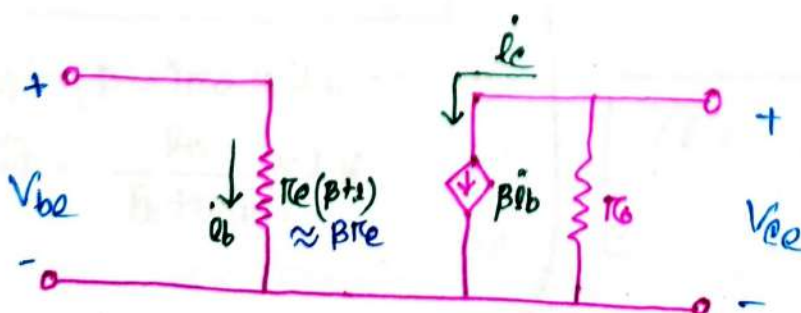
$$= (\beta + 1) I_b$$

$\therefore$ Drop Across r_e :-

$$V_{r_e} = r_e \cdot I_e$$

$$= r_e \cdot (\beta + 1) I_b$$

Resistance of r_e $\times$ Current flow, the r_e



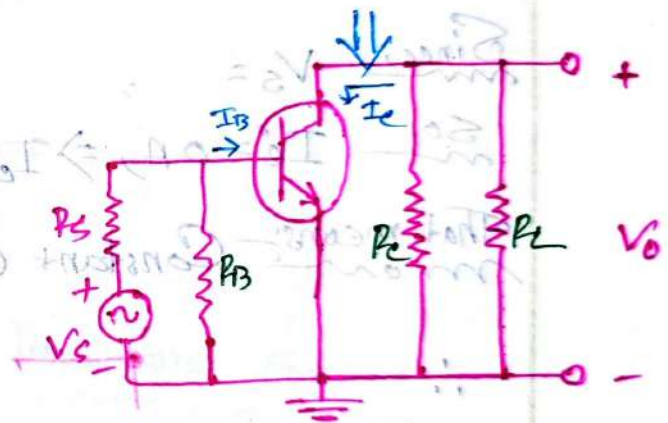
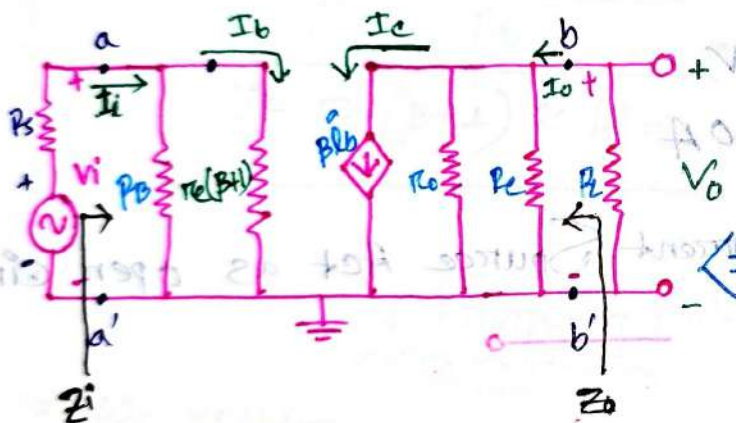
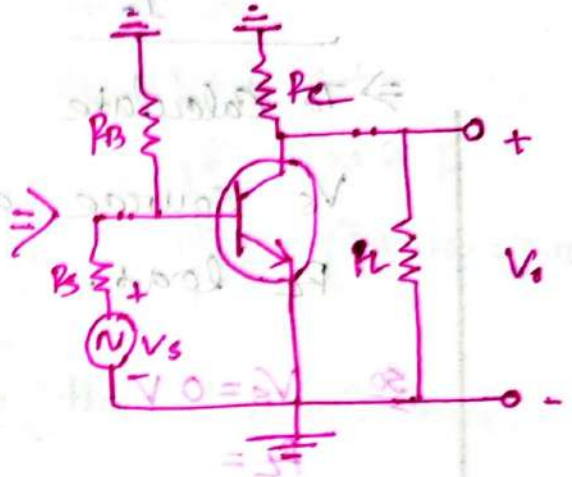
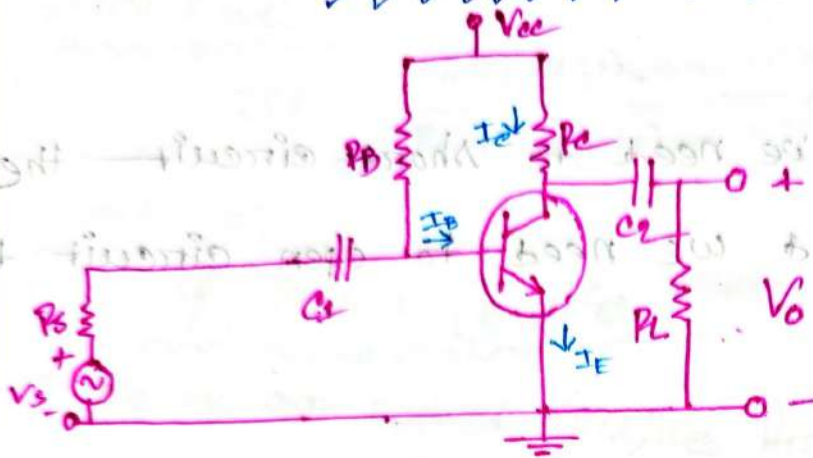
$$\beta \gg 1$$

$$\therefore \beta + 1 \approx \beta$$

So, $r_e(\beta + 1) \approx r_e \beta$

For simplification.

Fixed-Bias Configuration for r_e model:-



Input impedance:-

$$Z_i = \frac{V_i}{I_i} \quad \dots \quad (i)$$

OR

$$Z_i = \left(\frac{1}{R_B} + \frac{1}{r_e(\beta+1)} \right)^{-1} \quad \dots \quad (ii)$$

OR

$$V_i = (\beta+1)r_e \times I_b \quad \dots \quad (iii)$$

SO

$$I_b = \frac{R_B}{R_B + r_e(\beta+1)} \times I_i$$

put the I_b value in eq (iii) & then put the V_i value in eq (i)

$$\therefore Z_i = \frac{(\beta+1)r_e \times R_B I_i}{I_i \times (R_B + r_e(\beta+1))}$$

$$Z_i = \frac{(\beta+1)r_e R_B}{R_B + r_e(\beta+1)} \quad \dots \quad (iv)$$

Output Impedance:

$$Z_o = \frac{V_o}{I_o}$$

$\Rightarrow$ To calculate Z_o we need to short circuit the V_s source and we need to open circuit the R_L load.

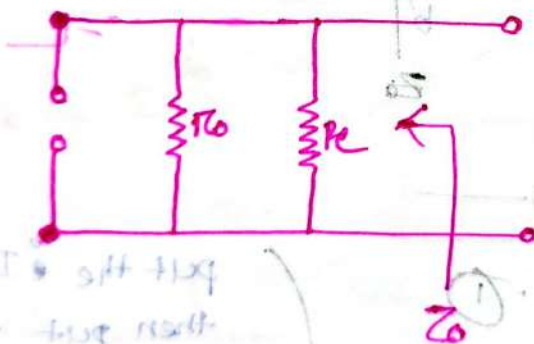
So $V_s = 0V$

$R_L = \infty \Omega$

Since, $V_s = 0V \Rightarrow V_i = 0V$

So, $I_i = 0A, \Rightarrow I_e = 0A$

That means: Constant Current Source Act as open circuit.



$$Z_o = \left(\frac{1}{r_o} + \frac{1}{R_e} \right)^{-1} \Omega$$

Simplify Input Impedance Exp:-

⇒ we have:-

$$Z_i = \frac{R_B r_e (\beta + 1)}{R_B + r_e (\beta + 1)} \quad \text{--- (1)}$$

with here
 $\beta \gg 1$

so $\beta + 1 \approx \beta$

Then, $(\beta + 1) r_e = \beta r_e$

Condition:-

If $R_B \geq 10 (\beta + 1) r_e$

⇒ so we can neglect $(\beta + 1) r_e$ from the denominator.

$$\therefore Z_i = \frac{R_B r_e (\beta + 1)}{R_B}$$

$$Z_i = r_e (\beta + 1) \approx \beta r_e \quad \Omega$$

Simplify Output Impedance Exp:-

⇒ we know:-

$$Z_o = \left(\frac{1}{r_o} + \frac{1}{R_e} \right)^{-1} = \frac{r_o R_e}{r_o + R_e} \quad \Omega$$

Condition:-

If $r_o \geq 10 R_e$

⇒ so, we can neglect R_e from the denominator.

$$\therefore Z_o = \frac{r_o R_e}{r_o + R_e} \approx \frac{r_o R_e}{r_o} = R_e \quad \Omega$$

$$\therefore Z_o = R_e \quad \Omega$$

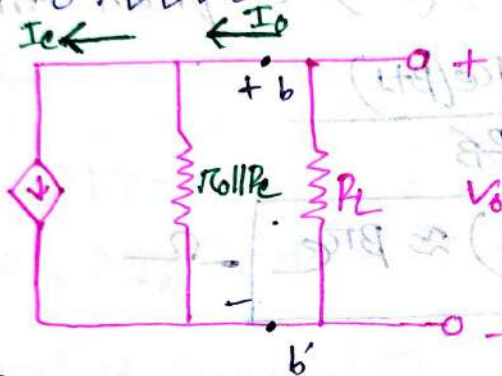
Current Gain (A_i):

⇒ Current Gain is defined as ratio of output current to the input current.

$$\therefore A_i = \frac{I_o}{I_i} \quad \text{--- (1)}$$

For Output Current (I_o):

Redraw the output side



I_o = Current Through the load Resistance R_L

Using CDR:

$$I_o = \frac{R_C || R_E}{R_C || R_E + R_L} \times I_e \quad A$$

$$\therefore I_o = \frac{R_C || R_E}{R_C || R_E + R_L} \times \beta I_b \quad \text{--- (1)}$$

$$I_b = \frac{R_B}{R_B + (\beta + 1)R_E} \times I_i$$

Now,

put the value of I_b in eq (1) =

$$\therefore I_o = \frac{R_C || R_E}{(R_C || R_E) + R_L} \times \beta \times \left(\frac{R_B I_i}{R_B + (\beta + 1)R_E} \right) \times 1$$

Now, put the value of I_o in eq (1) =

$$\therefore A_i = \frac{\beta \left(\frac{I_o R_B}{R_B + (\beta + 1) r_{ce}} \right) \left(\frac{r_{ce} \parallel R_e}{(r_{ce} \parallel R_e) + R_L} \right)}{I_o}$$

$$\underline{\text{So,}} \quad A_i = \beta \left(\frac{R_B}{R_B + (\beta + 1) r_{ce}} \right) \left(\frac{r_{ce} \parallel R_e}{(r_{ce} \parallel R_e) + R_L} \right)$$

Case: 1:-

1. if $r_{ce} \geq 10 R_e$

$$\underline{\text{So,}} \quad r_{ce} \parallel R_e = R_e$$

Then:-

$$\therefore A_i = \beta \left(\frac{R_B}{R_B + (\beta + 1) r_{ce}} \right) \left(\frac{R_e}{R_e + R_L} \right)$$

Case: 2:-

2. if $R_B \& R_e = \infty$

then $r_{ce} < R_e$

$$\underline{\text{So,}} \quad r_{ce} \parallel R_e = r_{ce}$$

$$\therefore A_i = \beta \left(\frac{R_B}{R_B + (\beta + 1) r_{ce}} \right) \left(\frac{r_{ce}}{r_{ce} + R_L} \right)$$

$$\therefore A_i = \beta \left(\frac{R_B}{R_B} \right) \left(\frac{r_{ce}}{r_{ce} + R_L} \right)$$

$$\underline{\text{So,}} \quad A_i = \beta \left(\frac{r_{ce}}{r_{ce} + R_L} \right)$$

Since

$$R_B = \infty$$

$$\therefore R_B \gg (\beta + 1) r_{ce}$$

Overall Current Gain (A_{is}):-

⇒ It is the ratio of output current to the source current

$$A_{is} = \frac{I_o}{I_s} = A_e \left(\frac{R_s}{R_s + r_i} \right)$$

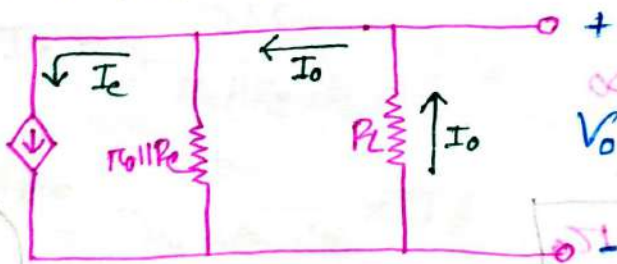
Voltage Gain (A_v):-

⇒ Voltage gain is defined as the ratio of output voltage to the input voltage.

$$A_v = \frac{V_o}{V_i}$$

For output Voltage:-

Draw the output Side:-



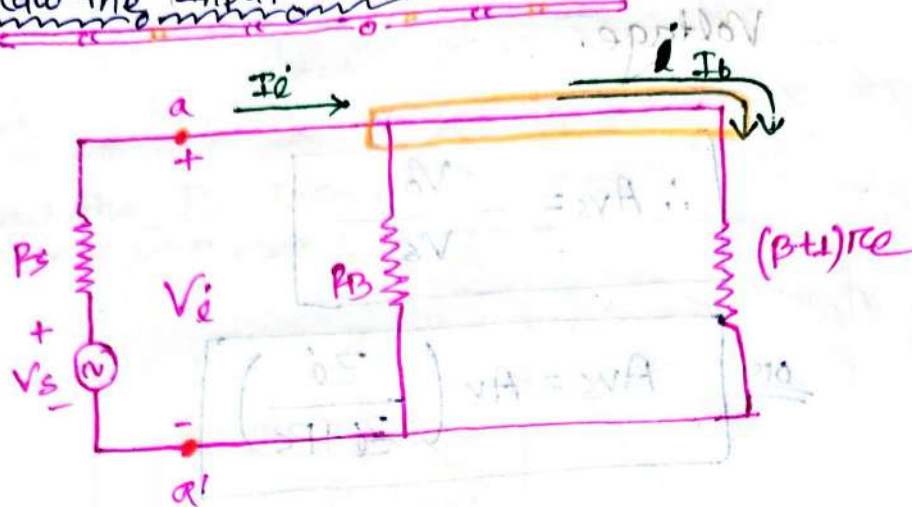
$$V_o = -I_o R_L$$

$$V_o = - \left\{ \beta I_b \times \frac{r_o || R_e}{r_o || R_e + R_L} \right\} \times R_L$$

$$\text{So, } V_o = - \left\{ \frac{r_o || R_e}{r_o || R_e + R_L} \times \beta I_b \right\} R_L$$

For Input Voltages—

Draw the Input Side—



$$\therefore V_i = (\beta + 1)r_e \times I_b \quad V$$

Now, put the value of V_i & V_o in eq (1) $\Rightarrow$

$$\therefore A_v = \frac{- \left\{ \frac{r_o \parallel R_L}{r_o \parallel R_L + R_L} \times \beta I_b \right\} R_L}{(\beta + 1)r_e \times I_b}$$

The Basic Exp

$$A_v = \frac{-\beta(r_o \parallel R_L \parallel R_L)}{(\beta + 1)r_e}$$

we know—

$$\frac{R_1 R_2}{R_1 + R_2} = R_1 \parallel R_2$$

here—

$$R_1 = r_o \parallel R_L$$

$$R_2 = R_L$$

Case 1—

Since, $\beta \gg 1$

so, $\beta + 1 \approx \beta$

$$\therefore A_v = - \frac{r_o \parallel R_L \parallel R_L}{r_e}$$

Case 2—

if $r_o \geq 10 R_L$

$$\therefore A_v = - \frac{R_L \parallel R_L}{r_e}$$

Case 3—

$R_L = \infty$ That means open ckt

Since $R_L \gg R_L$

$$\therefore A_v = - \frac{R_L \parallel r_o}{r_e}$$

NOTE:— The negative (—) sign in the exp represent 180° phase shift between the V_o & the V_i waveforms

Overall Voltage Gain (A_{vs})

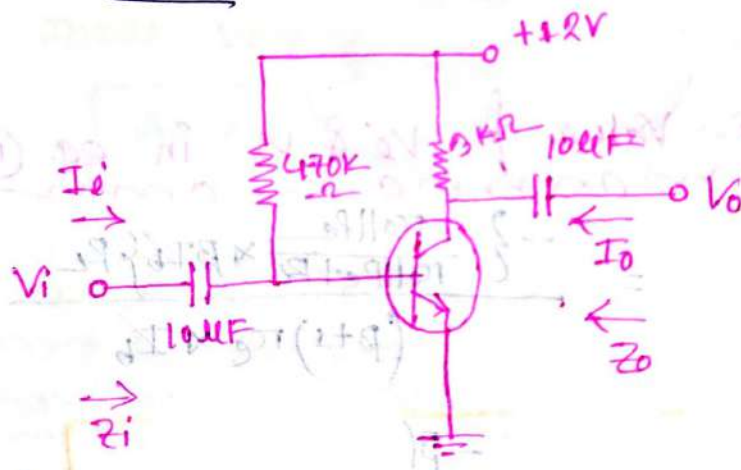
$\Rightarrow$ It is equal to the ratio of output Voltage to the Source Voltage.

$$\therefore A_{vs} = \frac{V_o}{V_s}$$

or

$$A_{vs} = A_v \left(\frac{Z_i}{Z_i + R_s} \right)$$

problems 5.1



$$\beta = 100$$

$\Rightarrow$ Determine

- (i) r_e
- (ii) Z_i (with $r_o = \infty$)
- (iii) Z_o (with $r_o = \infty$)
- (iv) A_v (with $r_o = \infty$)
- (v) Repeat (i) - (iv) with $r_o = 50k\Omega$

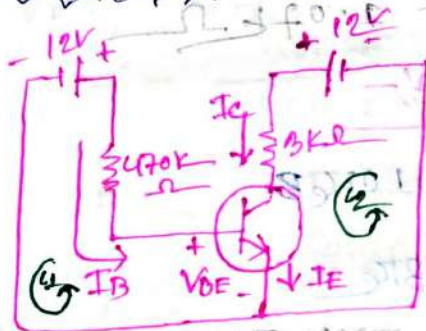
⇒

(i)

We know—
more

$$r_e = \frac{26mV}{I_E}$$

(1)

For I_E —
moreRedraw the DC Fixed Bias Ckt—
moreApply KVL in input loop 1—
more

$$-12 + 470kI_B + 0.7 = 0$$

$$\therefore I_B = \frac{12 - 0.7}{470k} = 24.04 \mu A$$

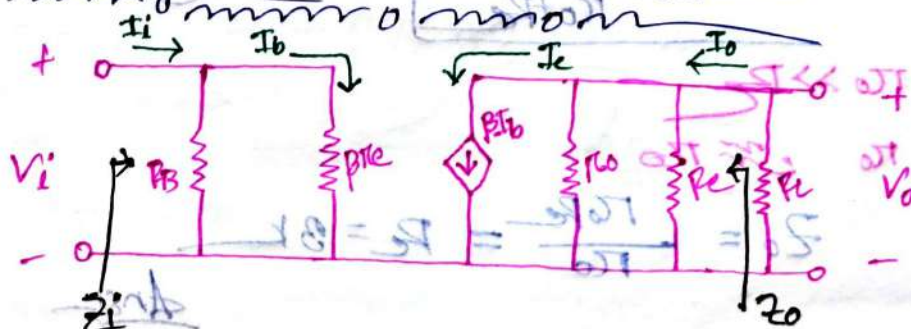
$$\therefore I_E = I_C + I_B$$

$$I_E = \beta I_B + I_B$$

$$I_E = I_B (\beta + 1) = 24.04 \mu A (100 + 1) = 2.42 mA$$

Put the value of I_E in eq (1) —
more

$$\therefore r_e = \frac{26m}{2.42m} = 10.74 \Omega$$

⇒ Draw the r_e equivalent model: —
more

here,

$$R_B = 470k\Omega$$

$$R_C = 3k\Omega$$

$$r_e = 10.74\Omega$$

$$\beta = 100$$

$$R_L = \infty\Omega$$

(ii)

∴ The input Impedance:—

$$Z_i = \left(\frac{1}{R_B} + \frac{1}{\beta r_e} \right)^{-1}$$

$$= \left(\frac{1}{470k} + \frac{1}{100 \times 10.74} \right)^{-1}$$

$$= 1.07k \Omega$$

OR Apply Condition:—

Since, $R_B \gg 10\beta r_e$

So, $Z_i = \beta r_e$

$$= 100 \times 10.74$$

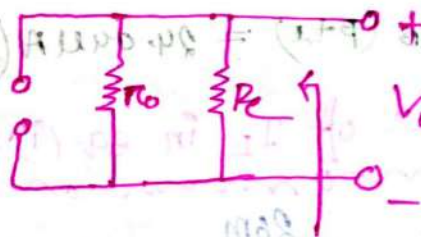
$$= 1.07k \Omega$$

Ans:—

(iii)

The Output Impedance:—

Redraw the Output Side:—



Since,

$$V_i = 0 \Rightarrow I_i = 0$$

$$\text{so } I_b = 0 \Rightarrow I_e = 0$$

$$\text{then } \beta I_b = 0$$

So, $Z_o = \left(\frac{1}{r_o} + \frac{1}{R_e} \right)^{-1}$

$$Z_o = \frac{r_o R_e}{r_o + R_e}$$

if

Then

$$r_o \gg R_e$$

$$r_o + R_e \approx r_o$$

$$Z_o = \frac{r_o R_e}{r_o} = R_e = 3k$$

Ans:—

(iv)

We know:-

$$A_v = \frac{V_o}{V_i} \quad \text{--- (1)}$$

For Output Voltage (V_o)

$$\begin{aligned} V_o &= -I_o R_L \\ &= - \frac{r_{o1} R_E}{r_{o1} R_E + R_L} \times \beta I_b \times R_L \\ &= - \frac{r_{o1} R_E}{r_{o1} R_E + R_L} \times \beta I_b R_L \end{aligned}$$

For Input Voltage (V_i)

$$V_i = \beta r_{E1} I_b$$

put the V_o & V_i in eq (1) $\Rightarrow$

$$A_v = \frac{V_o}{V_i} = - \frac{\frac{r_{o1} R_E}{r_{o1} R_E + R_L} \times \beta I_b R_L}{\beta r_{E1} I_b}$$

$$A_v = - \frac{\beta (r_{o1} R_E || R_L)}{\beta r_{E1}} \quad \text{--- (2)}$$

if $r_{o1} = R_L = \infty$ Then:-

$$\begin{aligned} A_v &= - \frac{R_E}{r_{E1}} \\ &= - \frac{24}{20.74} \end{aligned}$$

$$= -270.32$$

Ans:-

if $r_{o1} = 50k\Omega$

For Z_o :-

$\Rightarrow$ put the value of r_{o1} in ϕ (ii) eq (1) $\Rightarrow$

$$\begin{aligned}\therefore Z_o &= \frac{r_{o1} R_E}{r_{o1} + R_E} \\ &= \frac{50k\Omega \parallel 3k\Omega}{50k\Omega + 3k\Omega} \\ &= 2.83k\Omega\end{aligned}$$

Ans:-

For A_v :-

$\Rightarrow$ put the value of r_{o1} in ϕ (iv) eq (11) $\Rightarrow$

$$\begin{aligned}A_v &= \frac{-r_{o1} R_E \parallel R_L}{r_{e1}} \\ \therefore A_v &= \frac{-r_{o1} \parallel R_E}{r_{e1}}\end{aligned}$$

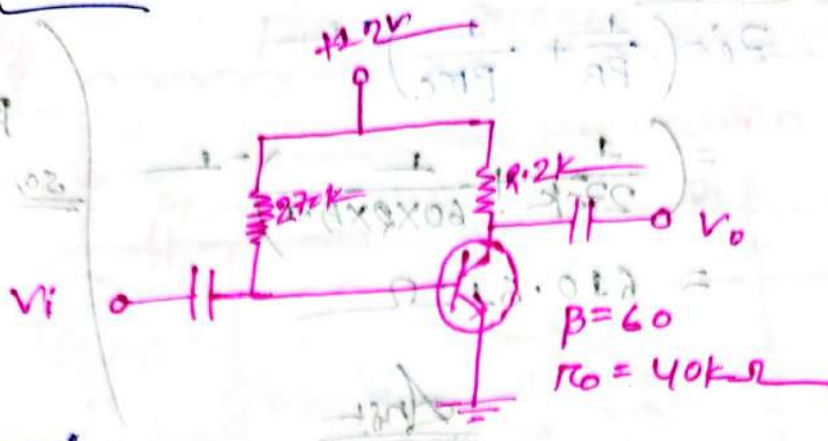
Since,
 $R_L = \infty$

$$= \frac{-50k\Omega \parallel 3k\Omega}{10.74}$$

$$A_v = -263.51$$

Ans:-

problem 1.2



⇒ Determine,

i Z_i, Z_o

ii A_v

iii A_i

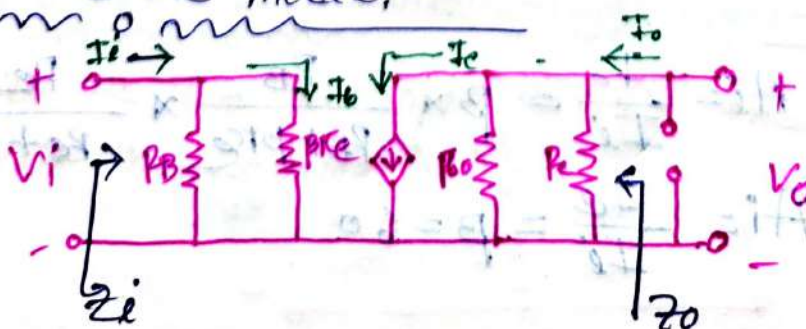
⇒ For I_E :-

we know:-

$$\begin{aligned} I_E &= I_B (\beta + 1) \\ &= \frac{V_{CC} - V_{BE}}{R_B} \times (\beta + 1) \\ &= \frac{12 - 0.7}{270k} \times (\beta + 1) \\ &= 2.55mA \end{aligned}$$

$$r_e = \frac{26mV}{I_E} = \frac{26m}{2.55mA} = 10.20 \Omega$$

Draw the rre model!



here
 $R_B = 270k$
 $R_C = 2.2k$
 $\beta = 60$
 $r_o = 40k$

we know:-

$$Z_i = \left(\frac{1}{R_B} + \frac{1}{R_{E1}} \right)^{-1}$$

$$= \left(\frac{1}{270k} + \frac{1}{60k \times 5.1} \right)^{-1}$$

$$= 610.61 \Omega$$

Ans:-

$$Z_o = \left(\frac{1}{R_{C1}} + \frac{1}{R_E} \right)^{-1}$$

$$= \left(\frac{1}{40k} + \frac{1}{2.2k} \right)^{-1}$$

$$= 2.08 k\Omega$$

Ans:-

ii

we know:-

$$A_v = \frac{V_o}{V_i} = \frac{-\beta R_{C1} R_E}{R_E}$$

$$= \frac{-2.08k}{10.2}$$

$$= -203.92$$

iii

we know:-

$$A_i = \frac{I_o}{I_i} = \beta \times \frac{R_B}{R_B + R_{E1}} \times \frac{R_E}{R_E + R_L}$$

$$\text{or, } A_i = \frac{I_o}{I_i} = \beta = 60$$

Since,

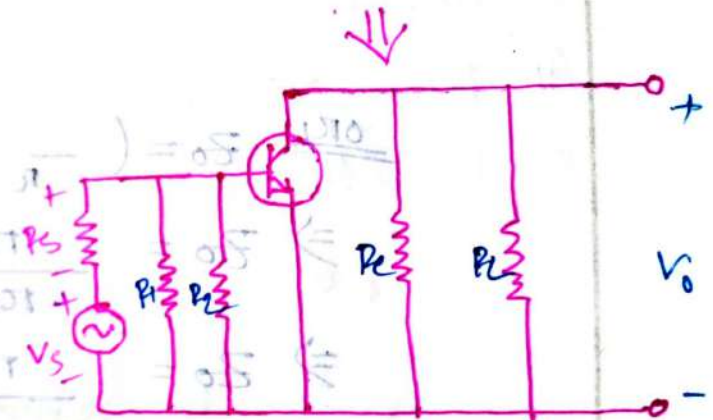
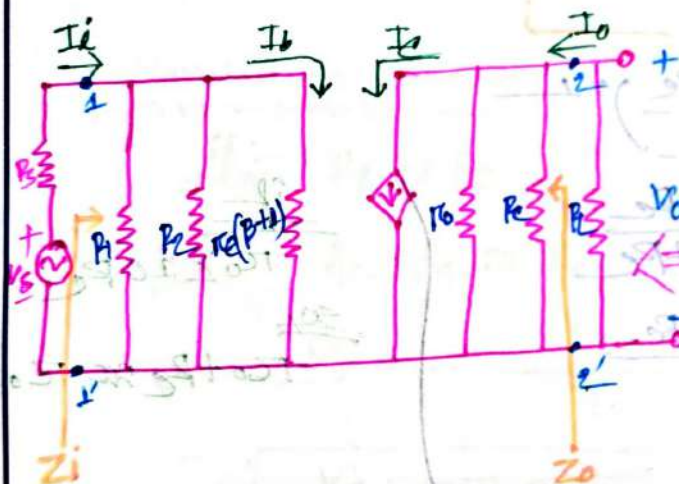
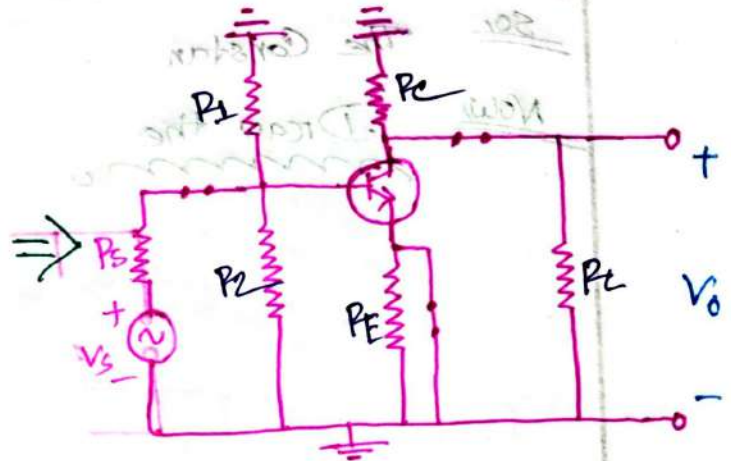
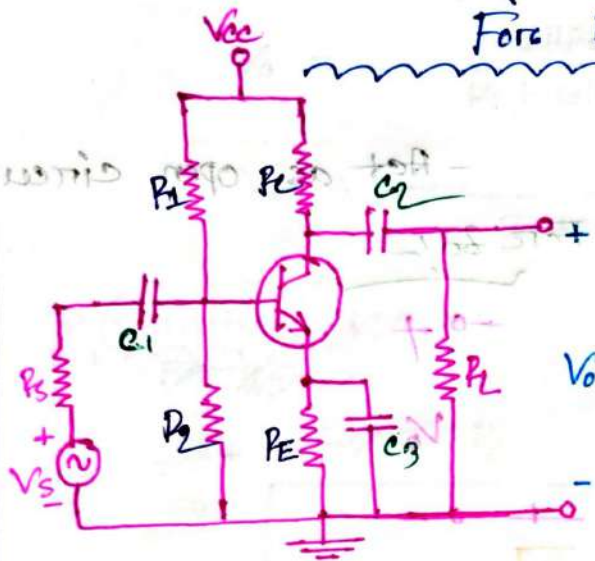
$$R_L = \infty$$

so R_L is open

ckt

$$\text{so } I_o = I_E$$

Voltage - Divider Bias For r_{ce} Model:-



∴ Input Impedance:-

$$Z_i = \left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{(\beta+1)r_{ce}} \right)^{-1}$$

$$\therefore Z_i = \left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{\beta r_{ce}} \right)^{-1}$$

Since,
 $\beta \gg 1$
So, $(\beta+1) \approx \beta$

∴ Output Impedance:-

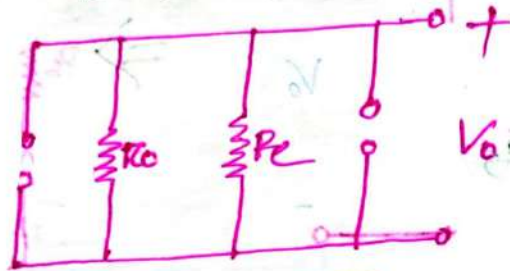
⇒ For Z_o we need to open circuit the R_L and short circuit the Source (V_s).

If $V_s = 0V \Rightarrow I_i = 0A \Rightarrow I_b = 0A$

Then $I_c = \beta I_b = 0$

So The Constant current Source Act as open circuit.

Now Draw the circuit For Z_o :-



$$\therefore Z_o = \left(\frac{1}{r_o} + \frac{1}{R_e} \right)^{-1} \Omega$$

OR $Z_o = \left(\frac{r_o + R_e}{r_o R_e} \right)^{-1} \Omega$

$$\Rightarrow Z_o = \frac{r_o R_e}{r_o + R_e}$$

$$\Rightarrow Z_o = \frac{r_o R_e}{r_o}$$

$$\therefore Z_o = R_e \Omega$$

If $r_o \geq 10 R_e$
So $r_o + R_e \approx r_o$

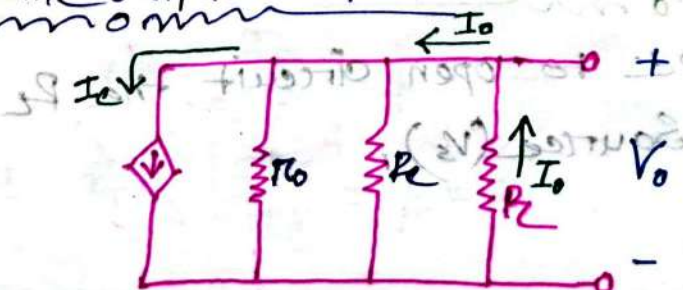
Voltage Gain (A_v) :-

$\Rightarrow$ we know:-

$$A_v = \frac{V_o}{V_i} \quad \text{--- (1)}$$

$\Rightarrow$ For Output voltage (V_o) :-

Draw the output side:-



$$\therefore V_o = -I_o R_L$$

$$V_o = - \frac{r_{o1} \parallel R_e}{R_L + r_{o1} \parallel R_e} \times I_o R_L$$

$$V_o = - \frac{(r_{o1} \parallel R_e) R_L}{R_L + r_{o1} \parallel R_e} \times \beta I_b$$

Condition: 01:-

if $r_{o1} \gg R_L$

sol
$$V_o = - \frac{R_L}{R_L + R_e} \times \beta I_b R_L$$

Condition: 02:-

if $R_L \gg R_e$

sol $R_L + R_e \approx R_L$

$$\therefore V_o = - \frac{R_e}{R_L} \times \beta I_b R_L$$

$$V_o = - R_e \beta I_b$$

For Input Voltage: (V_i) :-

$$V_i = (\beta + 1) r_{e1} I_b$$

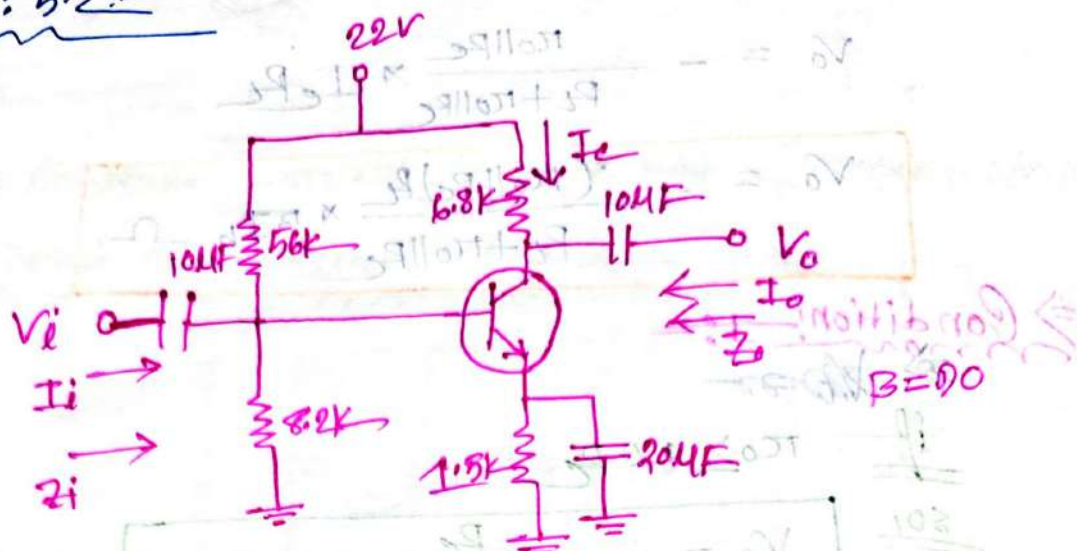
Now:- put the value of V_o & V_i in eq (1) $\Rightarrow$

$$A_v = \frac{-\beta I_b R_e}{(\beta + 1) r_{e1} I_b}$$

$$\therefore A_v = - \frac{R_e}{r_{e1}}$$

Since,
 $\beta \gg 1$
 $\therefore (\beta + 1) \approx \beta$

#problem: 5.2:-

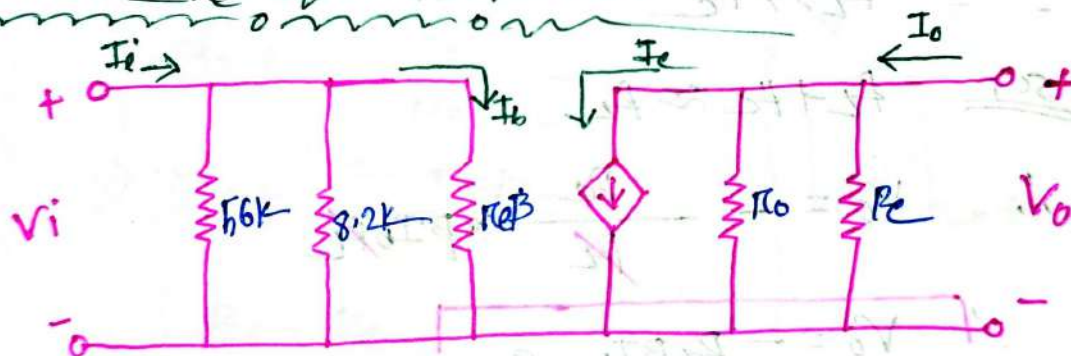


⇒ Determine

(i) r_e, z_i, z_o, A_v (with $r_o = \infty$)

(ii) Repeat (i) with $r_o = 50k\Omega$

⇒ Draw the r_e equivalent model:



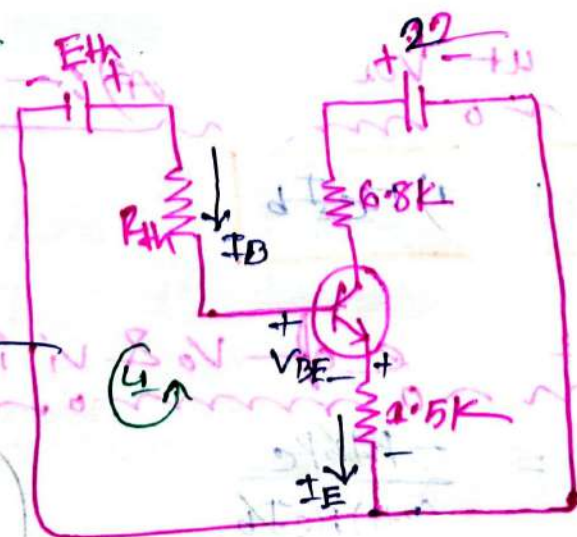
For I_E :-

$$R_{th} = \left(\frac{1}{56k} + \frac{1}{8.2k} \right)^{-1}$$

$$= 7.15k\Omega$$

$$E_{th} = \frac{8.2k}{(56 + 8.2)k} \times 22$$

$$= 2.8000V$$



#problem: 5.2: Contd:-

(i)

Now, Apply KVL in Loop (L1):

$$E_{Th} - I_{BTh} - V_{BE} - I_{E}R_E = 0$$

$$\Rightarrow E_{Th} - I_{BTh} - V_{BE} - (\beta + 1)I_{B}R_E = 0$$

$$\Rightarrow I_B = \frac{E_{Th} - V_{BE}}{R_{Th} + R_E(\beta + 1)}$$

$$\therefore I_B = \frac{2.8000 - 0.7}{7.15k + 1.5k \times 0.1}$$
$$= 14.61 \mu A$$

So,

$$I_E = (\beta + 1)I_B$$
$$= 61 \times 14.61 \mu A$$
$$= 0.9001 mA$$

Now,

$$r_{e} = \frac{26mV}{0.9001mA}$$
$$= 28.89 \Omega$$

(ii)

we know,

$$Z_i = \left(\frac{1}{56k} + \frac{1}{8.2k} + \frac{1}{28.89 \times 0.1} \right)^{-1}$$
$$= 1.02k \Omega$$

$$Z_o = \left(\frac{1}{r_o} + \frac{1}{R_E} \right)^{-1} = R_E$$

$$\therefore Z_o = 6.8k \Omega$$

Since
 $r_o = \infty$

Ans

(iii)

we know—

$$A_v = - \frac{R_c}{r_e}$$

$$= - \frac{8.2k}{(28.8\Omega)}$$

$$= -283.83$$

(b)

we know—

$$Z_o = \left(\frac{1}{r_o} + \frac{1}{R_c} \right)^{-1}$$

$$= \left(\frac{1}{50k} + \frac{1}{6.8k} \right)^{-1}$$

$$= 5.78k\Omega$$

Since,
 $r_o = 50k$

$$A_v = - \frac{R_c || Z_o}{r_e}$$

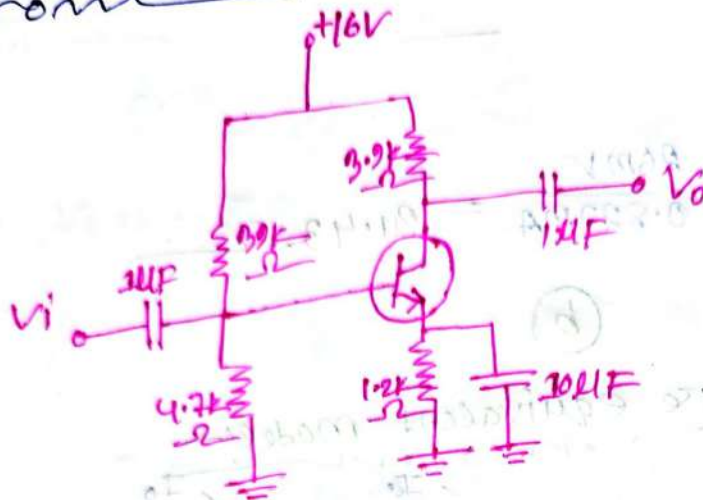
$$= - \frac{5.78k}{28.8\Omega}$$

$$= -200.77$$

Ans:

Subject: _____

problems 26



$\beta = 100$
 $r_o = 50k$

- ① Determine $r_{e'}$,
- ② Calculate Z_i, Z_o
- ③ Find A_v
- ④ Find $A_{v'}$

⇒ From DC Analysis: ①

For R_{th} —

$$R_{th} = \left(\frac{1}{9.9k} + \frac{1}{4.7k} \right)^{-1}$$

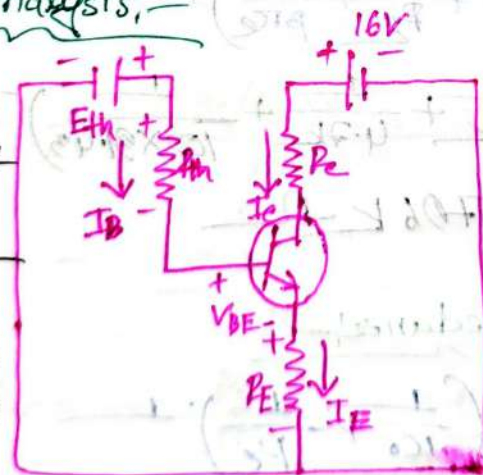
$$= 4.20k\Omega$$

For E_{th} —

$$E_{th} = \frac{4.7k}{(9.9k + 4.7k)} \times 16$$

$$= 1.72V$$

we know:—



here,

$$R_{th} = 4.20k\Omega$$

$$E_{th} = 1.72V$$

$$R_E = 1.2k\Omega$$

$$R_E = 1.2k\Omega$$

$$V_{BE} = 0.7V$$

$$\beta = 100$$

$$E_{th} - R_{th}I_B - V_{BE} - I_E R_E = 0$$

$$\therefore I_B = \frac{E_{th} - V_{BE}}{R_{th} + R_E(\beta + 1)}$$

$$\therefore I_B = \frac{1.72 - 0.7}{4.20k + 101 \times 1.2k} = 8.22\mu A$$

Since,

$$I_E = I_C + I_B$$

$$= \beta I_B + I_B$$

$$= I_B(\beta + 1)$$

$$\therefore I_E = (\beta + 1) I_B$$

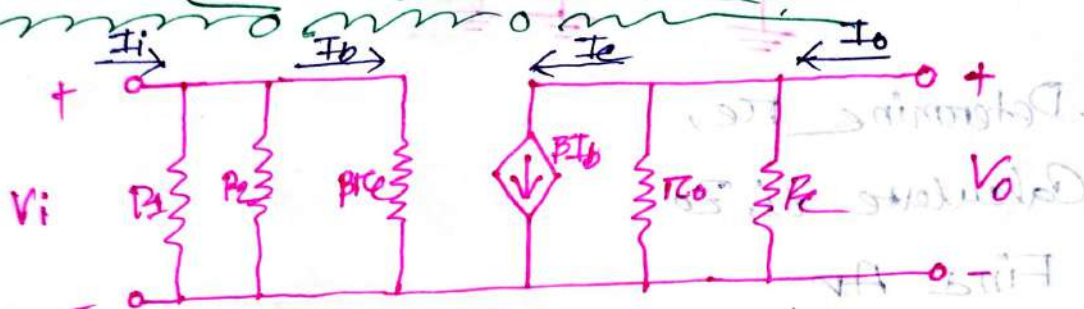
$$= 401 \times 8.10 \mu A$$

$$= 0.827 \text{ mA}$$

$$\therefore r_e = \frac{26 \text{ mV}}{0.827 \text{ mA}} = 31.43 \Omega$$

(b)

⇒ Draw the r_e equivalent model—



Input Impedance—

$$Z_i = \left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{\beta R_e} \right)^{-1}$$

$$= \left(\frac{1}{30k} + \frac{1}{4.7k} + \frac{1}{100 \times 31.43} \right)^{-1}$$

$$= 1.706 \text{ k}\Omega$$

Output Impedance—

$$Z_o = \left(\frac{1}{r_e} + \frac{1}{R_2} \right)^{-1}$$

$$= \left(\frac{1}{50k} + \frac{1}{3.9k} \right)^{-1}$$

$$= 3.61 \text{ k}\Omega$$

OR

Apply Condition: $r_e \gg 10R_e$

$$\therefore Z_o = R_e = 3.9 \text{ k}\Omega$$

(3)

We know—

$$A_v = \frac{V_o}{V_i} \quad \text{--- ①}$$

$$\therefore V_o = -I_o R_L$$

$$= - \frac{\beta I_b R_L}{\beta I_b R_L + R_L} \times R_L \times I_b$$

$$= - \frac{\beta I_b R_L}{\beta I_b R_L + R_L} \times R_L \times \beta I_b$$

$$V_i = I_b \times \beta R_L$$

put the value of V_o & V_i in eq ① $\Rightarrow$

$$\therefore A_v = - \frac{\beta (\beta I_b R_L) / (\beta I_b R_L + R_L)}{\beta I_b R_L}$$

$$\Rightarrow A_v = - \frac{R_L R_L / R_L + R_L}{R_L}$$

$$\Rightarrow A_v = - \frac{R_L R_L}{R_L + R_L}$$

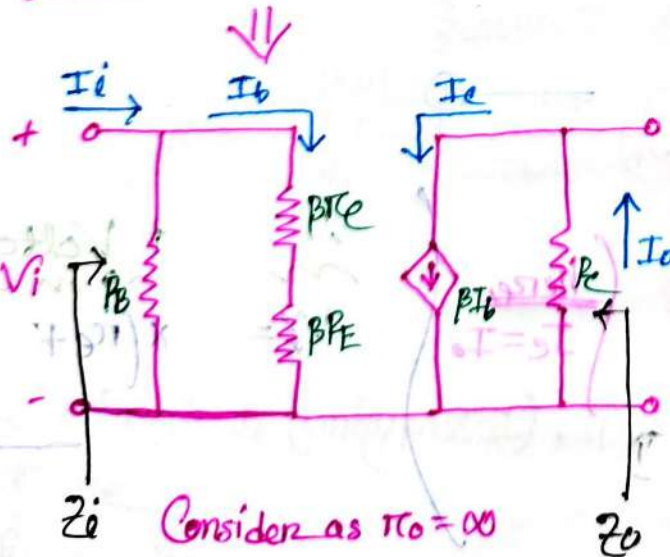
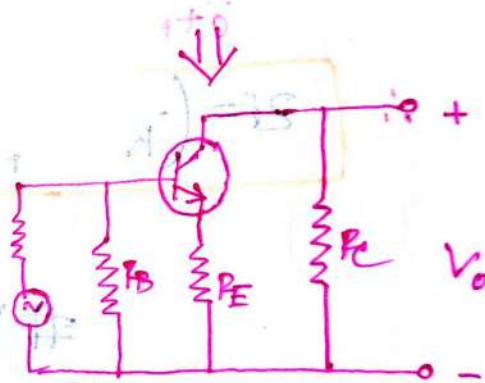
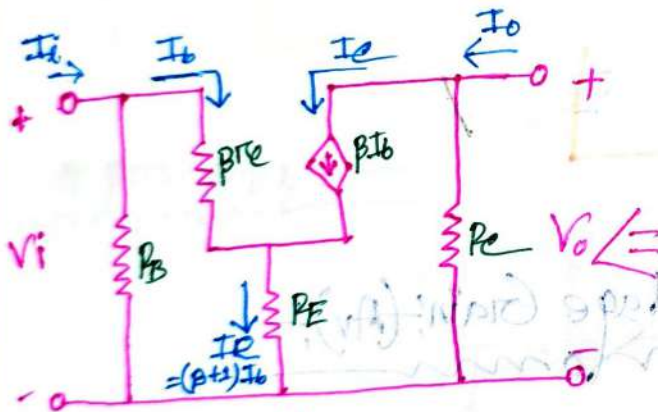
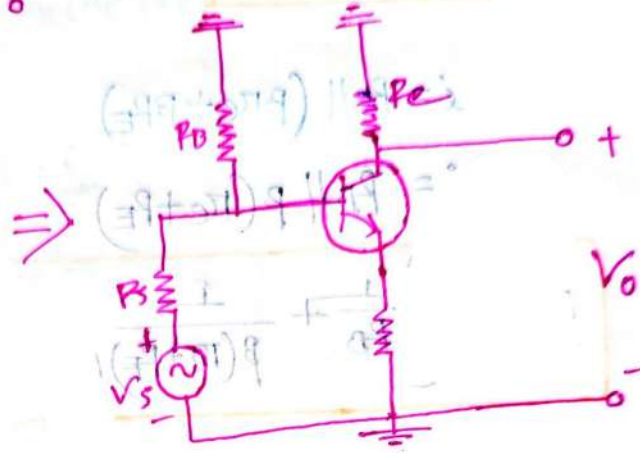
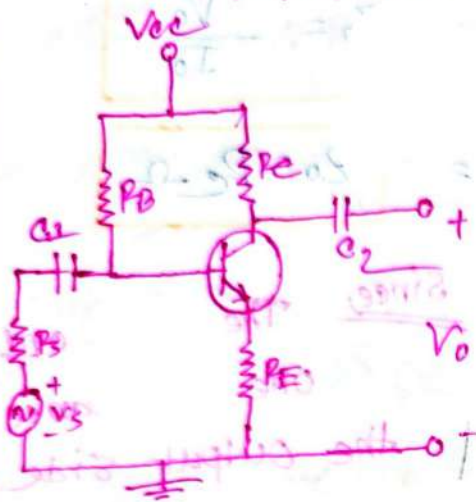
$$\underline{\text{So,}} \quad A_v = - \frac{R_L}{R_L} = - \frac{3.0K}{3.143}$$

$$= -124.08.$$

Ans:—

Since $R_L > 1R_L$ so we can neglect R_L
 Also, $R_L = \infty$
 $\therefore R_L + R_L = R_L$

Common - Emitter Bias Configuration of Unbypassed R_E



Consider as $\pi_o = \infty$

hence

$$I_E = I_C + I_B$$

$$\therefore I_E = I_C + I_B$$

$$= \beta I_B + I_B$$

$$= (\beta + 1) I_B$$

$\therefore$ Drop Across R_E :-

$$V_{RE} = R_E \times (\beta + 1) I_B$$

Resistance of R_E $\times$ Current flow through R_E

∴ Input Impedance:

$$Z_i = \frac{V_i}{I_i}$$

OR $Z_i = R_B \parallel (\beta r_e + \beta R_E)$

$$\Rightarrow Z_i = R_B \parallel \beta (r_e + R_E)$$

$$\Rightarrow Z_i = \left\{ \frac{1}{R_B} + \frac{1}{\beta (r_e + R_E)} \right\}^{-1} \Omega$$

Then, $R_E \gg r_e$

$r_e + R_E \approx R_E$

$$Z_i = \left(\frac{1}{R_B} + \frac{1}{\beta R_E} \right)^{-1} \Omega$$

∴ Output Impedance:

$$Z_o = \frac{V_o}{I_o}$$

OR $Z_o = R_E \Omega$

Since There is only one resistance into the output side.

Voltage Gain: (A_v):

We know:

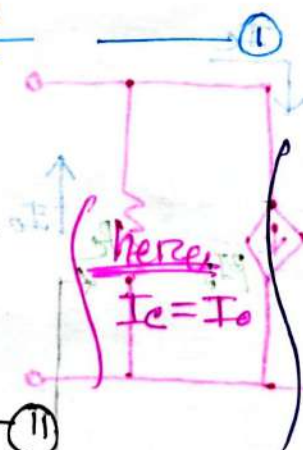
$$A_v = \frac{V_o}{V_i}$$

∴ Output Voltage:

$$V_o = -I_o R_E$$

$$\Rightarrow V_o = -I_e R_E$$

$$\Rightarrow V_o = -\beta I_b R_E \quad \text{--- (ii)}$$



∴ Input Voltage:

$$V_i = I_b \times (r_e + R_E) \beta$$

(iii)

Now! put the value of V_o & V_i in eq ① $\Rightarrow$

$$\therefore A_v = - \frac{\beta I_b R_c}{\beta I_b (r_e + R_E)}$$

$$A_v = - \frac{R_c}{r_e + R_E}$$

The Condition:

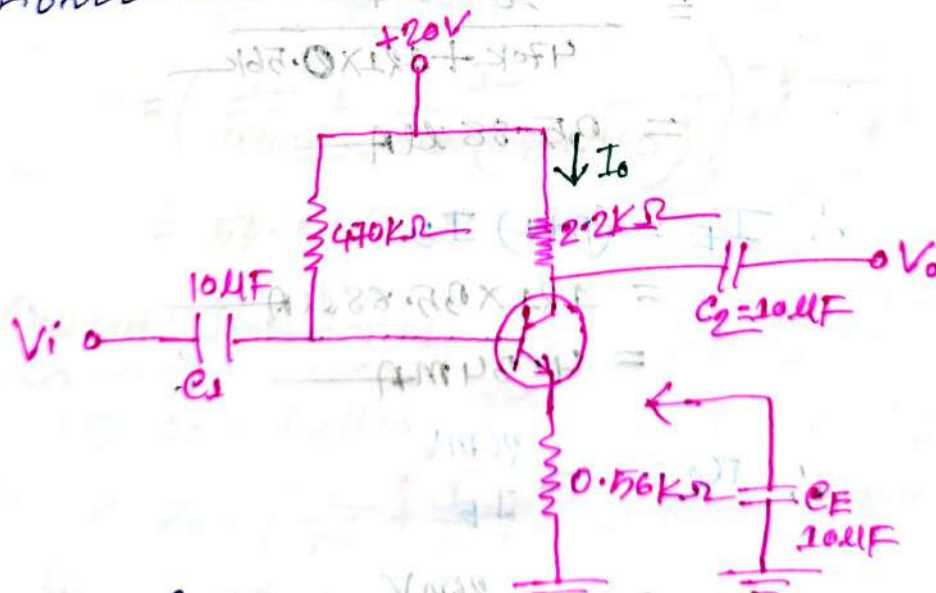
if $R_E \gg r_e$

Then $r_e + R_E \approx R_E$

So,

$$A_v = - \frac{R_c}{R_E}$$

problems 5.3:-

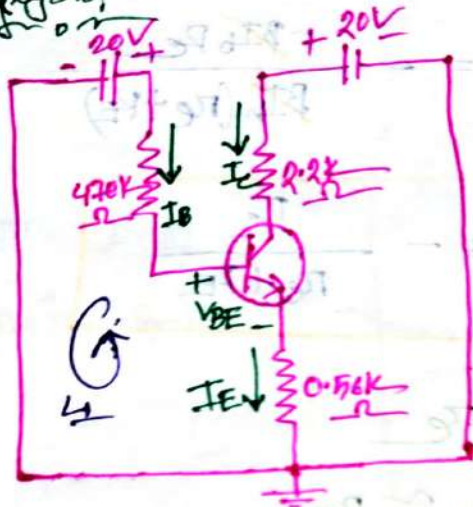


$\beta = 120$
 $r_o = 40K\Omega$

$\Rightarrow$ without C_E (unbypassed), Determine

- r_e
- Z_i, Z_o
- A_v

From DC Analysis: (a)



here,

$R_B = 470k\Omega$
 $R_E = 0.56k\Omega$
 $R_C = 2.2k\Omega$
 $\beta = 120$
 $V_{BE} = 0.7V$
 $V_{CC} = 20V$

Apply KVL in loop (L1):

$$V_{CC} - I_B R_B - V_{BE} - I_E R_E = 0$$

$$\Rightarrow I_B = \frac{V_{CC} - V_{BE}}{R_B + (\beta + 1) R_E}$$

$$= \frac{20 - 0.7}{470k + 121 \times 0.56k}$$

$$= 35.88 \mu A$$

$$\therefore I_E = (\beta + 1) I_B$$

$$= 121 \times 35.88 \mu A$$

$$= 4.34 mA$$

$$\therefore r_e = \frac{26mV}{I_E}$$

$$= \frac{26mV}{4.34 mA}$$

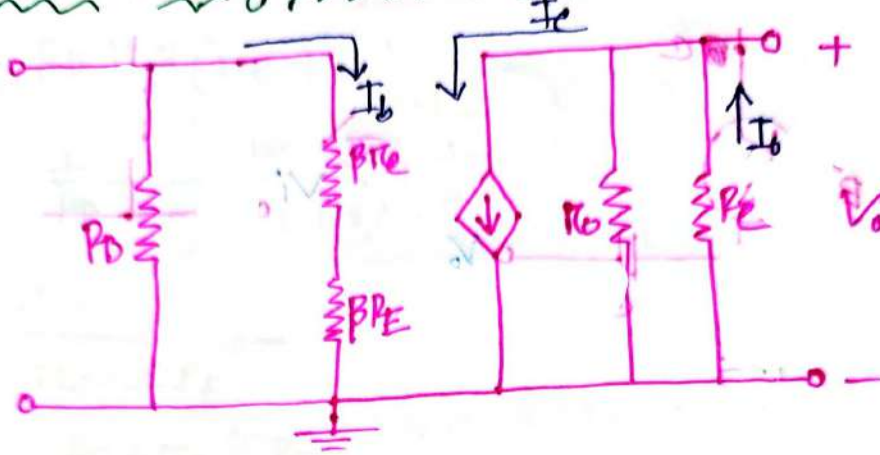
$$= 5.99 \Omega$$

Ans:-

(6)

we know, —

1st draw the re equivalent model —

∴ Input Impedance: —

$$\Rightarrow Z_i = R_B \parallel \beta(R_E + r_e)$$

$$\Rightarrow Z_i = \left(\frac{1}{R_B} + \frac{1}{\beta(R_E + r_e)} \right)^{-1}$$

$$= \left(\frac{1}{470k} + \frac{1}{120(5.79 + 0.56k)} \right)^{-1}$$

$$= 59.34 k\Omega$$

∴ Output Impedance: —

$$\Rightarrow Z_o = r_o \parallel R_E$$

$$\Rightarrow Z_o = \left(\frac{1}{r_o} + \frac{1}{R_E} \right)^{-1}$$

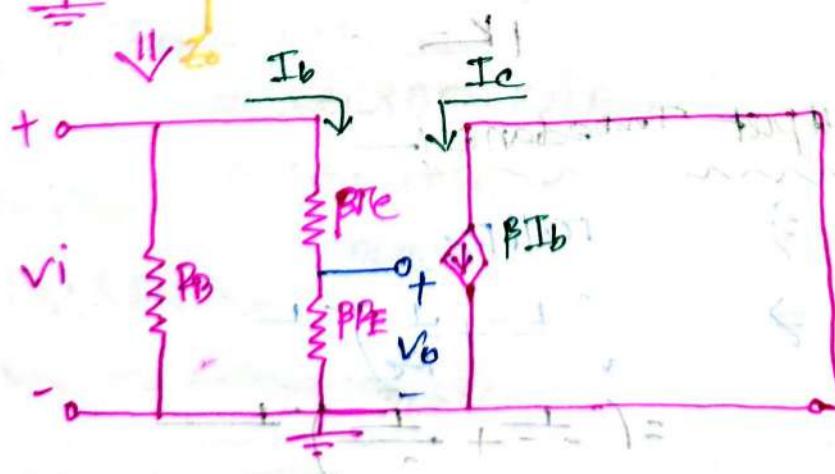
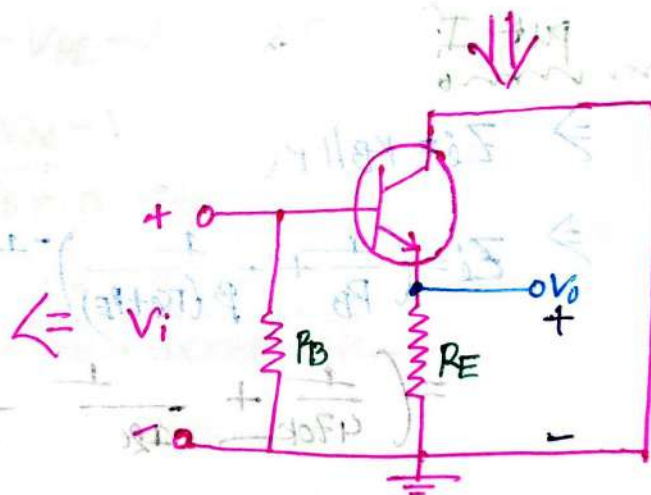
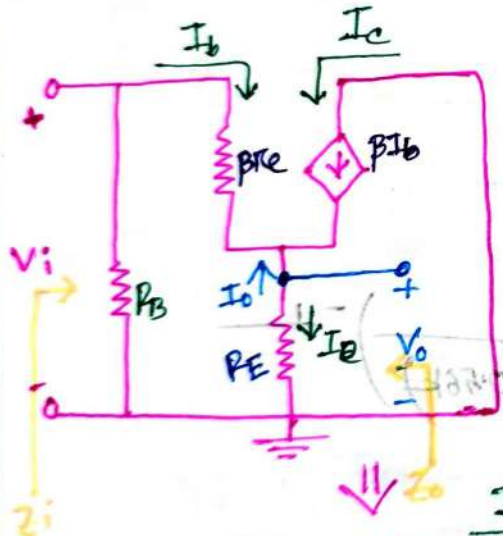
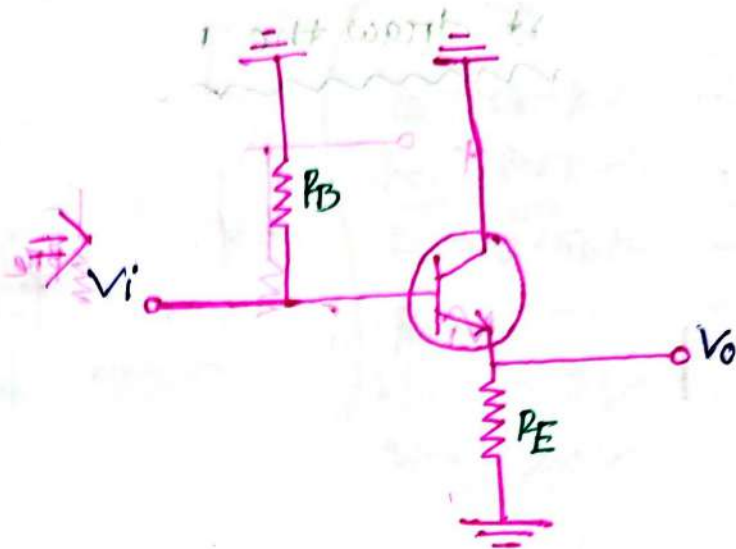
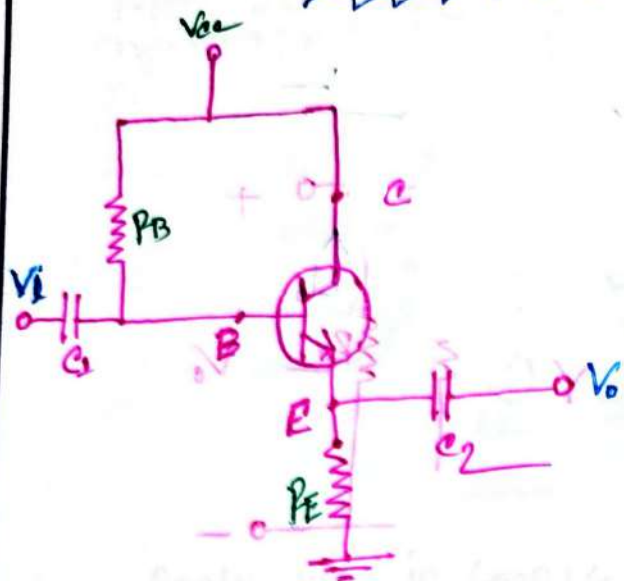
$$= \left(\frac{1}{40k} + \frac{1}{2.2k} \right)^{-1}$$

$$= 2.08 k\Omega$$

or Apply Condition ($r_o \gg 10 R_E$): —

$$\therefore Z_o = R_E = 2.2 k\Omega \approx 2.08 k\Omega$$

Emitter-Follower Configuration:-



Since

$$I_E = I_C + I_B$$

$$= \beta I_B + I_B$$

$$= (\beta + 1) I_B$$
 Drop Across V_{BE}

$$V_{BE} = I_E \times (\beta + 1) R_E$$
 Current flow the R_E
 Resistance of R_E

Input Impedance:-

$$Z_i = R_B \parallel (\beta r_e + \beta R_E)$$

$$\Rightarrow Z_i = R_B \parallel \beta (r_e + R_E)$$

$$\Rightarrow Z_i = \left(\frac{1}{R_B} + \frac{1}{\beta (r_e + R_E)} \right)^{-1}$$

Condition:-

if $r_e \ll R_E$

Then $R_E + r_e \approx R_E$

$$\text{So, } Z_i = \left(\frac{1}{R_B} + \frac{1}{\beta R_E} \right)^{-1}$$

Output Impedance:-

we know:-

$$I_b = \frac{V_i}{\beta (r_e + R_E)}$$

$$I_e = (\beta + 1) I_b$$

$$= \frac{(\beta + 1) V_i}{\beta (r_e + R_E)}$$

$$= \frac{\beta (V_i)}{\beta (r_e + R_E)}$$

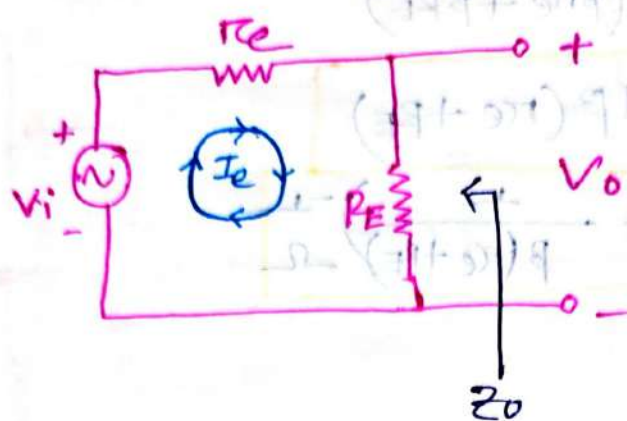
Since,

$\beta \gg 1$
So, $\beta + 1 \approx \beta$

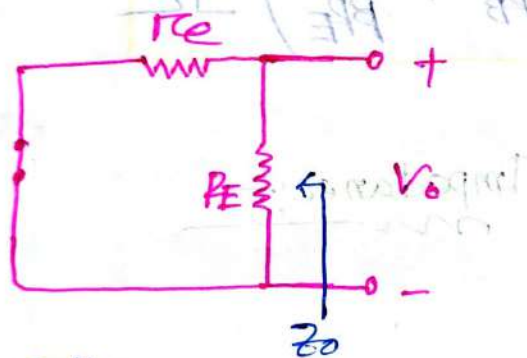
$$\therefore I_e = \frac{V_i}{r_e + R_E}$$

①

⇒ By using the equation (1) we can draw the network.



→ Now To calculate the output Impedance we have to short the input Source V_i .



$$\therefore Z_o = r_e \parallel R_E$$

$$\Rightarrow Z_o = \left(\frac{1}{r_e} + \frac{1}{R_E} \right)^{-1}$$

Condition:
mon

if $R_E \gg r_e$
then $R_E + r_e \approx R_E$

$$\therefore Z_o = \left(\frac{r_e + R_E}{r_e R_E} \right)^{-1}$$

$$Z_o = \frac{r_e R_E}{R_E} = r_e \Omega$$

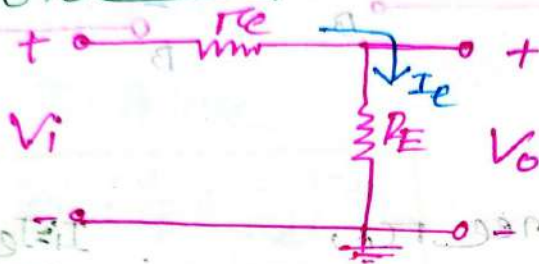
$$\therefore Z_o \approx r_e \Omega$$

Voltage Gain, (A_v) :-

We know:-

$$A_v = \frac{V_o}{V_i} \quad \text{--- (1)}$$

∴ output voltage:-



$$\therefore V_o = I_e \times R_E$$

$$\Rightarrow V_o = \frac{V_i}{R_E + r_e} \times R_E$$

$$\Rightarrow V_o = \frac{V_i}{R_E} \times R_E$$

$$\therefore V_o = V_i$$

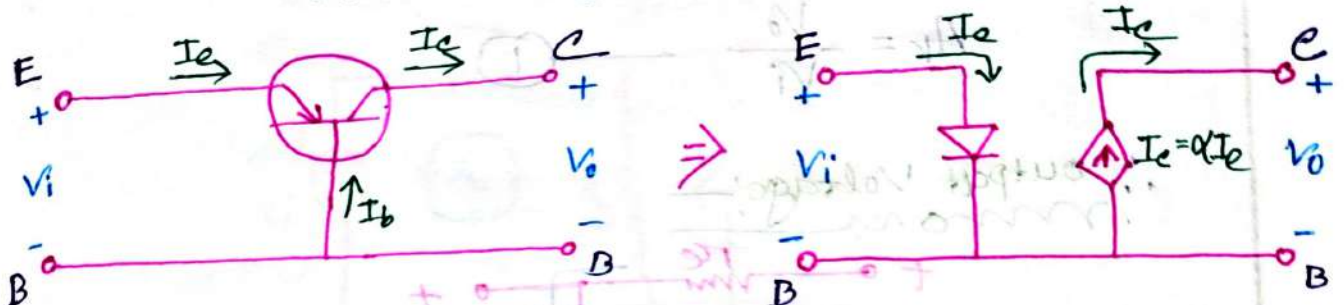
Put the value of V_o in eq. (1) $\Rightarrow$

$$\therefore A_v = \frac{V_o}{V_i} = 1$$

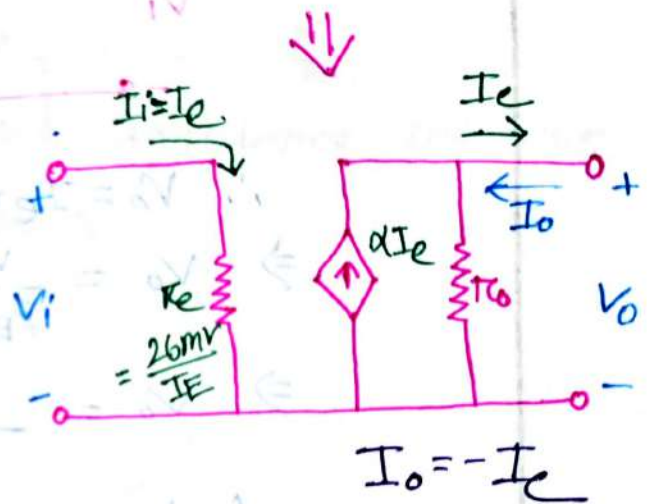
$$\therefore \boxed{A_v \approx 1}$$

Since, $R_E \gg r_e$
So, $R_E + r_e \approx R_E$

r_e model for Common Base - Configuration:

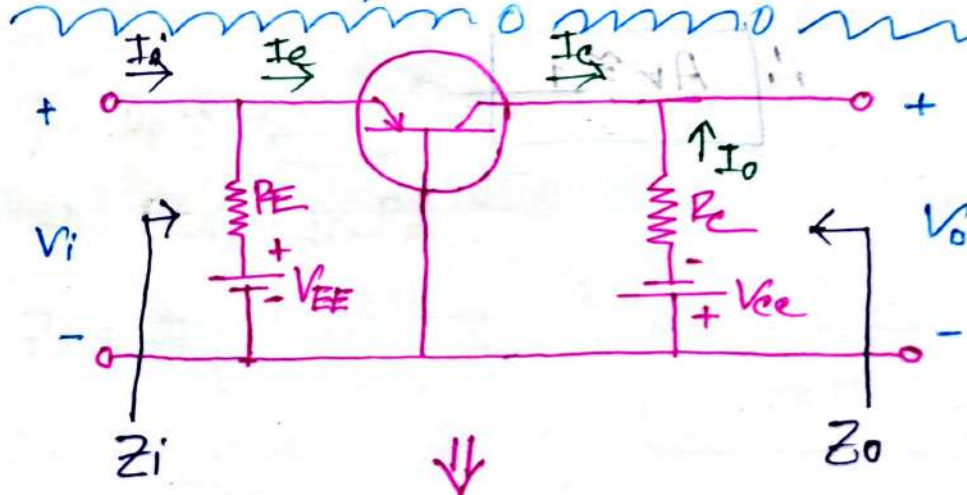


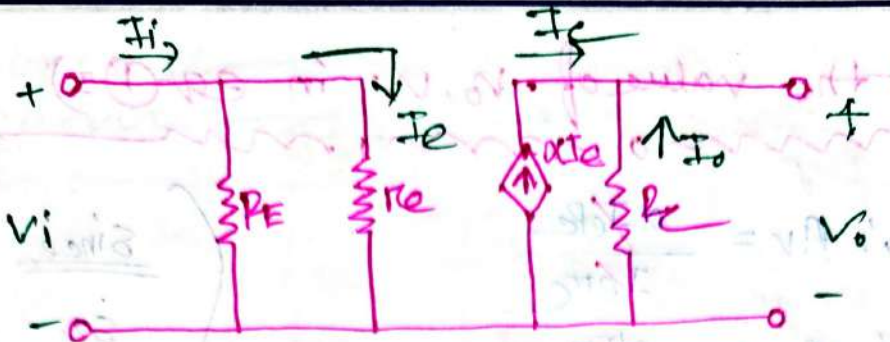
⇒ The output resistance r_o is quite high. Typically extend into the $M\Omega$ range. So we can neglect r_o from the circuit, or we can denote that r_o as open circuit.



This is the r_e equivalent model.

Calculation of Input & Output Impedance, Voltage Gain, Current Gain for Common - Base





∴ Input Impedance: (Z_i) :-

$$Z_i = R_E \parallel r_e$$

$$\Rightarrow Z_i = \left(\frac{1}{R_E} + \frac{1}{r_e} \right)^{-1}$$

∴ Output Impedance: (Z_o) :-

$$Z_o = R_C$$

Now:- Voltage Gain (A_v) :-

we know:-

$$A_v = \frac{V_o}{V_i}$$

∴ Output Voltage:-

$$V_o = -I_o R_C$$

$$\Rightarrow V_o = -(-I_e R_C)$$

$$= I_e R_C$$

$$= \alpha I_e R_C$$

Input Voltage:-

$$V_i = I_e r_e$$

we know:-

$$\Rightarrow I_o = -I_c$$

$$\Rightarrow I_c = \alpha I_e$$

Now put the value of V_o, V_i in eq (1) $\Rightarrow$

$$\therefore A_v = \frac{\alpha I_e R_e}{I_e R_e}$$

$$\therefore A_v = \frac{\alpha R_e}{R_e}$$

$$\underline{\text{So, } A_v = \frac{R_e}{R_e}}$$

Since $R_e \gg \alpha$

So, $R_e \alpha \approx R_e$

Now Current Gain (A_i) :-

we know:-

$$A_i = \frac{I_o}{I_i}$$

$\therefore$ output current:-

$$I_o = -I_e$$

$$\Rightarrow I_o = -\alpha I_e$$

$$\Rightarrow I_o = -\alpha I_i$$

Since

$R_e \gg R_e$

So

$I_i \approx I_e$

Now put the value of I_o in eq (2) $\Rightarrow$

$$A_i = \frac{-\alpha I_i}{I_i}$$

$$\underline{A_i = -\alpha \approx -1}$$

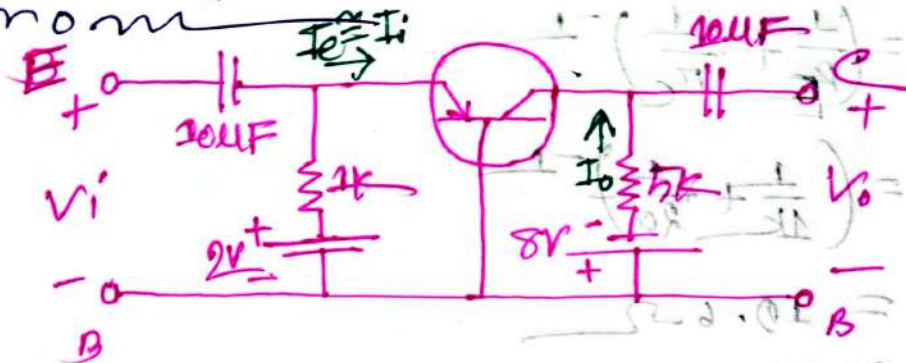
Since

$I_i \gg \alpha$

So

$I_i \alpha \approx I_i$

problem 5.8:—



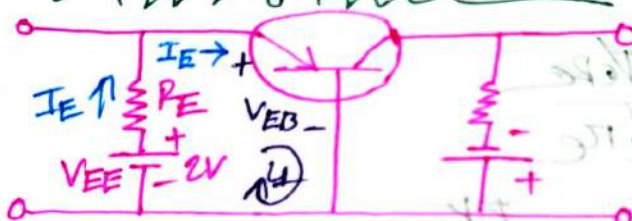
$$\alpha = 0.98$$

$$r_o = 1M\Omega$$

⇒ Determine, r_e , z_i , z_o , A_v , A_i

①

For I_E r_e draw the circuit:—



⇒ Now apply KVL in Loop 1:—

$$V_{EE} - I_E R_E - V_{EB} = 0$$

$$\therefore I_E = \frac{V_{EE} - V_{EB}}{R_E}$$

$$\therefore I_E = \frac{2 - 0.7}{1k} = 1.3mA$$

$$\therefore r_e = \frac{26mV}{1.3mA}$$

$$r_e = 20\Omega$$

here,

$$V_{EE} = 2V$$

$$R_E = 1k\Omega$$

$$V_{EB} = 0.7V$$

we know:-

$$Z_i = \left(\frac{1}{R_E} + \frac{1}{r_e} \right)^{-1}$$

$$= \left(\frac{1}{1k} + \frac{1}{20} \right)^{-1}$$

$$= 10.6 \Omega$$

$$\approx r_e$$

$$Z_o = R_E = 5k\Omega \quad (1)$$

$$A_v = \frac{V_o}{V_i}$$

$$= \frac{\alpha R_E}{R_E + r_e}$$

$$\approx \frac{R_E}{R_E} = \frac{5k}{20} = 250$$

$$A_i \approx -\alpha = 0.98 \approx 1$$

Ans:-

Field-Effect
Transistors

on

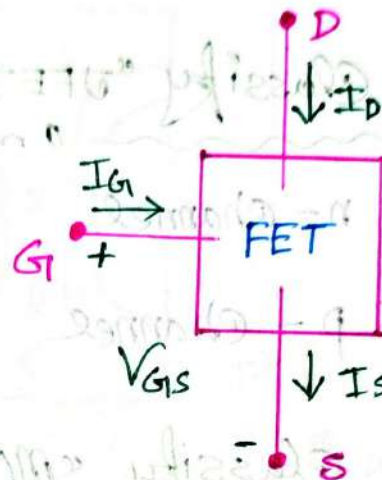
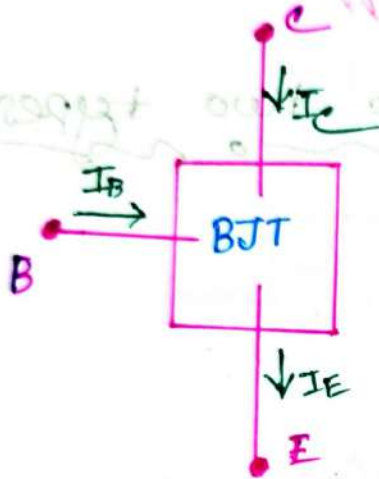
Chopin

Introduction to FET

⇒ FET = Field-Effect Transistor.

FET's Vs BJT's

1. Like BJT's, FET's also three terminal Device.



D = Drain
G = Gate
S = Source.

2. BJT's = Current Control Device.

2. FET's = Voltage Control Device.

3. $I_C = \beta(I_B)$

3. $I_D = f(V_{GS})$

4. BJT's = Bipolar Device

4. FET's = Unipolar Device.

5. Basically 2 types.

5. Also 2 types.

i) npn → e^-

i) n-channel → e^-

ii) pnp → holes

ii) p-channel → holes.

6. BJT's > FET's
Sensitive & size

FET: mon

⇒ There is two types of Classification:-

1. JFET (Junction Field-Effect Transistor)

2. MOSFET (Metal-Oxide-Semiconductor Field-Effect Transistor)

⇒ We can classify "JFET" Into two types:-

i) n-channel

ii) p-channel

⇒ We can classify "MOSFET" Into two types:-

i) D. MOSFET

→ n-channel

→ p-channel

ii) E. MOSFET

→ n-channel

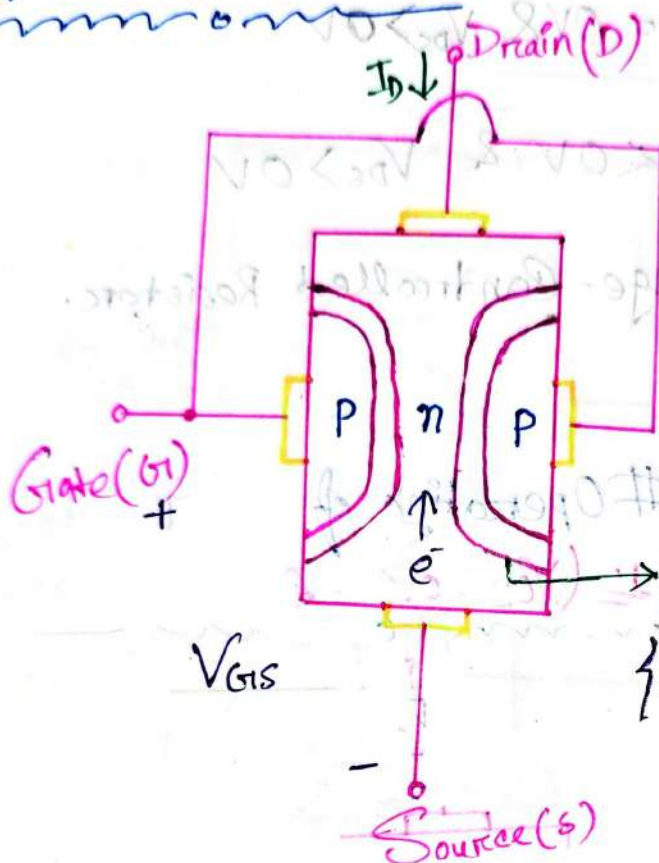
→ p-channel

⇒ D. MOSFET = Depletion type MOSFET

⇒ E. MOSFET = Enhancement type MOSFET

Construction and Working of JFET

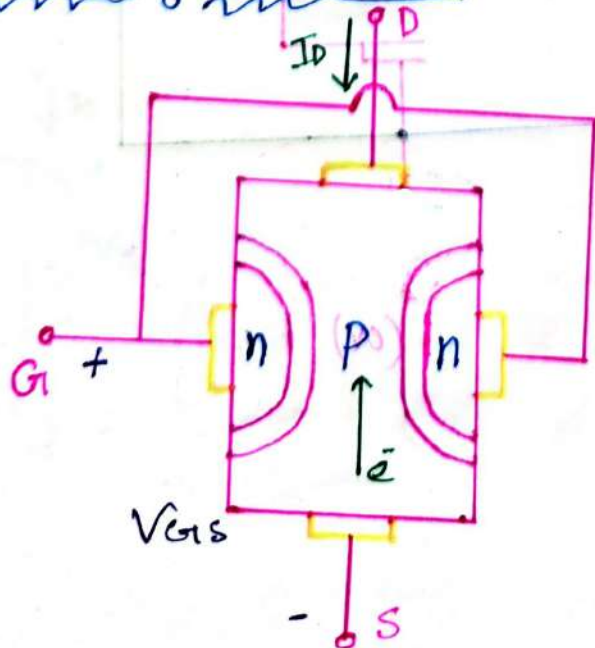
n-channel JFET:



Depletion Regions (D.R.)

if we increasing the D.R. it reduces the n-channel width obstructing e^- flow. The Gate-Source Voltage (V_{GS}) Controls the D.R. there by Controlling the output current I_D

p-channel JFET:



JFET Operating Conditions:-

⇒ there are three basic operating condition.

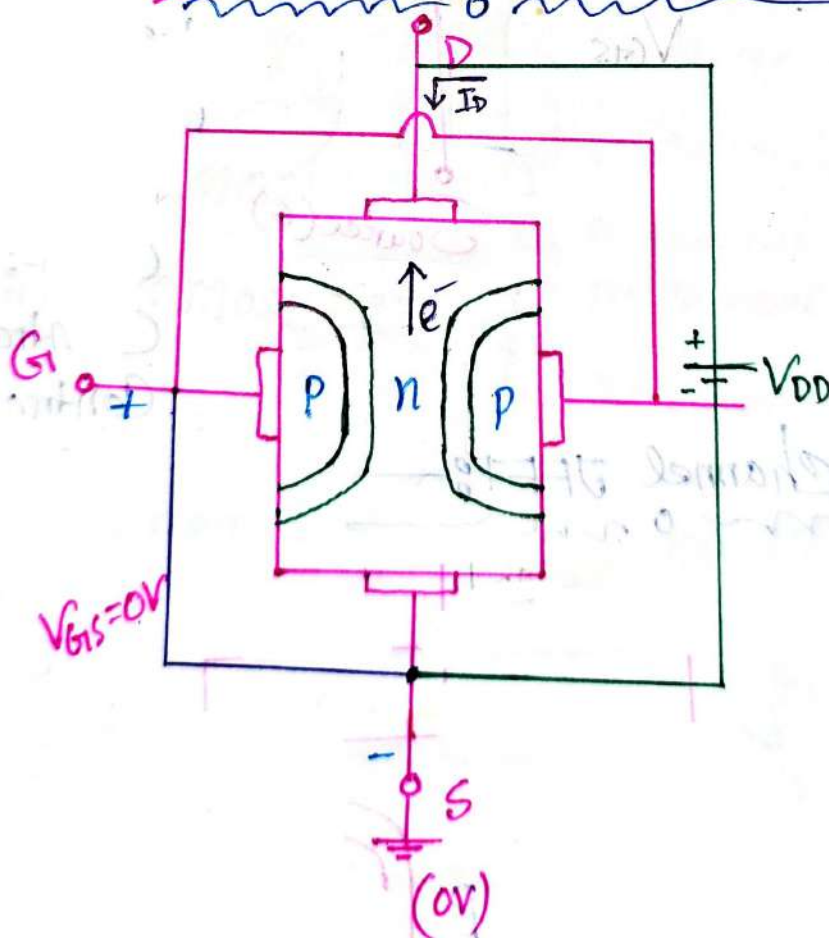
1. $V_{GS} = 0V$ & $V_{DS} > 0V$

2. $V_{GS} < 0V$ & $V_{DS} > 0V$

3. Voltage-Controlled Resistor.

Operating of JFET

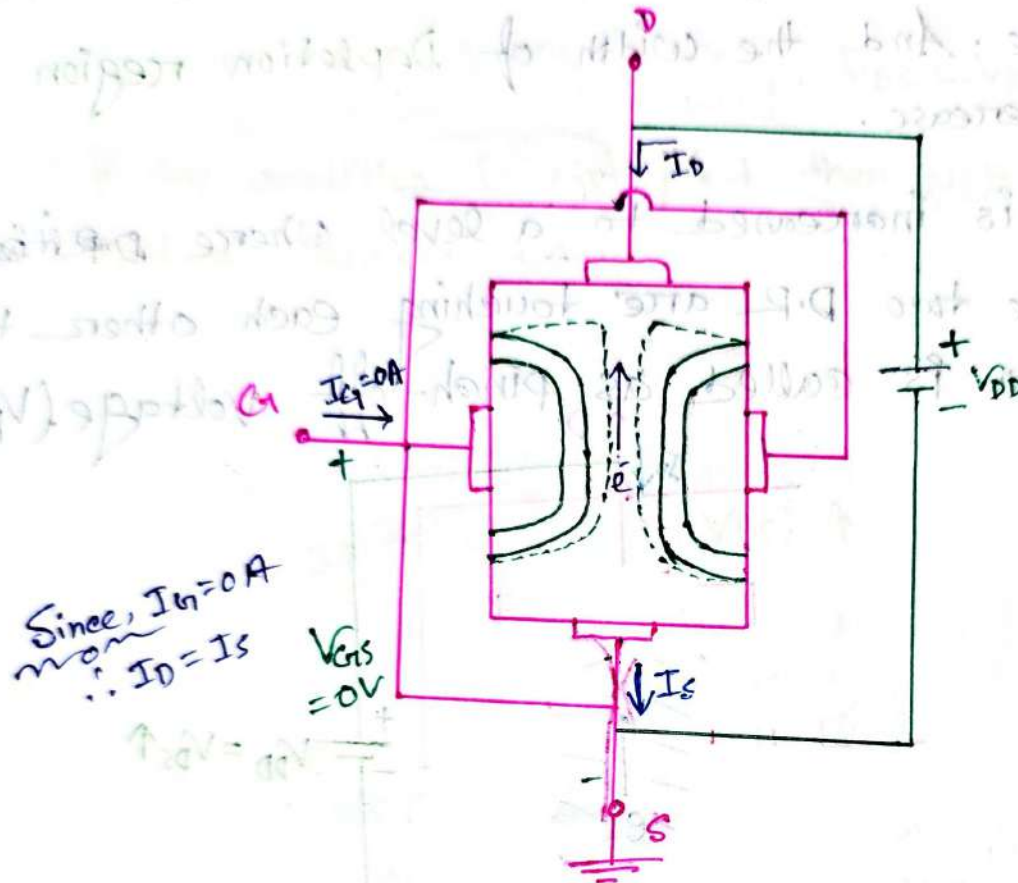
1. ($V_{GS} = 0V$ & $V_{DS} > 0V$)



Since,
 $V_{GS} = 0V$

Since
 $V_{DD} = V_{DS}$
 $\downarrow$
 $\oplus V_e$

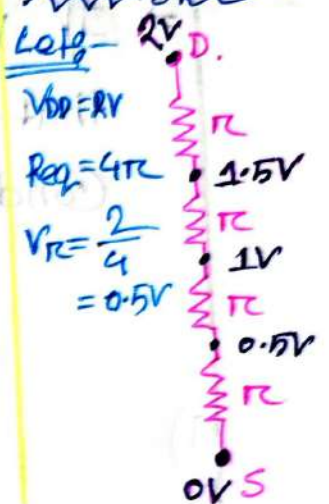
⇒ When we connect V_{DD} in the circuit the width of depletion layer will change.



Since, $I_{G1} = 0A$
 $\therefore I_D = I_S$

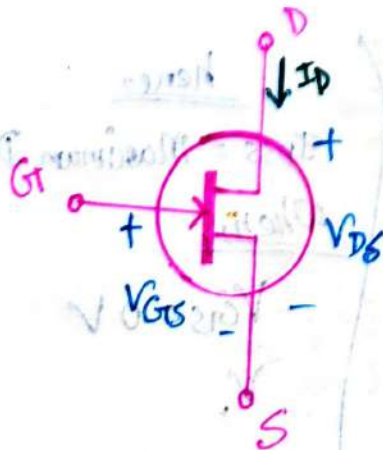
$V_{GS} = 0V$

This happens because of mon

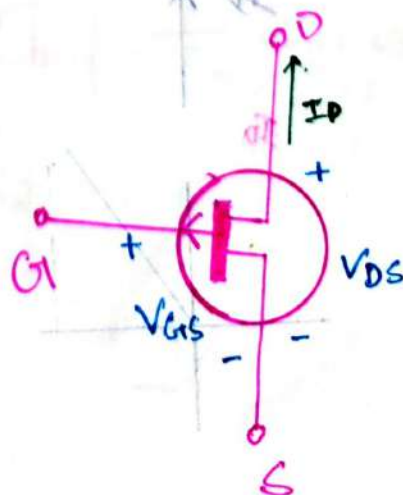


Symbol of JFET

n-channel:



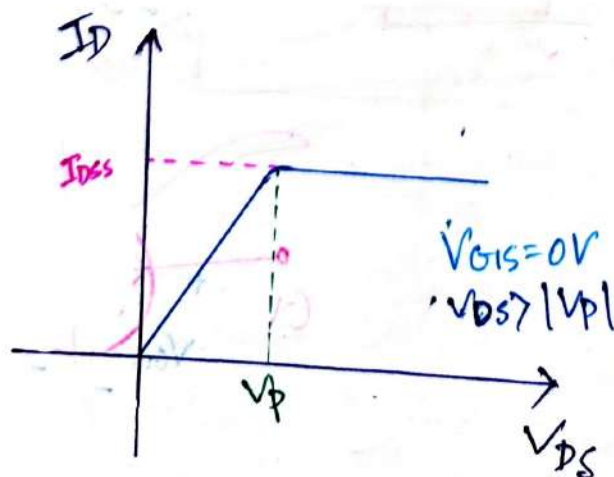
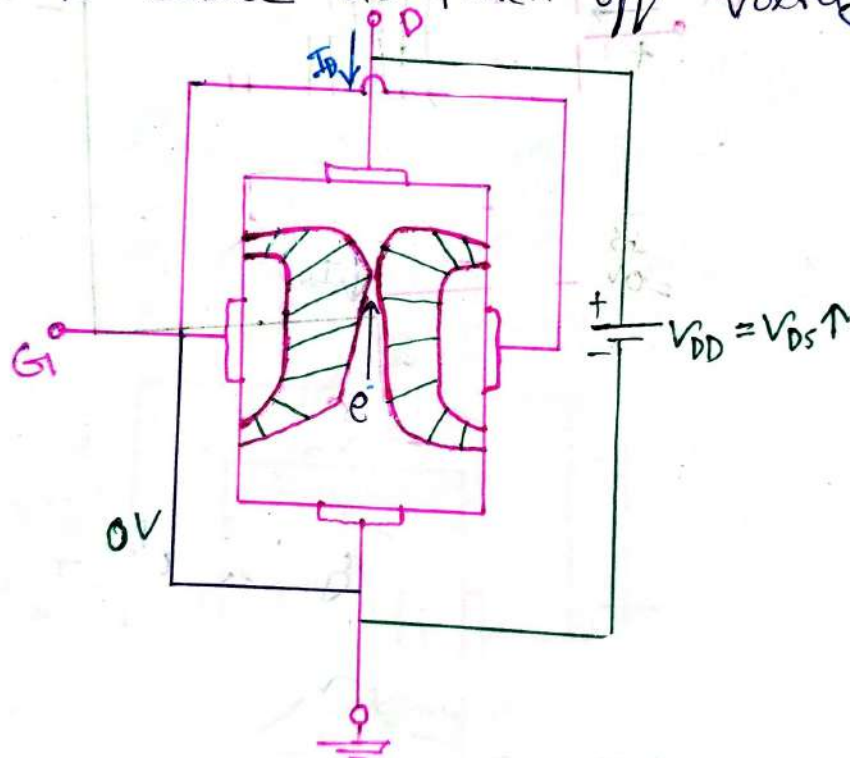
p-channel:



Pinch-off Voltage:-

⇒ If we Increasing V_{DD} the Current I_D will also increase. And the width of Depletion region will also increase.

⇒ If V_{DD} is increased to a level where it appears that the two D.R are touching each other the Condition is called as pinch-off voltage (V_p)



here

I_{DSS} = Maximum Drain Current

When,

$$V_{GS} = 0V$$

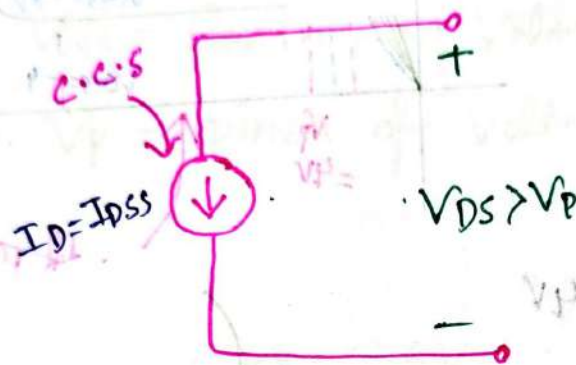
$$V_{DS} > |V_p|$$

JFET as a Current Source

The Condition:-

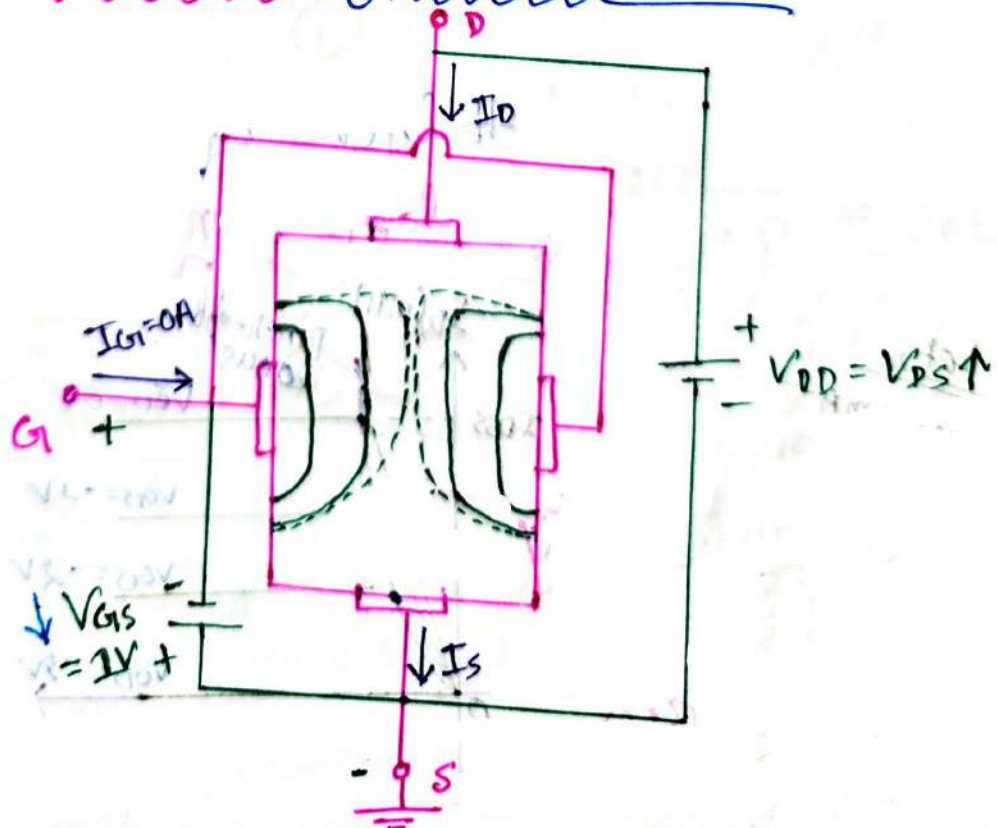
$$I_D = I_{DSS}, \quad V_{GS} = 0V, \quad V_{DS} > V_P, \quad V_{DS} < V_{DS(max)}$$

⇒ if the condition is satisfied then JFET acting as a Constant Current Source.



Operating of JFET

$$\cong V_{GS} \leq 0V \text{ \& } V_{DD} > 0V$$

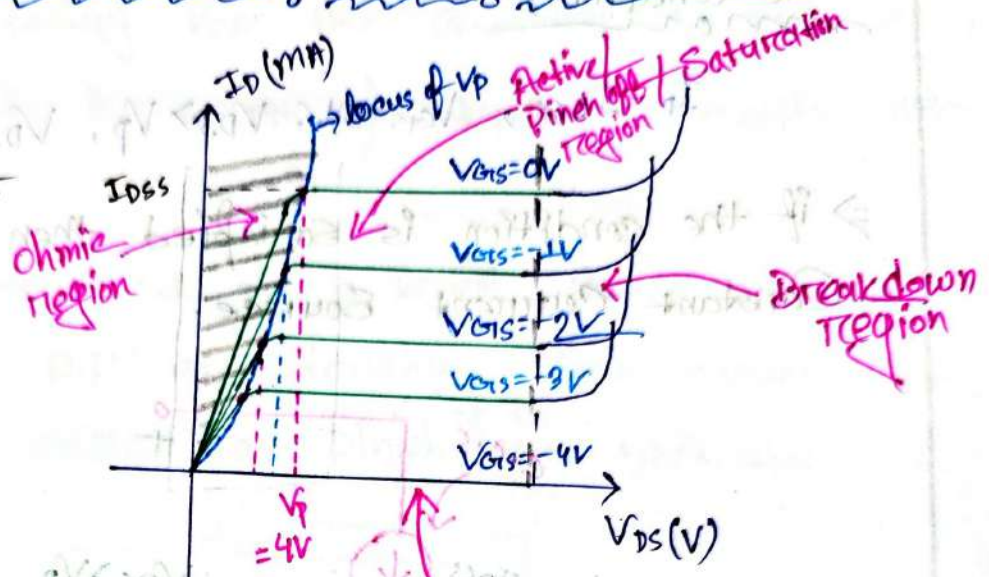


Output of JFET OR Drain Characteristics of JFET

here,

I_{DS} = Output Current

V_{DS} = Output Voltage.



Let,

$$V_P = -4V$$

$$I_{DSS} = 8mA$$

$$V_{GS} = 0V, \&$$

$$V_{DS} > |V_P|$$

Cut-off region

n-ch $\rightarrow \oplus V_e$
p-type $\rightarrow \ominus V_e$ } $\pi.b$

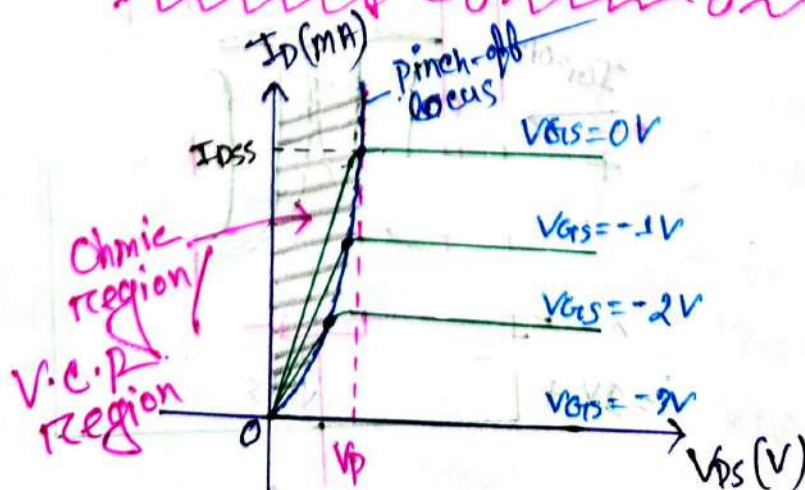
Operating of JFET

2. Voltage Controlled Resistor (V.C.R)

Let,

$$I_{DSS} = 6mA$$

$$V_P = -3V$$



if

$$V_{GS} \gg 0V$$

Then: $r_{DS} \uparrow$

∴ the resistance at a particular level of V_{GS}

$$r_d = \frac{r_{co}}{\left(1 - \frac{V_{GS}}{V_p}\right)^2}$$

where,

r_{co} = Resistance at $V_{GS} = 0V$.

V_{GS} = Controlling Voltage

V_p = pinch of voltage.

problem 1:—

⇒ if $r_{co} = 10k\Omega$, $V_p = -6V$, $V_{GS} = 0V$

① find r_d

② Repeat (1) for $V_{GS} = -3V$

①

we know:—

$$r_d = \frac{r_{co}}{\left(1 - \frac{V_{GS}}{V_p}\right)^2} = \frac{10k}{\left(1 - \frac{0}{-6}\right)^2} = 10k\Omega$$

②

For $V_{GS} = -3V$ —

$$r_d = \frac{r_{co}}{\left(1 - \frac{V_{GS}}{V_p}\right)^2} = \frac{10k}{\left(1 - \frac{-3}{-6}\right)^2} = 40k\Omega$$

Transfer Characteristics of JFET

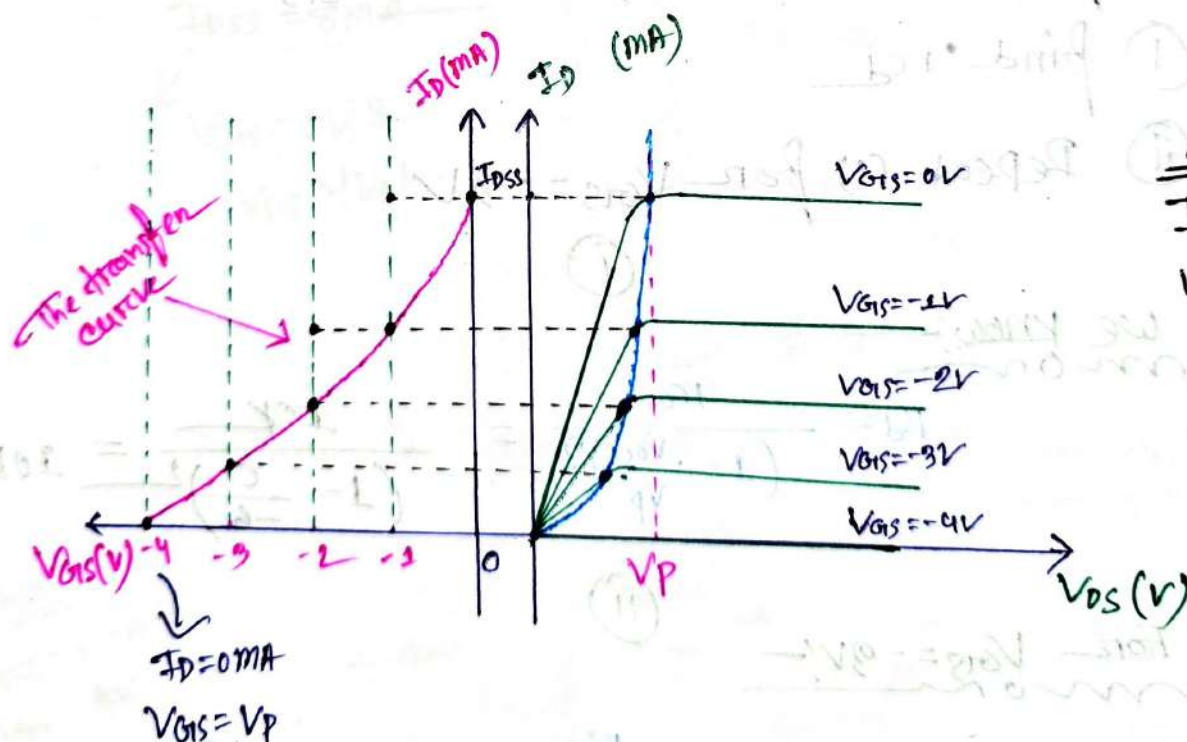
⇒ It is the I_D vs V_{GS} plot keeping V_{DS} constant.

$$\therefore I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_P} \right)^2$$

$\downarrow$ Constants $\downarrow$ Control Variable

if $V_{GS} \uparrow \leq 0V$
 Then $I_D \uparrow$ (exp)

Transfer Curve



Let,
 $I_{DSS} = 8mA$
 $V_P = -4V$

Subject _____

Date: / /

problems:-

⇒ If $I_{DSS} = 10 \text{ mA}$, $V_p = -4 \text{ V}$. Then calculate Drain current for.

① $V_{GS} = 0 \text{ V}$

② $V_{GS} = -1 \text{ V}$

③ $V_{GS} = -4 \text{ V}$

①

⇒ we know:-

$$I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_p} \right)^2$$

$$\therefore I_D = 10 \text{ mA} \left(1 - \frac{0}{-4} \right)^2$$

$$= 10 \text{ mA}$$

②

For, $V_{GS} = -1 \text{ V}$:-

$$\therefore I_D = 10 \text{ mA} \left(1 - \frac{-1}{-4} \right)^2 = 5.625 \text{ mA}$$

③

For, $V_{GS} = -4 \text{ V}$:-

$$\therefore I_D = 10 \text{ mA} \left(1 - \frac{-4}{-4} \right)^2$$

$$= 0 \text{ A}$$

problem 2

⇒ A JFET produces gate current of 2nA when is reversed biased with 8V . Determine the resistance between gate and source.

⇒ In given:-

$$I_{G1} = 2\text{nA}$$

$$V_{GS} = 8\text{V}$$

∴ we know:-

$$R_{GS} = \frac{V_{GS}}{I_{G1}}$$

$$= \frac{8\text{V}}{2\text{nA}} = 4 \times 10^9 \Omega$$

$$= 4\text{G}\Omega \quad \underline{\underline{\text{Ans:-}}}$$

problem 6.1:-

⇒ Sketch the transfer curve defined by $I_{DSS} = 12\text{mA}$

$$V_p = -6\text{V}$$

⇒ In given:-

$$I_{DSS} = 12\text{mA}$$

$$V_p = -6\text{V}$$

we know:-

$$I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_p}\right)^2$$

At $V_{GS} = 0\text{V}$:-

$$I_D = 12\text{mA} \left(1 - \frac{0}{-6}\right)^2 = 12\text{mA}$$

At $V_{GS} = -1V$:-

$$I_D = 12m \left(1 - \frac{-1}{-6} \right)^2$$

$$= 8.33mA$$

At $V_{GS} = -2V$:-

$$I_D = 12m \left(1 - \frac{-2}{-6} \right)^2$$

$$= 5.33mA$$

At $V_{GS} = -3V$:-

$$I_D = 12m \left(1 - \frac{-3}{-6} \right)^2$$

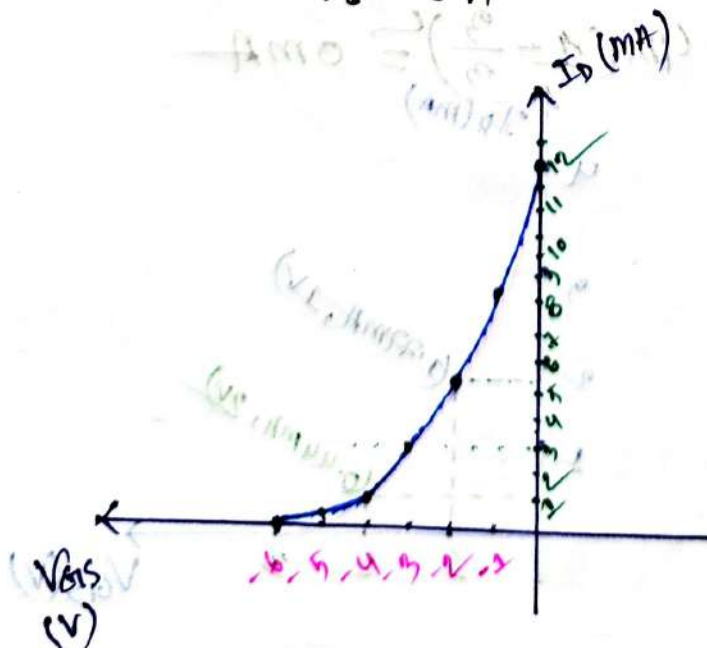
$$= 3mA$$

Similar way At $V_{GS} = -4, -5, -6V$:-

$$I_{D4} = 1.33mA$$

$$I_{D5} = 0.33mA$$

$$I_{D6} = 0A$$



#problem 8.2

⇒ Sketch the transfer curve for a p-channel device with $I_{DSS} = 4\text{mA}$ and $V_p = 3\text{V}$

⇒ Is given

$$I_{DSS} = 4\text{mA}$$

$$V_p = 3\text{V}$$

we know:-

$$I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_p}\right)^2$$

At $V_{GS} = 0\text{V}$:-

$$I_D = 4\text{m} \left(1 - \frac{0}{3}\right)^2 = 4\text{mA}$$

At $V_{GS} = 1\text{V}$:-

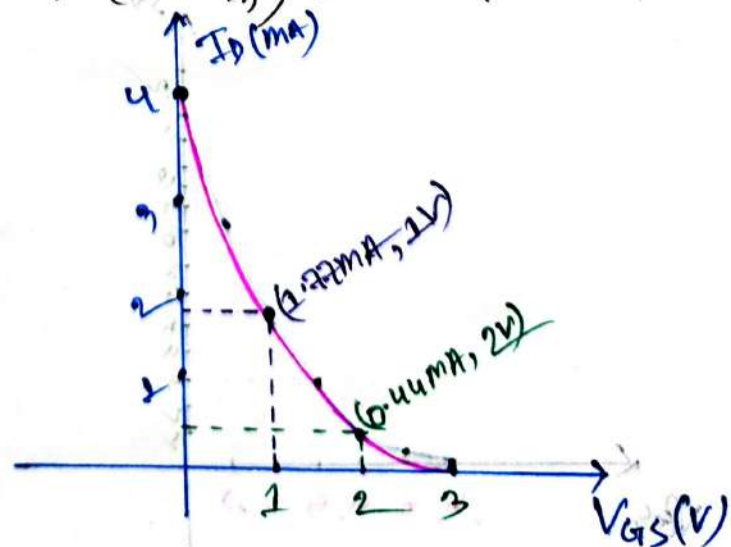
$$I_D = 4\text{m} \left(1 - \frac{1}{3}\right)^2 = 1.77\text{mA}$$

At $V_{GS} = 2\text{V}$:-

$$I_D = 4\text{m} \left(1 - \frac{2}{3}\right)^2 = 0.44\text{mA}$$

At $V_{GS} = 3\text{V}$:-

$$I_D = 4\text{m} \left(1 - \frac{3}{3}\right)^2 = 0\text{mA}$$

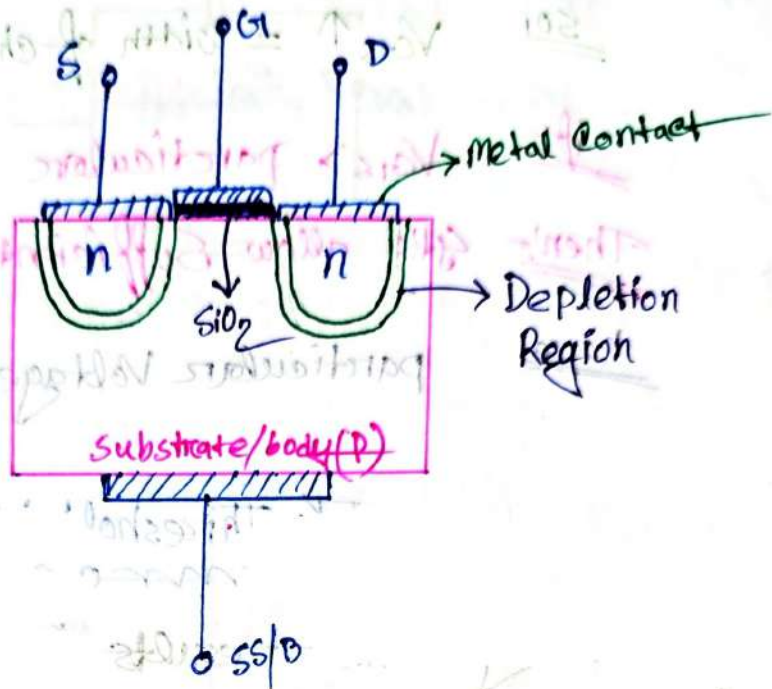


#MOSFET
(Metal-Oxide Semiconductor Field-Effect Transistor)

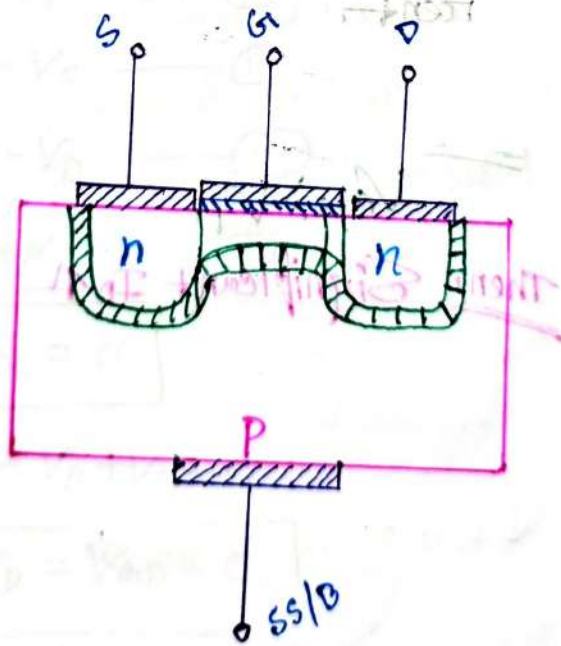
Enhancement
Type - MOSFET

Construction of Enhancement Type MOSFET

n-channel



⇒ Now try to form the channel between the Source & Drain. "For this we have to make Gate more positive"



Note:-

⇒ The Source & Substrate/body terminal are connected internally, so V_{GS} OR V_{GB} write as V_{GS} . This is actually Gate-to-Source Voltage. Because Source & Body

are on the same potential. So that the potential Difference between the Source & Body is zero.

So $V_{GS} \uparrow = \text{width of channel} \uparrow$

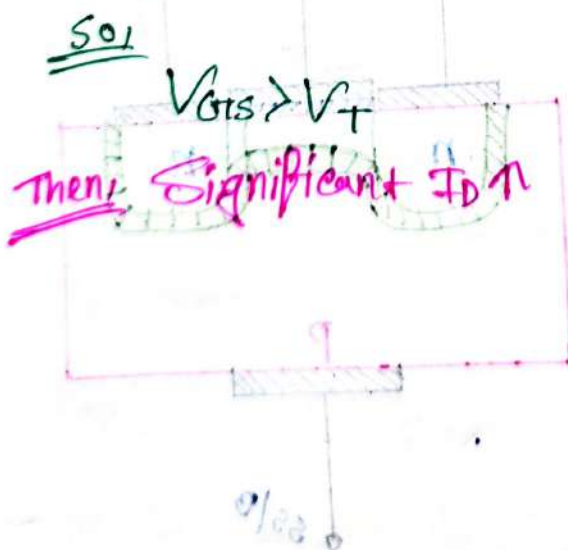
if $V_{GS} > \text{particular voltage}$

Then - It's allow sufficient flow of current I_D

here particular voltage = Threshold voltage.

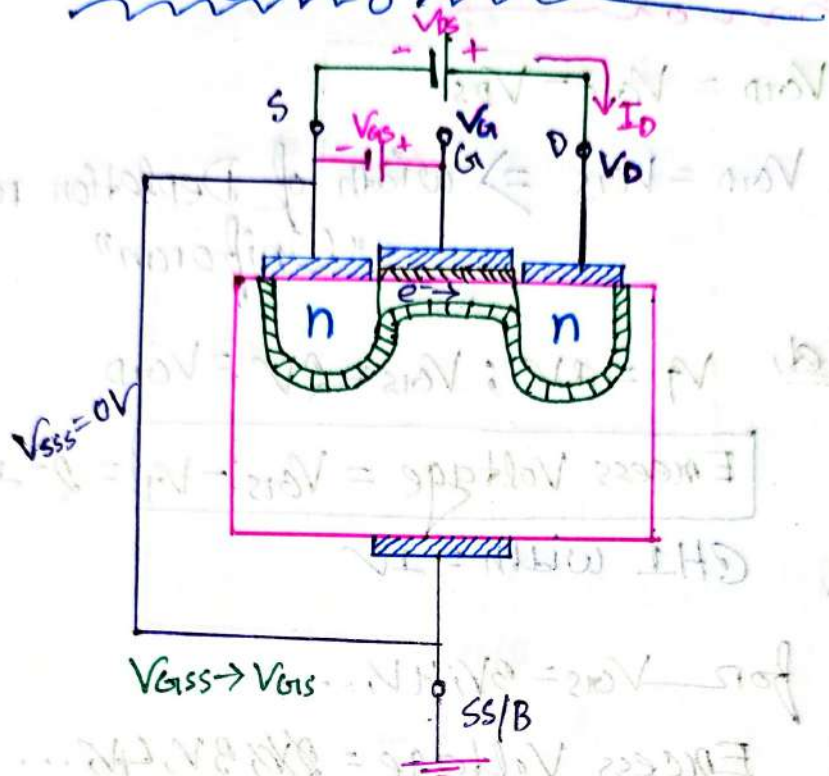
Threshold Voltage (V_T):-

$\Rightarrow$ The voltage that results in this significant amount of current to flow from source to drain is known as threshold voltage. The current is the Drain Current.



Note $\Rightarrow$ The source & substrate/body terminal are connected together, so V_{GS} is written as V_{GS} . Because source & substrate/body terminal are connected together.

Working of Enhancement Type MOSFET



Now:-

We know:-

$$V_{GS} = V_G - V_S \quad \text{--- (i)}$$

$$V_{DS} = V_D - V_S \quad \text{--- (ii)}$$

$$V_{GD} = V_G - V_D \quad \text{--- (iii)}$$

From S, G, D Loop:-

$$V_{GS} - V_{DS} = 0$$

$$\Rightarrow V_G - V_S - V_D + V_S = 0$$

$$\Rightarrow V_G - V_D = V_{GD} = 0$$

Sol

$$V_{GD} = V_G - V_D = V_{GS} - V_{DS}$$

Case 1:
mon

$\Rightarrow V_{DS} = 0V$
mon

So,

$V_{GD} = V_{GS} - V_{DS}$

$\therefore V_{GD} = V_{GS} \Rightarrow$ width of Depletion region
"Uniform"

Now:

Let $V_T = 1V$; $V_{GS} = 2V = V_{GD}$

so, $\text{Excess Voltage} = V_{GS} - V_T = 2 - 1 = 1V$

$\therefore$ CH1 width = 1V

Again:

for $V_{GS} = 3V, 4V, \dots$

Excess Voltage = 2V, 3V, 4V, ...

$\therefore$ CH2, 3 width = 2V, 3V

$\therefore CH1 < CH2 < CH3$

so The width of channel is dependent on
Excess Voltage

Case 2:
mon

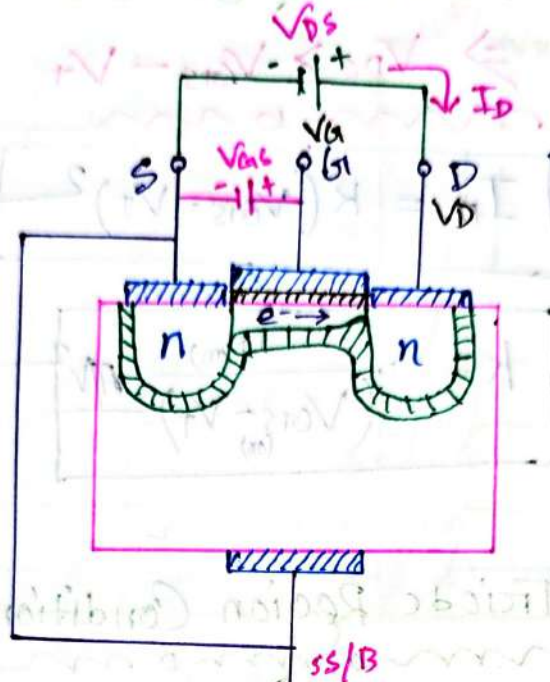
$\Rightarrow V_{DS} > 0V$
mon

so,

$\downarrow V_{GD} = V_{GS} - V_{DS} \uparrow$

This Implies Drain is more positive than the gate.

That means: The width of channel will reduce and the
width of D.R will increase.



Note - When the channel will become very narrow the width will be almost zero. This condition known as "pinch off".
 $V_{DS} = V_{DS_{sat}} = V_{GS} - V_T$

Drain & Transfer Characteristics of Enhancement-type MOSFET

Drain Ch:-

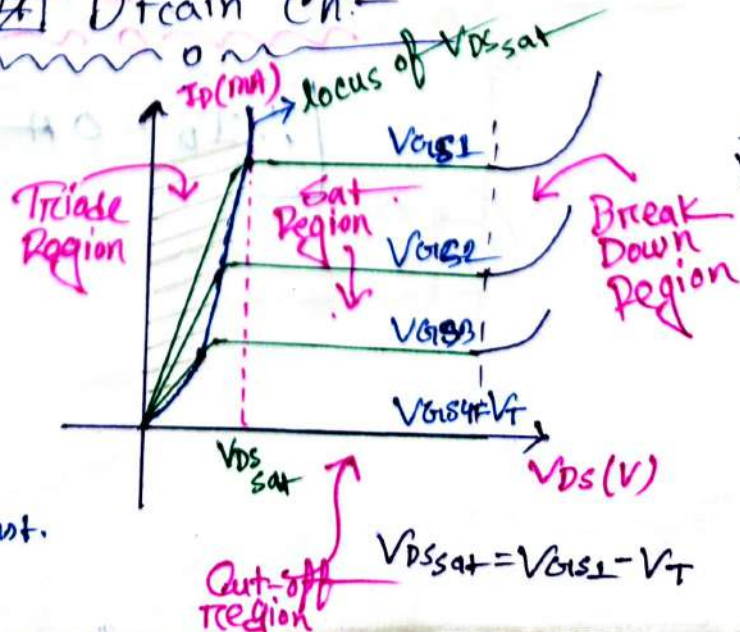
⇒ In case of MOSFET:-
 V_T & k is given

if $V_{GS1} > V_T$

Then! $V_{eff} = V_{GS1} - V_T$

if $V_{DS} \uparrow \rightarrow I_D \uparrow$

When! $V_{DS} = V_{DS_{sat}} \Rightarrow I_D$ const.
 (pinch off)



Now!
 $V_{GS2} < V_{GS1}$

Cut-off
 region

$$V_{DS_{sat}} = V_{GS1} - V_T$$

☒ Saturation Region Condition:-

$$\Rightarrow V_{DS} \geq V_{GS} - V_T$$

$$\therefore I_D = K (V_{GS} - V_T)^2$$

$$\therefore K = \frac{I_{D(on)}}{(V_{GS(on)} - V_T)^2 A/V^2}$$

☒ Triode Region Condition:-

$$\Rightarrow V_{DS} < V_{GS} - V_T$$

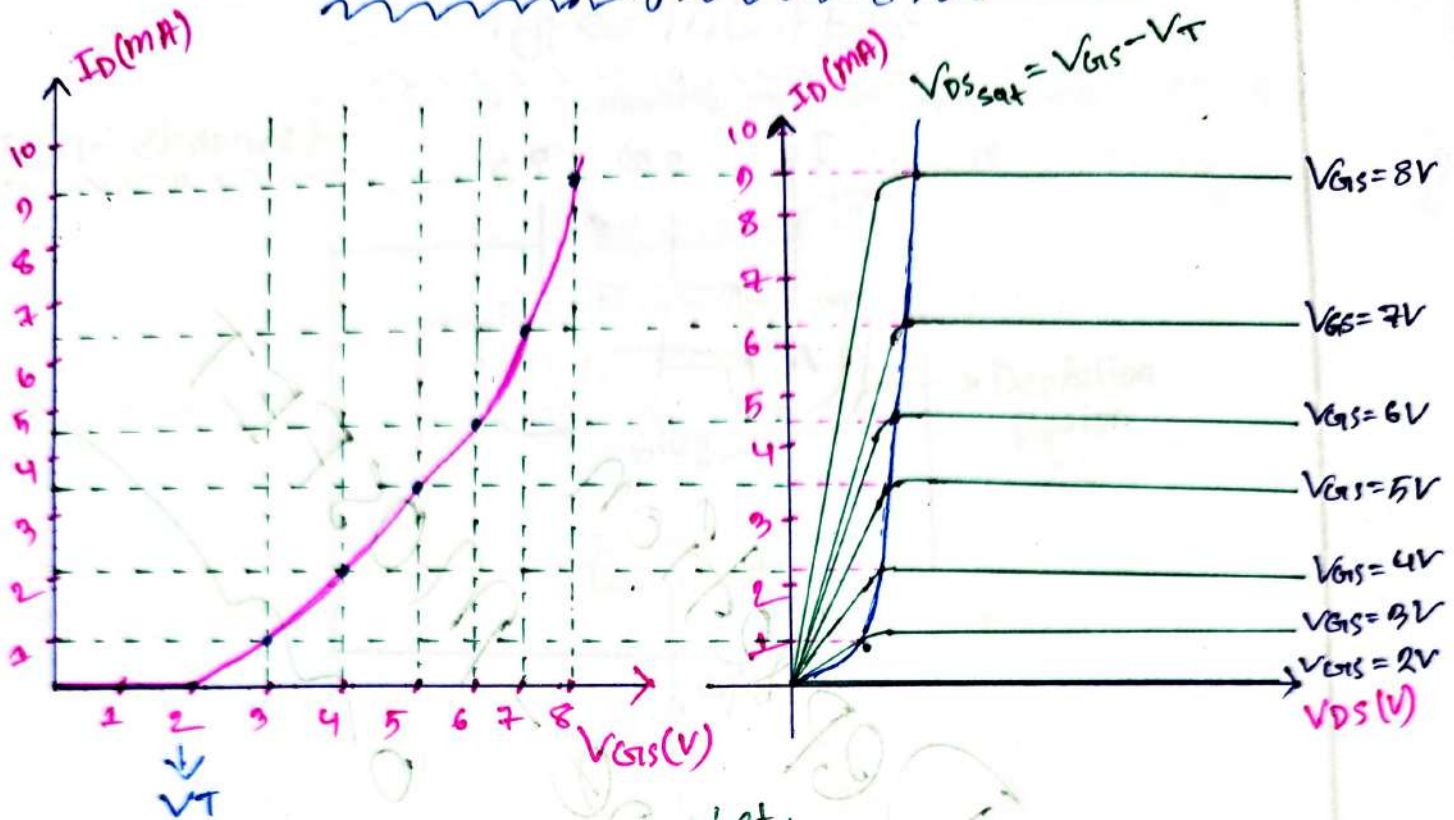
$$\therefore I_D = 2K \left[(V_{GS} - V_T) V_{DS} - \frac{V_{DS}^2}{2} \right]$$

☒ Cut-off Region:- Condition:-

$$\Rightarrow V_{GS} \leq V_T$$

$$\therefore I_D = 0A$$

Transfer characteristics



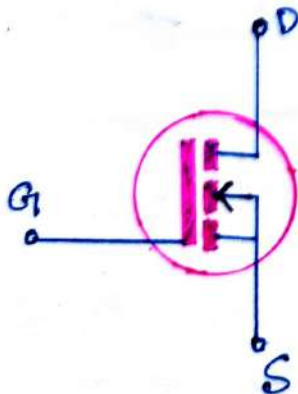
Let

$$V_T = 2V$$

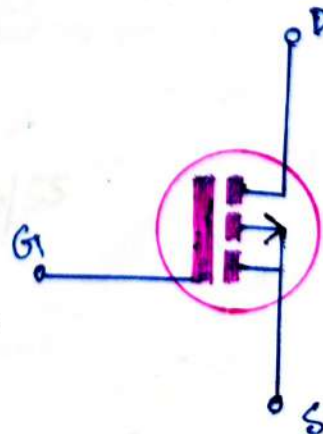
$$K = 0.27 \times 10^{-3} A/V^2$$

Symbol of Enhancement type - MOSFET

n-channel:-



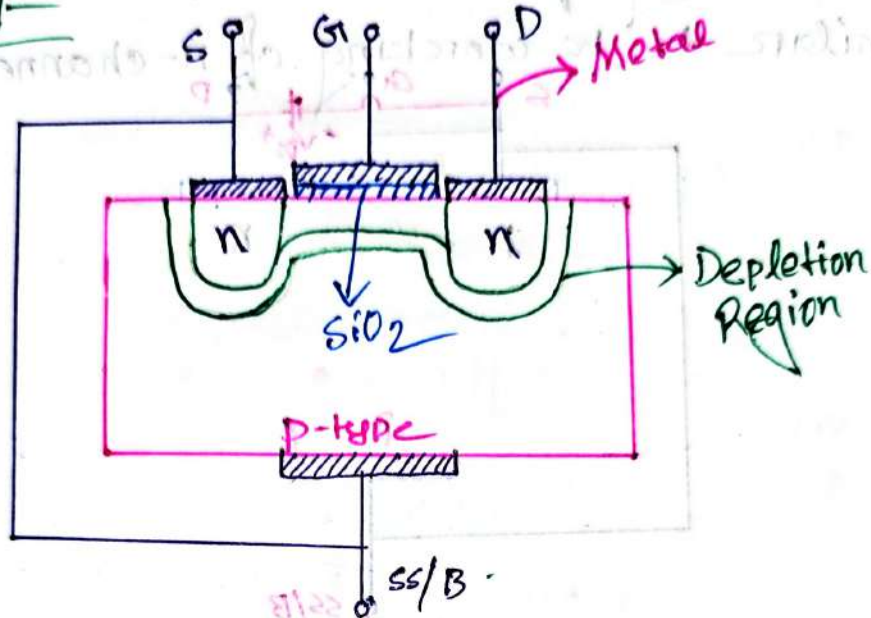
p-channel:-



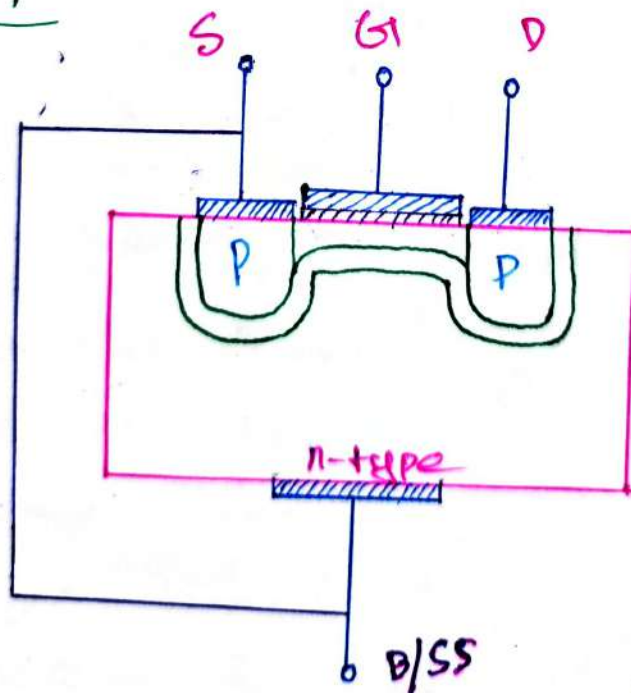
Depletion
Type MOSFET

Construction of Depletion Type - MOSFET:-

n-channel:-



p-channel:-

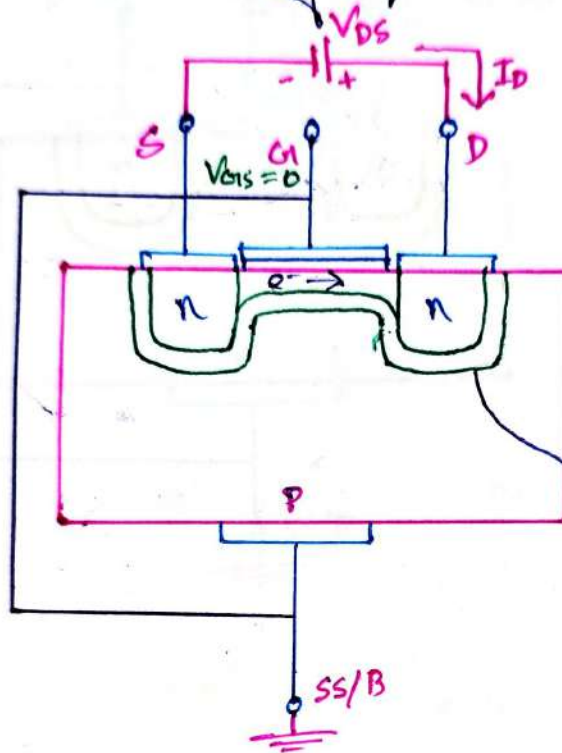


Working of D-MOSFET

⇒ The working of n-channel D-MOSFET is somewhat similar to the working of n-channel JFET

Case 1:
mon
 $V_{GS} = 0$
mon

Let,
 $V_{DS} = V_{DD}$
 $I_D = I_{D1}$



if $V_{DS} \uparrow \rightarrow I_D \uparrow$
Since, $V_{DS} = V_D - V_S$
So, $V_{DS} \uparrow \Rightarrow V_D \uparrow$

if $V_D \uparrow \rightarrow \pi \cdot b \uparrow$
The ch. width $\downarrow$
So, $I_D = \text{Const.}$

$$\therefore I_D = I_{DSS}$$

$$I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_P} \right)^2$$

Case 2:
mon
 $V_{GS} < 0$
mon

Let, $V_{DS} = V_{DD}$
 $I_D = I_{D2}$

Since, $V_{GS} < 0$
So, $I_{D1} > I_{D2}$

Case 3:
mon
 $V_{GS} > 0$
mon

Let, $V_{DS} = V_{DD}$
 $I_D = I_{D3}$

Since, $V_{GS} > 0V$

So, $I_{D3} > I_{D1}$
& $I_D \uparrow$ rapidly

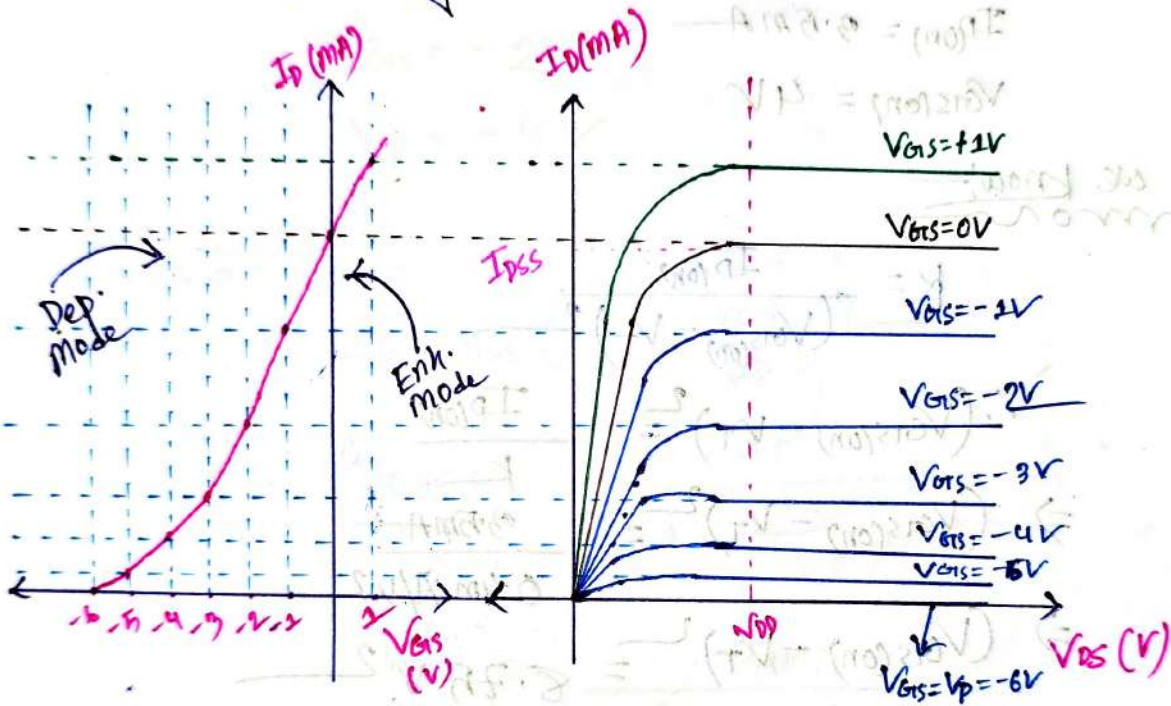
Drain Characteristics & Transfer CH. of D-MOSFET

O/P current

I_D vs V_{DS} for various V_{GS}

O/P voltage.
 I/P voltage.

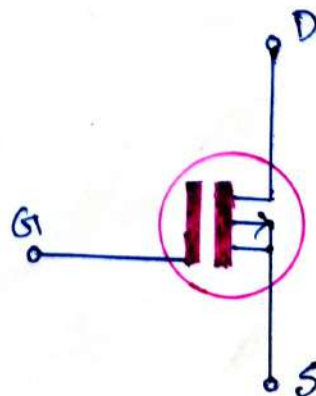
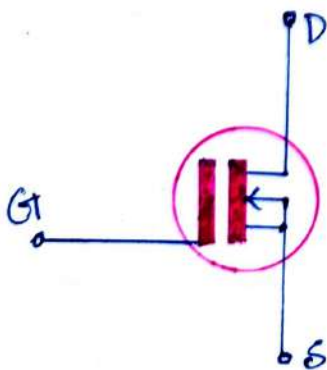
I_D vs V_{GS} for fixed V_{DS}



Symbol of D-MOSFET

n-channel

p-channel



problem 1:-

⇒ Given $k = 0.4 \text{ mA/V}^2$ and $I_{D(on)} = 0.5 \text{ mA}$ with $V_{GS(on)} = 4 \text{ V}$. Determine V_T

⇒ In given:-

$$k = 0.4 \text{ mA/V}^2$$

$$I_{D(on)} = 0.5 \text{ mA}$$

$$V_{GS(on)} = 4 \text{ V}$$

we know:-

$$k = \frac{I_{D(on)}}{(V_{GS(on)} - V_T)^2}$$

$$\therefore (V_{GS(on)} - V_T)^2 = \frac{I_{D(on)}}{k}$$

$$\Rightarrow (V_{GS(on)} - V_T)^2 = \frac{0.5 \text{ mA}}{0.4 \text{ mA/V}^2}$$

$$\Rightarrow (V_{GS(on)} - V_T)^2 = 8.75 \text{ V}^2$$

$$\Rightarrow V_{GS(on)} - V_T = 2.95 \text{ V}$$

$$\therefore V_T = 4 - 2.95 = 1.05 \text{ V}$$

Ans:-

Subject _____

problem: 2:—

⇒ Given $I_D = 4 \text{ mA}$ at $V_{GS} = -2 \text{ V}$, determine the Saturation Current if $V_P = -4 \text{ V}$

⇒ In given:—

$$I_D = 4 \text{ mA}$$

$$V_{GS} = -2 \text{ V}$$

$$V_P = -4 \text{ V}$$

We know:—

$$I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_P} \right)^2$$

$$\therefore I_{DSS} = \frac{I_D}{\left(1 - \frac{V_{GS}}{V_P} \right)^2}$$

$$= \frac{4 \text{ mA}}{\left(1 - \frac{-2}{-4} \right)^2}$$

$$= 16 \text{ mA}$$

Ans:—

At $V_{DS} = 10V$, determine the
 saturation current I_{DSS} if $V_{GS} = 0V$
 and $V_{GS} = -2V$, determine the

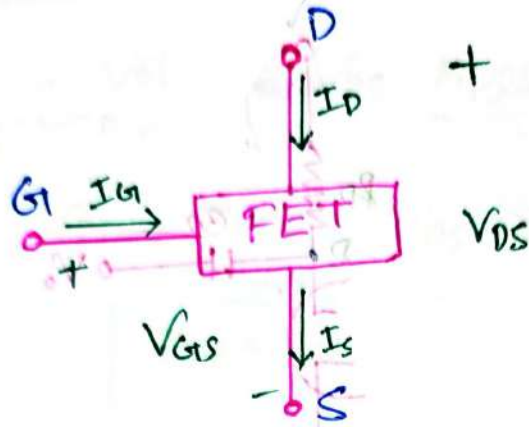
$$V_{GS} = -2V$$

$$V_{DS} = 10V$$

FET Biasing

Ch-02

General Relationships



$$\therefore I_{G1} \cong 0A \quad \text{due to } Z_i \uparrow$$

$$\therefore I_D = I_S$$

For JFET & D-MOSFET:

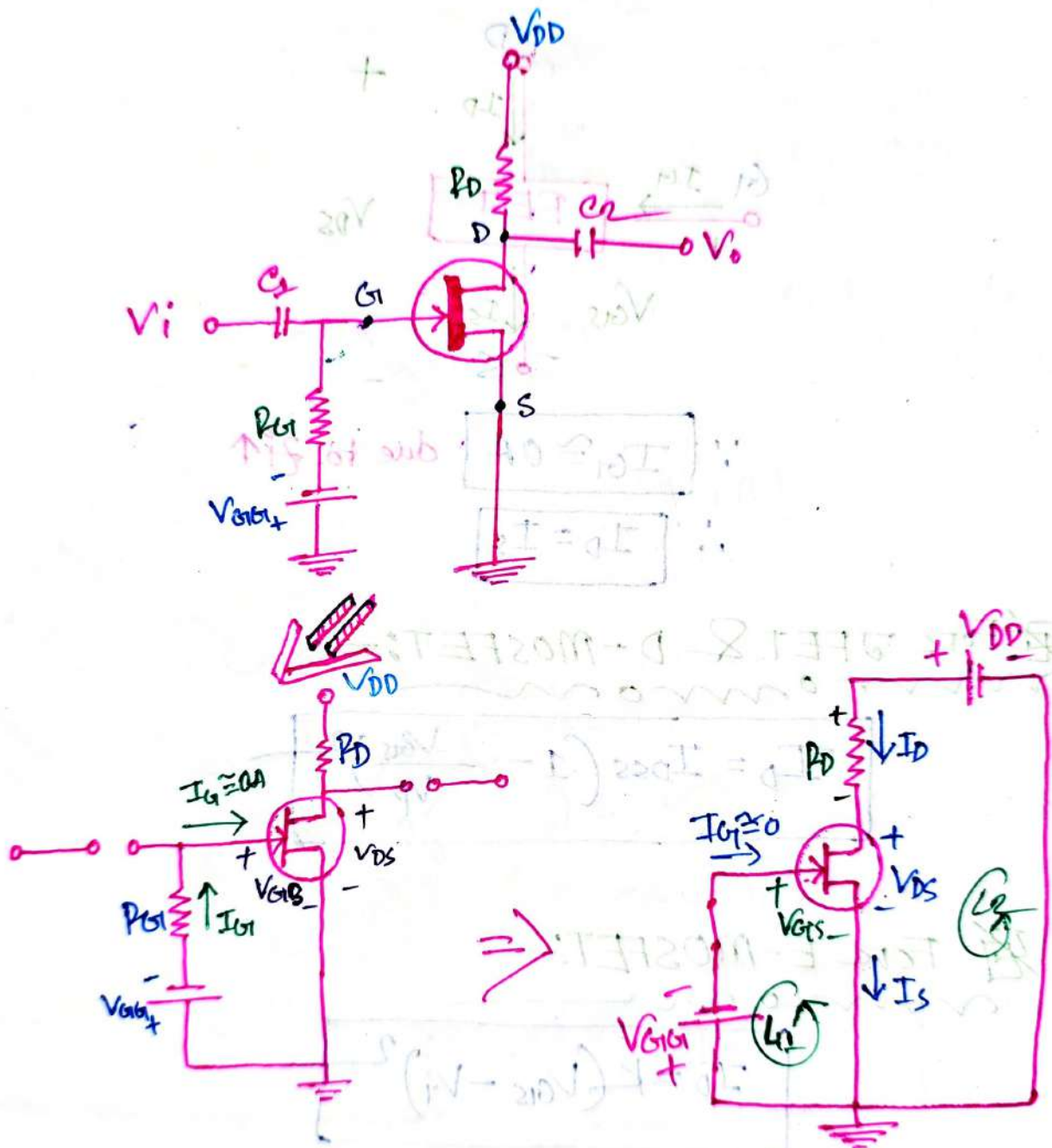
$$I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_P} \right)^2$$

For E-MOSFET:

$$I_D = K (V_{GS} - V_T)^2$$

Fixed-Bias Configuration of JFET

Mathematical Approach:-



⇒ Now Apply KVL in the Input loop (L1):-

$$V_{G1} + V_{GS} = 0$$

$$\therefore V_{GS} = -V_{G1}$$

⇒ The Output Current I_D :-

$$I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_P}\right)^2$$

⇒ For Output Voltage V_{DS} Apply KVL in eq Output loop (22):-

$$V_{DD} - I_D R_D - V_{DS} = 0$$

$$\therefore V_{DS} = V_{DD} - I_D R_D$$

Since: $V_S = 0V$

$$\therefore V_{DS} = V_D - V_S$$

$$V_D = V_{DS}$$

$$\Rightarrow V_{GS} = V_G - V_S$$

$$\therefore V_G = V_{GS}$$

Graphical Approach:-

We know:-

$$I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_P}\right)^2$$

Case 1:-

$$V_{GS} = 0$$

$$\therefore I_D = I_{DSS}$$

$$P_1 \equiv (0, I_{DSS})$$

$$P_2 \equiv (V_P, 0)$$

$$P_3 \equiv \left(\frac{V_P}{2}, \frac{I_{DSS}}{4}\right)$$

Case 2:-

$$I_D = 0$$

$$\therefore V_{GS} = V_P$$

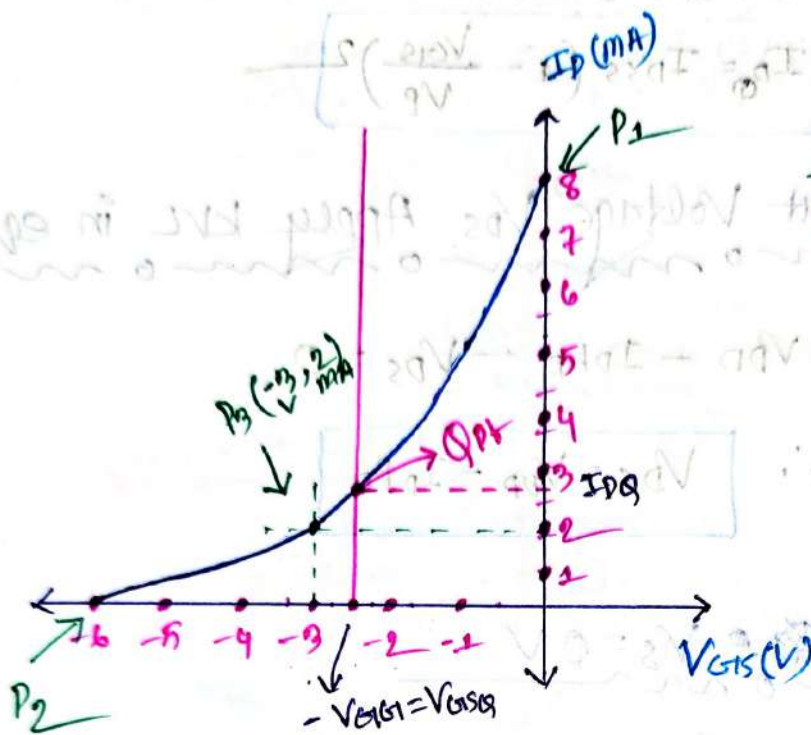
Case 3:-

$$V_{GS} = V_P/2$$

$$\Rightarrow I_D = I_{DSS} \left(1 - \frac{V_P/2}{V_P}\right)^2$$

$$\Rightarrow I_D = I_{DSS} \left(1 - \frac{1}{2}\right)^2$$

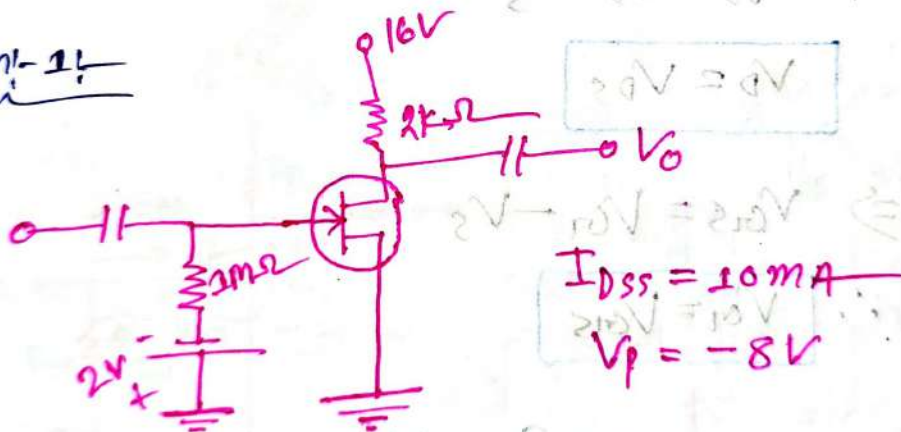
$$\therefore I_D = I_{DSS}/4$$



Let,
 $I_{DSS} = 8 \text{ mA}$
 $V_P = -6 \text{ V}$

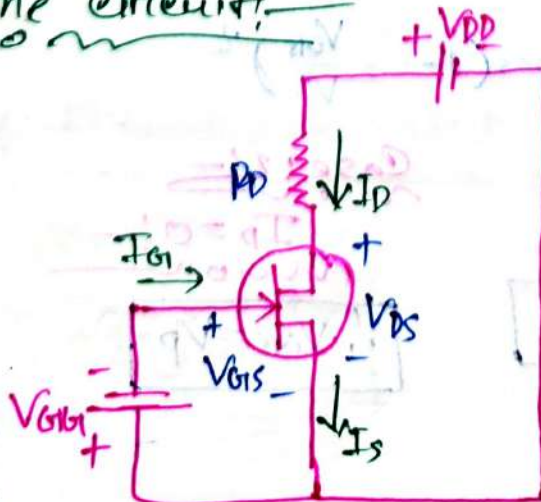
Q point = operating point
 $Q \equiv (-V_{GSQ}, I_{DQ})$

problem - 1



⇒ Determine $V_{GSQ}, I_{DQ}, V_{DS}, V_D, V_G, V_S$

⇒ re-draw the circuit!



here,
 $I_{DSS} = 10 \text{ mA}$
 $V_P = -8 \text{ V}$
 $R_D = 2 \text{ k}\Omega$
 $V_{GS} = 2 \text{ V}$
 $V_{DD} = 16 \text{ V}$

⇒ Apply KVL in Input Loop:-

$$V_{GS} + V_{GS} = 0$$

$$\therefore V_{GS} = -V_{GS}$$

$$\therefore V_{GS} = -2V$$

⇒ we know:-

$$\begin{aligned} I_D &= I_{DSS} \left(1 - \frac{V_{GS}}{V_P} \right)^2 \\ &= 10\text{m} \left(1 - \frac{-2}{-8} \right)^2 \\ &= 5.62\text{mA} \end{aligned}$$

⇒ For V_{DS} ; Apply KVL in Output Loop:-

$$V_{DD} - I_D R_D - V_{DS} = 0$$

$$V_{DS} = V_{DD} - I_D R_D$$

$$= 16 - 5.62\text{mA} \times 2\text{k}\Omega$$

$$= 4.76\text{V}$$

⇒ For V_D, V_G, V_S :-

$$\boxed{V_S = 0\text{V}} \quad [\because \text{Connected to ground}]$$

$$\therefore V_{DS} = V_D - V_S$$

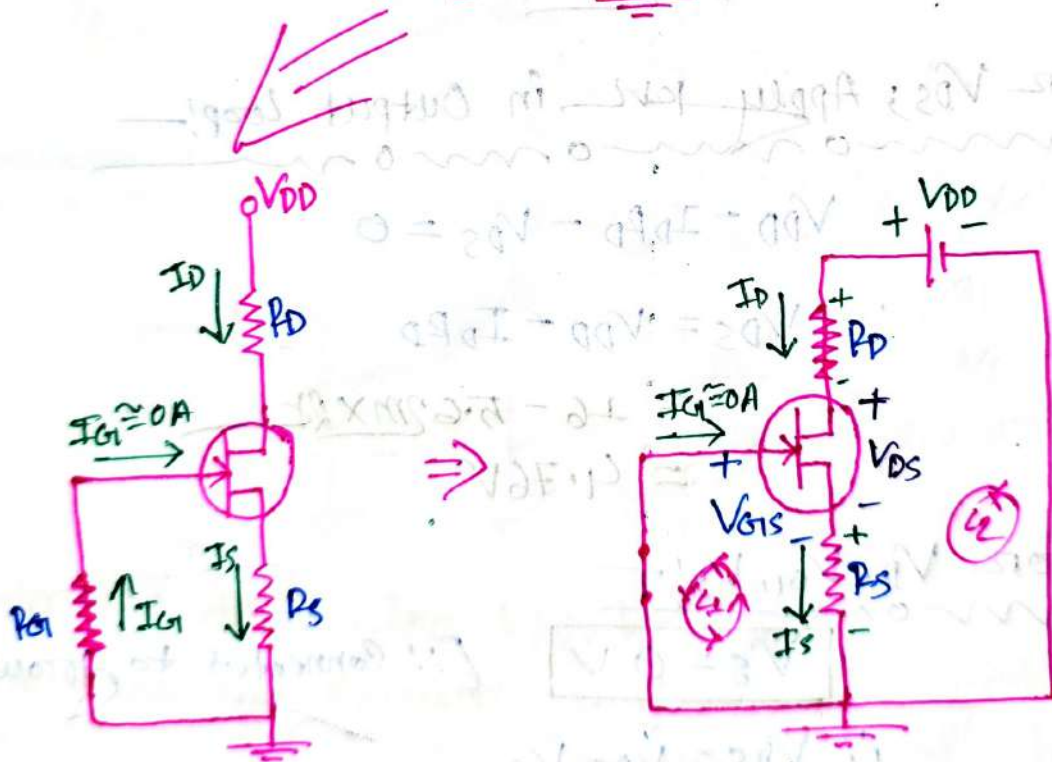
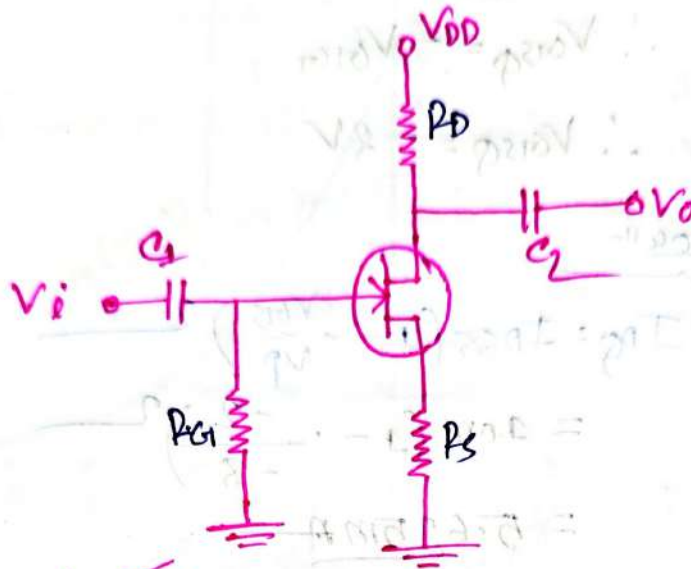
$$\boxed{V_D = V_{DS} = 4.76\text{V}}$$

$$\therefore V_{GS} = V_G - V_S$$

$$\boxed{V_G = V_{GS} = -2\text{V}}$$

Self-Bias Configuration of JFET

Mathematical Approach:-



⇒ Apply KVL in Input Loop (L1):-

$$V_{GS} + I_S R_S = 0$$

$$\therefore V_{GS} = -I_S R_S$$

$$V_{GS} = -I_D R_S$$

Since:-
 $I_D = I_S$

⇒ The Output Current I_D :-

$$I_{DQ} = I_{DSS} \left(1 - \frac{V_{GS}}{V_P}\right)^2$$

⇒ Apply KVL in Output Loop (L2):-

$$V_{DD} - I_D R_D - V_{DS} - I_S R_S = 0$$

$$\Rightarrow V_{DS} = V_{DD} - I_D (R_D + R_S)$$

Since:-
 $I_D = I_S$

⇒ For V_{GS} , V_D , V_S :-

$$V_S = I_S R_S - 0 = I_S R_S$$

$$\therefore V_{GS} = V_{G1} - V_S$$

$$\therefore V_{G1} = V_{GS} + V_S$$

$$\text{OR, } V_{G1} = V_{GS} + I_S R_S$$

Again:-

$$\therefore V_{DS} = V_D - V_S$$

$$\therefore V_D = V_{DS} + V_S$$

$$\text{OR, } V_D = V_{DS} + I_S R_S$$

Graphical Approach:-

We know:-

$$I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_P}\right)^2$$

Case: 1:-

$$I_D = 0A$$

$$V_{GS} = V_P$$

$$P1 \equiv (V_P, 0)$$

Case: 2:-

$$V_{GS} = 0V$$

$$I_D = I_{DSS}$$

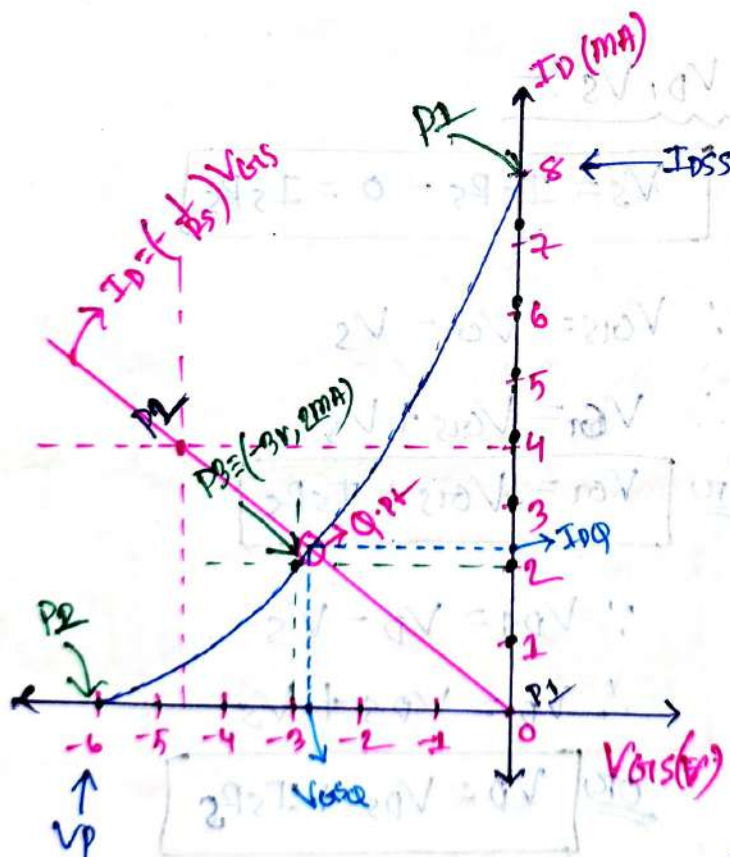
$$P2 \equiv (0, I_{DSS})$$

Case: 3:-

$$V_{GS} = \frac{V_P}{2}$$

$$I_D = \frac{I_{DSS}}{4}$$

$$P3 \equiv \left(\frac{V_P}{2}, \frac{I_{DSS}}{4}\right)$$



Let,

$$I_{DSS} = 8mA$$

$$V_P = -6V$$

We know:-

$$V_{GS} = -I_D R_S$$

$$\Rightarrow I_D = \left(-\frac{1}{R_S}\right) V_{GS}$$

The operating point = Q.Pt

$$Q \equiv (V_{GSQ}, I_{DQ})$$

Now we need point to plot the straight line

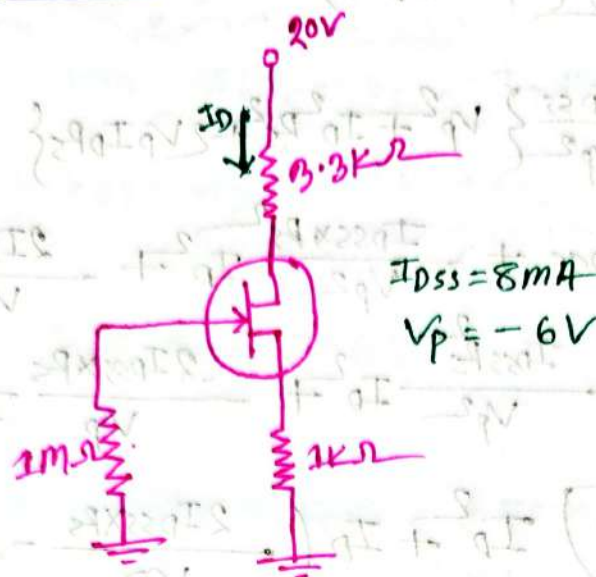
$$P1 \equiv (V_{GS} = 0, I_D = 0)$$

For P2:-

$$I_D = I_{DSS}/2$$

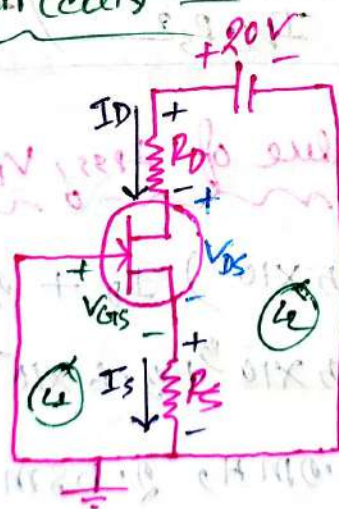
$$\therefore V_{GS} = -I_{DSS} \times R_S / 2$$

problem: 1! —



- (a) Determine V_{GSQ} , I_{DQ} , V_{DS} , V_S , V_{G1} , V_D
 (b) Plot I_D vs V_{GS} and draw a line from the origin

⇒ re draw the circuit: (a)



here,
 $R_D = 3.3k\Omega$
 $V_{DD} = 20V$
 $I_D = I_S$
 $R_S = 1k\Omega$
 $I_{DSS} = 8mA$
 $V_P = -6V$

⇒ For V_{GSQ} : Apply KVL in Loop 1: —

$$V_{GS} + I_S R_S = 0$$

$$\therefore V_{GS} = -I_D R_S \quad \text{--- (1)}$$

we know: —

$$I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_P} \right)^2$$

$$\Rightarrow I_D = I_{DSS} \left(1 + \frac{I_D R_S}{V_P} \right)^2$$

$$\Rightarrow I_D = I_{DSS} \left(\frac{V_P + I_D R_S}{V_P} \right)^2$$

$$[\because V_{GS} = -I_D R_S]$$

$$\Rightarrow I_D = \frac{I_{DSS}}{V_p^2} (V_p + I_D R_s)^2$$

$$\Rightarrow I_D = \frac{I_{DSS}}{V_p^2} \{ V_p^2 + I_D^2 R_s^2 + 2V_p I_D R_s \}$$

$$\Rightarrow I_D = I_{DSS} + \frac{I_{DSS} R_s^2}{V_p^2} I_D^2 + \frac{2I_{DSS} R_s}{V_p} I_D$$

$$\Rightarrow I_{DSS} + \frac{I_{DSS} R_s^2}{V_p^2} I_D^2 + \frac{2I_{DSS} R_s}{V_p} I_D - I_D = 0$$

$$\Rightarrow \left(\frac{I_{DSS} R_s^2}{V_p^2} \right) I_D^2 + I_D \left(\frac{2I_{DSS} R_s}{V_p} - 1 \right) + I_{DSS} = 0$$

$$\Rightarrow I_D^2 + \left(\frac{2I_{DSS} R_s - V_p}{V_p} \times \frac{V_p^2}{I_{DSS} R_s^2} \right) I_D + \frac{I_{DSS} V_p^2}{I_{DSS} R_s^2} = 0$$

$$\Rightarrow I_D^2 + \left(\frac{(2I_{DSS} R_s - V_p) V_p}{I_{DSS} R_s^2} \right) I_D + \frac{V_p^2}{R_s^2} = 0$$

Now:- put the value of I_{DSS} , V_p , R_s :-

$$\Rightarrow I_D^2 + (-16.5 \times 10^3) I_D + 3.6 \times 10^{-5} = 0$$

$$\Rightarrow I_D^2 - 16.5 \times 10^3 I_D + 3.6 \times 10^{-5} = 0$$

$$\Rightarrow I_D = 13.0 \text{ mA}, 2.58 \text{ mA}$$

$$\therefore I_{DQ} = 2.58 \text{ mA}$$

Since,
 $I_D = 13.0 \text{ mA} > I_{DSS}$
 which is not possible.

put the value of I_{DQ} in eq (1):-

$$\therefore V_{GSQ} = -I_{DQ} \times R_s$$

$$= -2.58 \text{ mA} \times 1 \text{ k}$$

$$V_{GSQ} = -2.58 \text{ V}$$

Ans:-

⇒ Apply KVL in Output loop (L2):—

$$V_{DD} - I_D R_D - V_{DS} - I_S R_S = 0$$

$$\therefore V_{DS} = V_{DD} - I_D (R_D + R_S)$$

$$= 20 - 2.58 \text{ m} (3.3 \text{ k} + 1 \text{ k})$$

$$V_{DS} = 8.906 \text{ V}$$

we know:—

$$\Rightarrow V_S = I_S R_S = I_D R_S = 2.58 \text{ m} \times 1 \text{ k} \\ = 2.58 \text{ V}$$

$$\Rightarrow V_{DS} = V_D - V_S$$

$$\therefore V_D = V_{DS} + V_S \\ = 8.906 + 2.58 \text{ V}$$

$$= 11.486 \text{ V}$$

$$\Rightarrow V_{GS} = V_G - V_S$$

$$\therefore V_G = V_{GS} + V_S$$

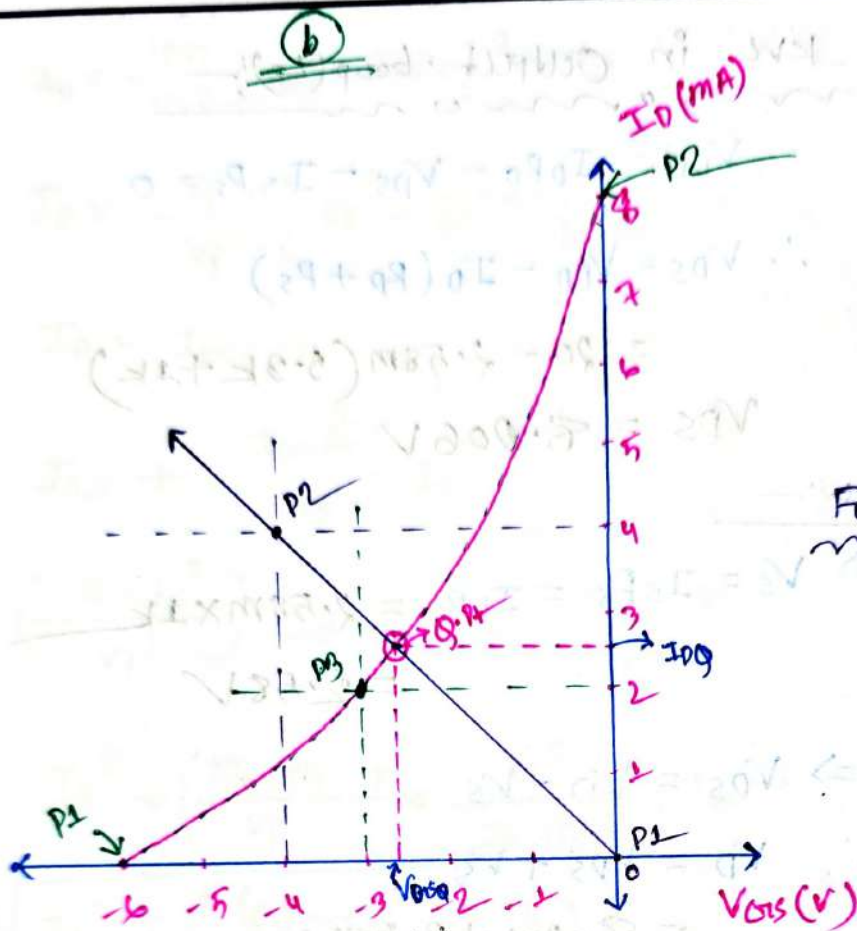
$$= -2.58 + 2.58$$

$$= 0 \text{ V}$$

Ans:—

Subject _____

Date : / /



$$P_1 \equiv (V_P, 0) \equiv (-6, 0)$$

$$P_2 \equiv (0, I_{DSS}) \equiv (0, 8)$$

$$P_3 \equiv \left(\frac{V_P}{2}, \frac{I_{DSS}}{4} \right) \equiv (-3, 2)$$

For Straight Line:-

$$I_D = \left(\frac{1}{R_S} \right) V_{GS} \quad \text{--- (1)}$$

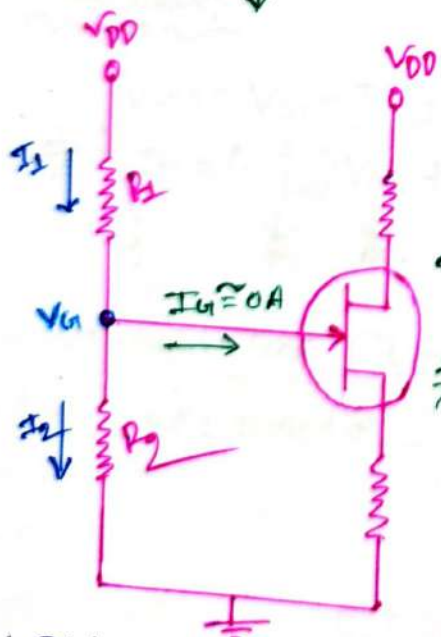
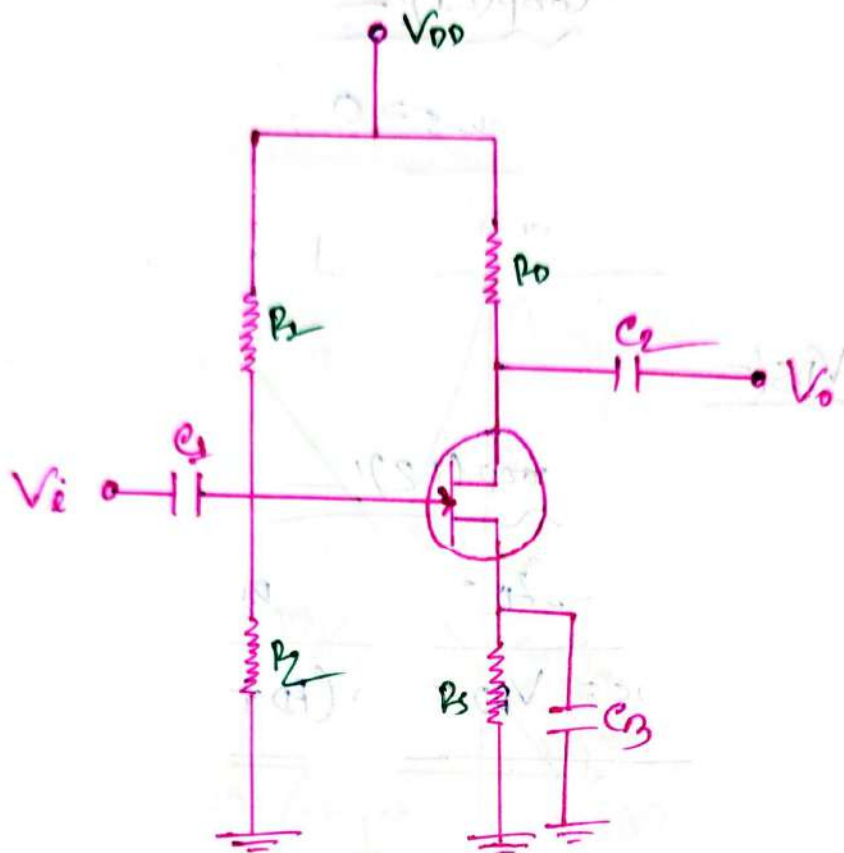
$$P_1 \equiv (0, 0 \text{ A})$$

$$P_2 \equiv \left(\frac{-I_{DSS} R_S}{2}, \frac{I_{DSS}}{2} \right) \equiv (-3, 2)$$

$$\therefore Q \equiv (-2.6 \text{ V}, 2.6 \text{ mA})$$

Ans:-

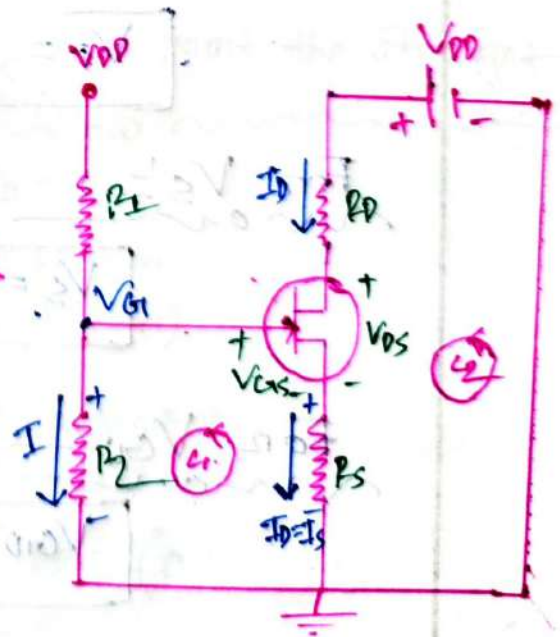
Voltage-Divider Biasing of JFET



Voltage Drop across R_2 :-

$$V_{G1} = I R_2$$

$$\Rightarrow V_{G1} = \frac{V_{DD} R_2}{R_1 + R_2}$$



KCL:-

$$I_1 = I_2 + I_{G1}$$

$$\therefore I_1 = I_2 = I$$

$$R_T = R_1 + R_2$$

$$\therefore I = \frac{V_{DD} - 0}{R_T}$$

$$I = \frac{V_{DD}}{R_T}$$

For V_{GS} :-

$\Rightarrow$ Apply KVL in loop (L1):-

$$I_{R2} - V_{GS} - I_{D}R_S = 0$$

$$\Rightarrow \boxed{V_{GS} = V_{G1} - I_{D}R_S}$$

Since
 $I_{R2} = V_{G1}$

For V_{DS} :-

$\Rightarrow$ Apply KVL in loop (L2):-

$$V_{DD} - I_{D}R_D - V_{DS} - I_{D}R_S = 0$$

$$\therefore \boxed{V_{DS} = V_{DD} - I_D(R_D + R_S)}$$

For V_D :-

$$V_{DS} = V_D - V_S$$

$$\therefore \boxed{V_D = V_{DS} + V_S}$$

For V_S :-

$$\boxed{V_S = I_{D}R_S}$$

For V_{GD} :-

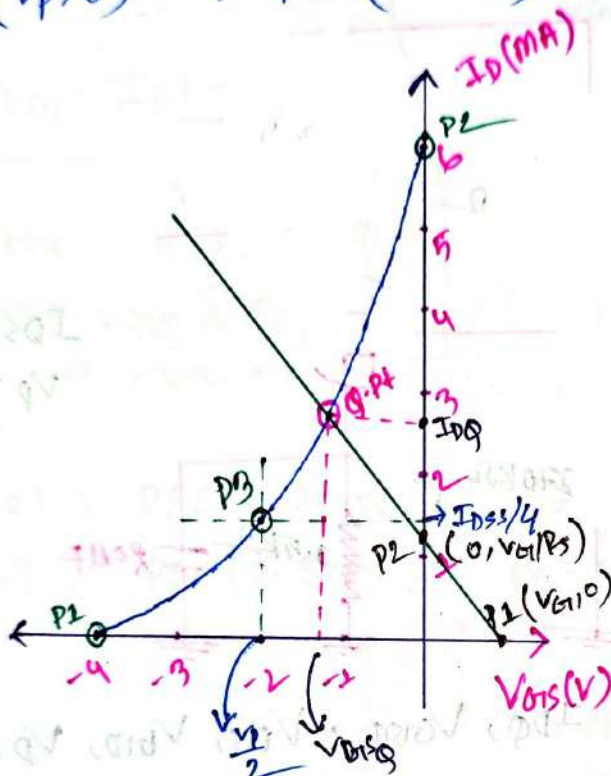
$$\boxed{V_{GD} = V_{G1} - V_D}$$

Subject : _____

Graphical Approach:-

we know:-

$$P_1 \equiv (V_p, 0) \quad ; \quad P_2 \equiv (0, I_{DSS}) \quad ; \quad P_3 \equiv \left(\frac{V_p}{2}, \frac{I_{DSS}}{4} \right)$$



we have:-

$$V_{GS} = V_G - I_D R_S$$

$$\Rightarrow I_D = \left(-\frac{1}{R_S} \right) V_{GS} + \frac{V_G}{R_S}$$

Straight line eqn:-

$I = mx + c$

Now, we need point the Straight Line:-

Let, $I_D = 0$

$$\therefore V_{GS} = V_G$$

$$P_1 \equiv (V_G, 0)$$

Now, Let, $V_{GS} = 0$

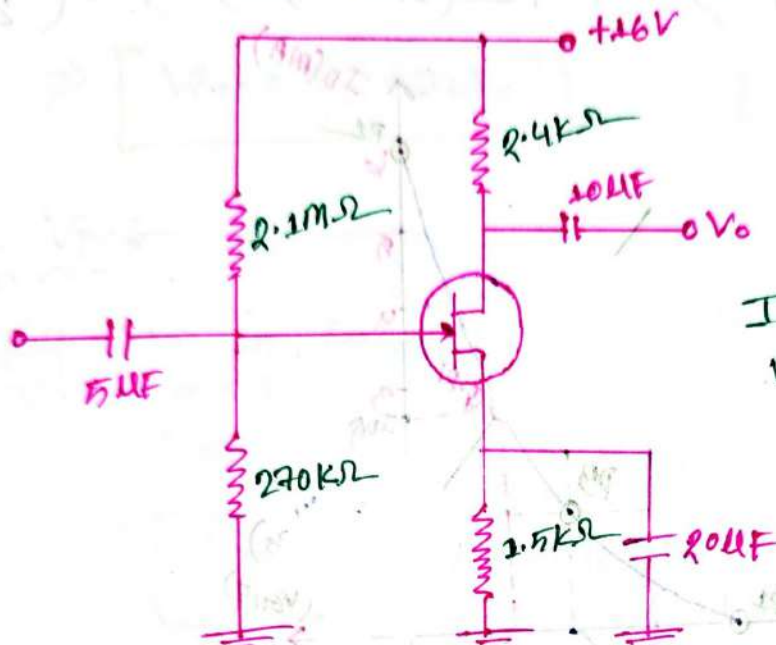
$$I_D = \frac{V_G}{R_S}$$

$$P_2 \equiv \left(0, \frac{V_G}{R_S} \right)$$

Q = operating point

$$Q \equiv (V_{GSQ}, I_{DQ})$$

problems 1! —

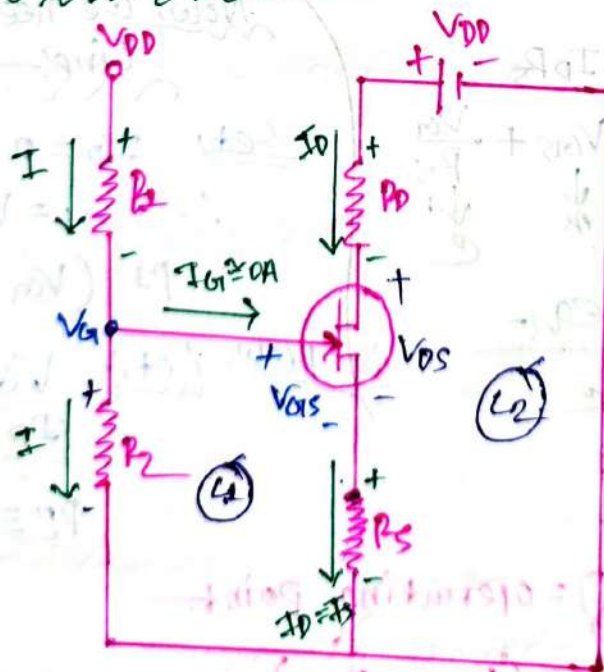


$$I_{DSS} = 8 \text{ mA}$$

$$V_p = -4 \text{ V}$$

⇒ Determine I_{DQ} , V_{GSQ} , V_{DS} , V_{GD} , V_p , V_s , V_G

⇒ re-draw the circuit! —



here,

$$V_{DD} = 16 \text{ V}$$

$$R_1 = 2.1 \text{ M}\Omega$$

$$R_2 = 270 \text{ K}\Omega$$

$$R_S = 1.5 \text{ K}\Omega$$

$$R_D = 2.4 \text{ K}\Omega$$

$$I_{DSS} = 8 \text{ mA}$$

$$V_p = -4 \text{ V}$$

Since $I_G \approx 0$

$$\therefore R_T = R_1 + R_2 = 2.1 \times 10^6 + 270 \times 10^3 = 2.37 \text{ M}\Omega$$

$$I = \frac{V_{DD}}{R_T} = \frac{16}{2.37 \text{ M}} = 6.75 \text{ }\mu\text{A}$$

For V_{GS} :- $\Rightarrow$ Apply KVL in loop 1:-

$$V_{G1} - V_{GS} - I_D R_S = 0$$

$$\therefore V_{GS} = V_{G1} - I_D R_S$$

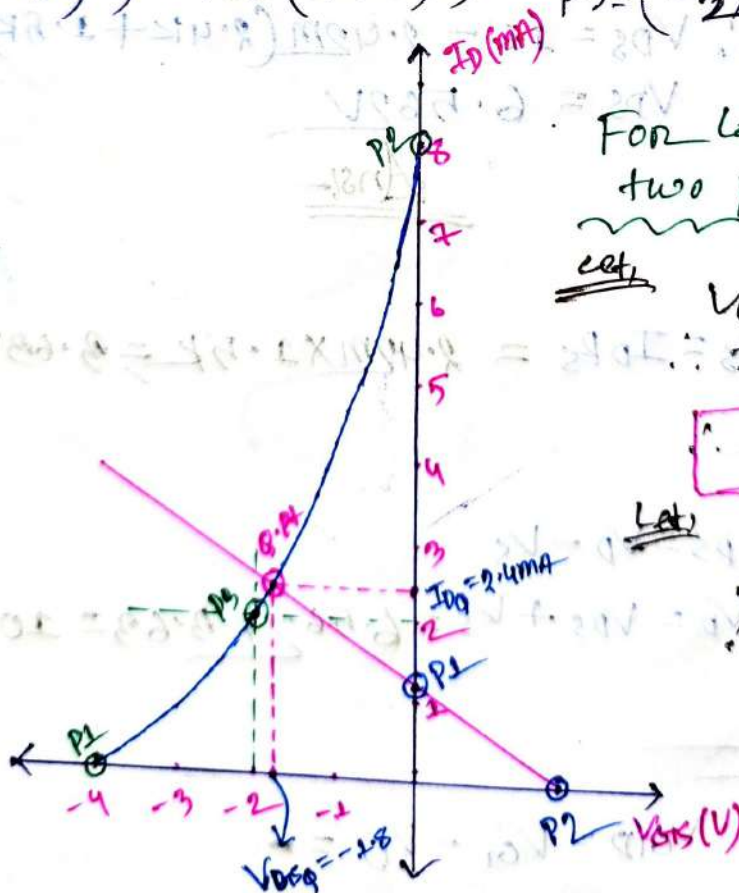
For V_{G1} :-V. Drop Across R_2 :-

$$V_{G1} = I_{R_2} = 6.25 \text{ mA} \times 270 \text{ k}\Omega$$
$$= 1.822 \text{ V}$$

Ans:-Now, plot the V_{GS} vs I_D graph to find the V_{GSQ} & I_{DQ} We know:-

$$P1 = (V_P, 0); \quad P2 = (0, I_{DSS}); \quad P3 = \left(\frac{V_P}{2}, \frac{I_{DSS}}{4} \right)$$

$$\therefore P1 = (-4, 0); \quad P2 = (0, 8 \text{ mA}); \quad P3 = (-2, 2)$$

For Load Line we need two pointLet,

$$V_{GS} = 0$$

$$\therefore I_D = \frac{V_{G1}}{R_S}$$

$$\therefore P1 = \left(0, \frac{V_{G1}}{R_S} \right)$$

Let,

$$I_D = 0$$

$$\therefore V_{GS} = V_{G1}$$

$$\therefore P2 = (V_{G1}, 0)$$

$$\therefore P1 = (0, 1.22 \text{ mA})$$

$$\therefore P2 = (1.822 \text{ V}, 0)$$

$$\therefore Q = (-1.8 \text{ V}, 2.4 \text{ mA})$$

Ans:-

For V_{DS} :-
mon

⇒ Apply KVL in loop (L2):-

$$V_{DD} - I_D R_D - V_{DS} - I_D R_S = 0$$

$$\therefore V_{DS} = V_{DD} - I_D (R_D + R_S) \quad \text{--- (1)}$$

We know:-
mon

$$I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_P}\right)^2$$

$$= 8 \text{ mA} \left(1 - \frac{-1.8}{-4}\right)^2$$

$$= 2.42 \text{ mA}$$

put the value of I_D in eq. (1) =)

$$\therefore V_{DS} = 16 - 2.42 \text{ m} (2.4 \text{ k} + 1.5 \text{ k})$$

$$V_{DS} = 6.562 \text{ V}$$

Ans:-

For V_S :-
mon

$$V_S = I_D R_S = 2.42 \text{ m} \times 1.5 \text{ k} = 3.63 \text{ V}$$

For V_D :-
mon

$$V_{DS} = V_D - V_S$$

$$\therefore V_D = V_{DS} + V_S = 6.562 + 3.63 = 10.192 \text{ V}$$

For V_{GD} :-
mon

$$V_{GD} = V_G - V_D$$

$$= 1.822 - 10.192$$

$$= -8.37 \text{ V}$$

$$\therefore V_{GD} = -8.37 \text{ V}$$

Ans:-

Subject _____

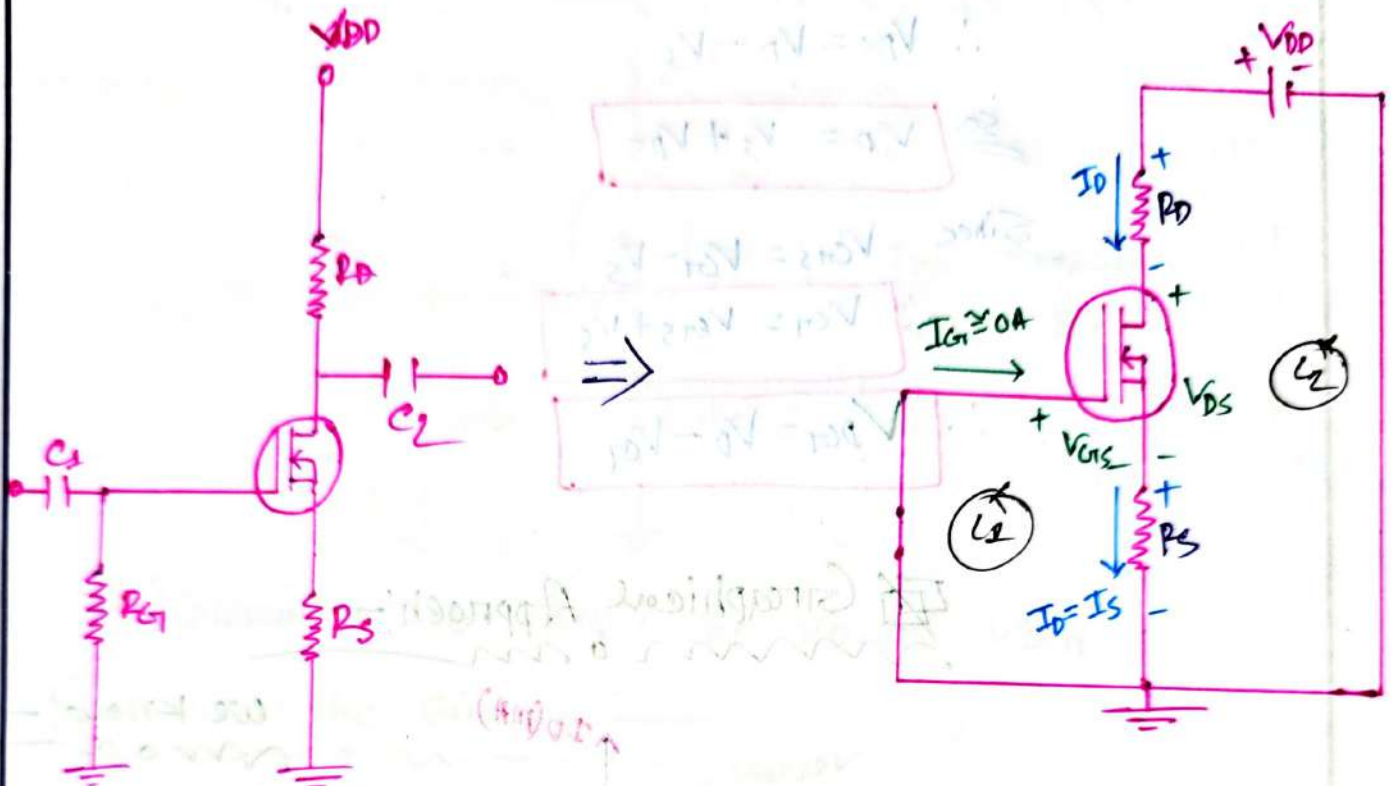
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#1-D-MOSFET
Biasing

Self-Bias Configuration of D-MOSFET



⇒ For V_{GS} :-

⇒ Apply KVL in Loop L_1 :-

$$V_{GS} + I_D R_S = 0$$

$$V_{GS} = -I_D R_S$$

⇒ For I_D :-

⇒ we know:-

$$I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_P} \right)^2$$

⇒ For V_{DS} :-

⇒ Apply KVL in Loop L_2 :-

$$V_{DD} - I_D R_D - V_{DS} - I_D R_S = 0$$

$$V_{DS} = V_{DD} - I_D (R_D + R_S)$$

Now:- For V_S, V_D, V_{GS}, V_{GS} :-

$$\therefore V_S = I_S R_S = I_D R_S$$

$$\therefore V_{DS} = V_D - V_S$$

$$\text{So, } V_D = V_S + V_{DS}$$

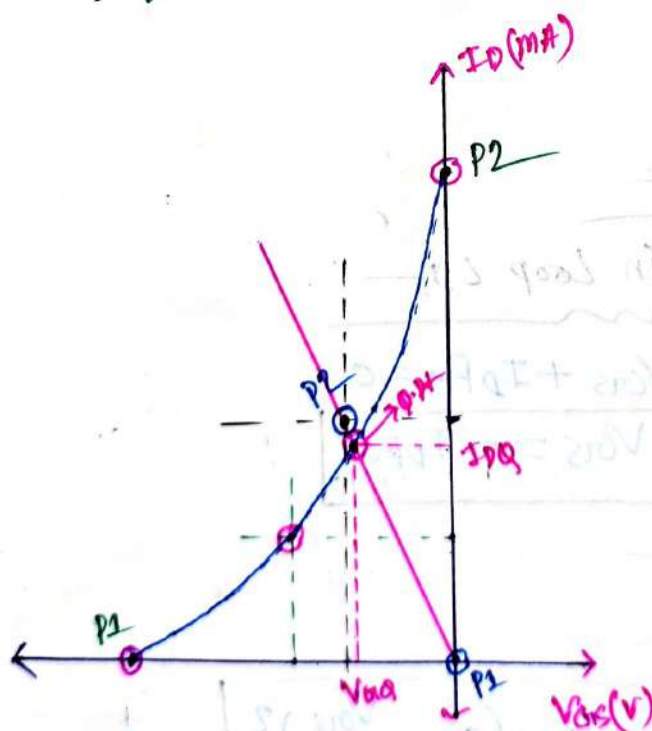
$$\text{Since } V_{GS} = V_{G1} - V_S$$

$$\therefore V_{G1} = V_{GS} + V_S$$

$$\therefore V_{DGS} = V_D - V_{G1}$$

Graphical Approach:-

Let,
 $I_{DSS} = 6\text{mA}$
 $V_P = -4\text{V}$



We know:-

$$P1 \equiv (V_P, 0)$$

$$P2 \equiv (0, I_{DSS})$$

$$P3 \equiv \left(\frac{V_P}{2}, \frac{I_{DSS}}{4} \right)$$

$V_{GS} = -I_D R_S \rightarrow$ straight line passing through the origin.

Now To Draw Load line we need two points:-

if

$$I_D = 0\text{A}$$

$$\therefore V_{GS} = 0\text{V}$$

$$P1 \equiv (0, 0)$$

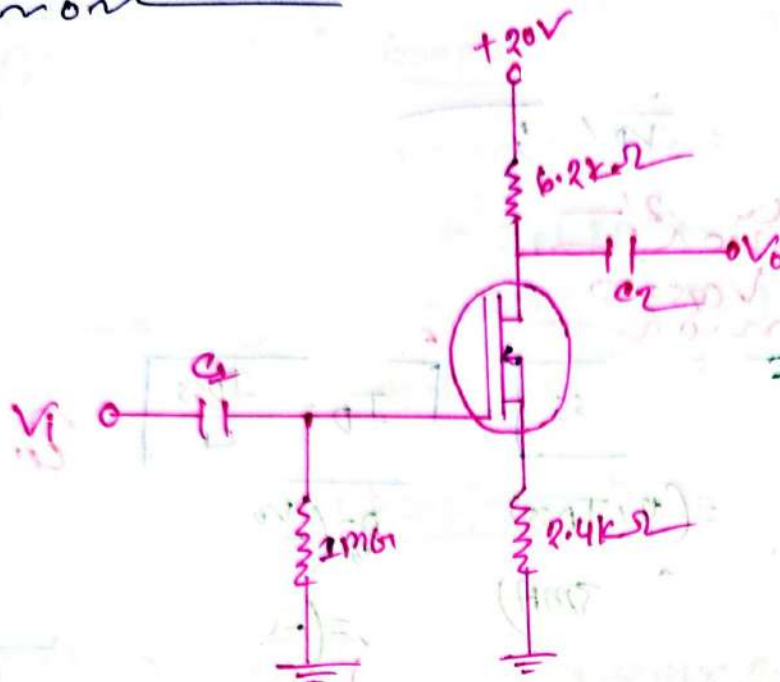
if

$$I_D = \frac{I_{DSS}}{2}$$

$$V_{GS} = -\frac{I_{DSS} R_S}{2}$$

$$P2 \equiv \left(-\frac{I_{DSS} R_S}{2}, \frac{I_{DSS}}{2} \right)$$

problem: 1:—
~~~~~

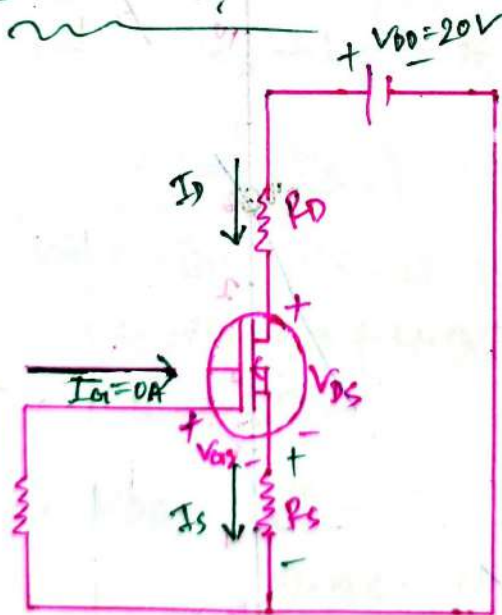


$$I_{DSS} = 8\text{mA}$$

$$V_p = -8\text{V}$$

⇒ Determine  $I_{DQ}$ ,  $V_{GSQ}$ ,  $V_{DS}$ ,  $V_D$ ,  $V_S$ ,  $V_{DG}$

⇒ Redraw the circuit:—  
~~~~~



here,

$$V_{DD} = 20\text{V}$$

$$R_D = 6.2\text{k}\Omega$$

$$R_S = 2.4\text{k}\Omega$$

$$I_{DSS} = 8\text{mA}$$

$$V_p = -8\text{V}$$

For V_{GS} :—
~~~~~

$$V_{GS} + I_S R_S = 0$$

$$\therefore V_{GS} = -I_D R_S$$



For  $V_{GS}$  &  $I_D$ :-we know:-

$$I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_P}\right)^2 \quad ; \quad V_{GS} = -I_D R_S$$

Case 1:-

$I_D = 0A$

$$V_{GS} = V_P$$

$$P_1 \equiv (V_P, 0)$$

$$P_1 \equiv (-8, 0)$$

Case 2:-

$V_{GS} = 0$

$$I_D = I_{DSS}$$

$$P_2 \equiv (0, I_{DSS})$$

$$P_2 \equiv (0, 8mA)$$

Case 3:-

$V_{GS} = \frac{V_P}{2}$

$$I_D = \frac{I_{DSS}}{4}$$

$$P_3 \equiv \left(\frac{V_P}{2}, \frac{I_{DSS}}{4}\right)$$

$$P_3 \equiv (-4V, 2mA)$$

Case 4:-

$V_{GS} \Rightarrow 0V$  &

$V_{GS} < \left|\frac{V_P}{2}\right|$

$V_{GS} = \left|\frac{V_P}{4}\right|$

$$I_D = I_{DSS} \left(\frac{5}{4}\right)^2$$

$$P_4 \equiv \left(\left|\frac{V_P}{4}\right|, I_{DSS} \frac{25}{16}\right)$$

$$\therefore P_4 \equiv (2, 12.5mA)$$

For load line:-

$$P_1 \equiv (0, 0)$$

if  $I_D = \frac{I_{DSS}}{2}$

Then  $V_{GS} = -\frac{I_{DSS} R_S}{2}$

$$\therefore P_2 \equiv \left(-\frac{I_{DSS} R_S}{2}, \frac{I_{DSS}}{2}\right)$$

$$\therefore P_2 \equiv (-9.6, 4)$$

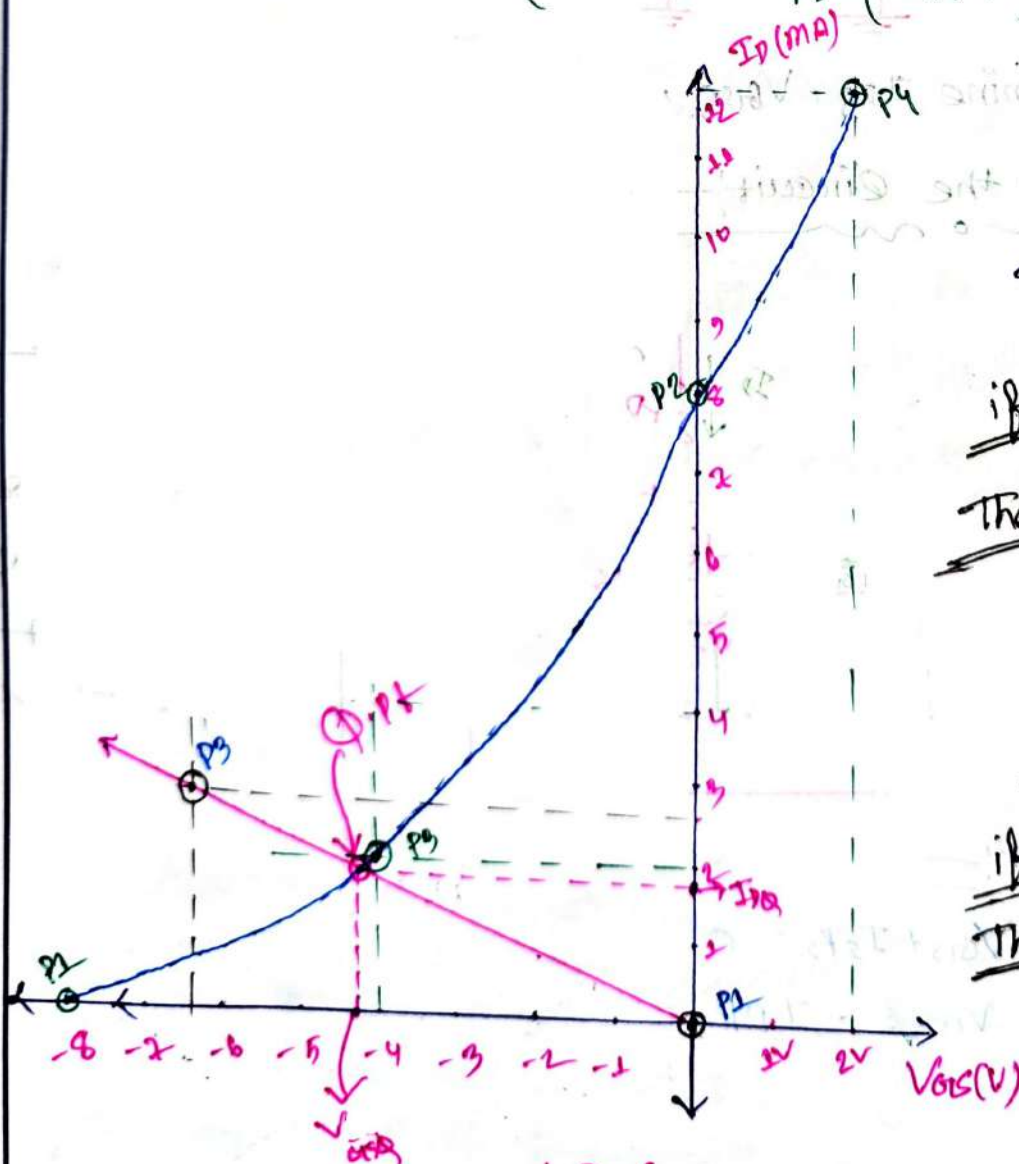
if  $I_D = \frac{I_{DSS}}{3}$

Then  $V_{GS} = -\frac{I_{DSS} R_S}{3}$

$$\therefore P_3 \equiv \left(-\frac{I_{DSS} R_S}{3}, \frac{I_{DSS}}{3}\right)$$

$$\therefore P_3 \equiv (-6.4, 2.6)$$

$$\therefore Q \equiv (-4.3V, 1.8mA)$$





Subject \_\_\_\_\_

Date : / /

For  $V_{DS}$ :-

$\Rightarrow$  Apply KVL in Loop 2:-

$$V_{DD} - I_D R_D - V_{DS} - I_D R_S = 0$$

$$\therefore V_{DS} = V_{DD} - I_D (R_D + R_S)$$

$$= 20 - 1.8 \text{m} (2.4 \text{k} + 6.2 \text{k})$$

$$= 5.06 \text{V}$$

For  $V_S, V_D, V_{GS}, V_{GD}$ :-

we know:-

$$V_S = I_D R_S = 1.8 \text{m} \times 2.4 \text{k} = 4.32 \text{V}$$

$$\therefore V_{DS} = V_D - V_S$$

$$\therefore V_D = V_{DS} + V_S = 5.06 + 4.32 = 9.38 \text{V}$$

$$\therefore V_{GS} = V_{G1} - V_S$$

$$\therefore V_{G1} = V_{GS} + V_S$$

$$= -4.33 + 4.33 = 0 \text{V}$$

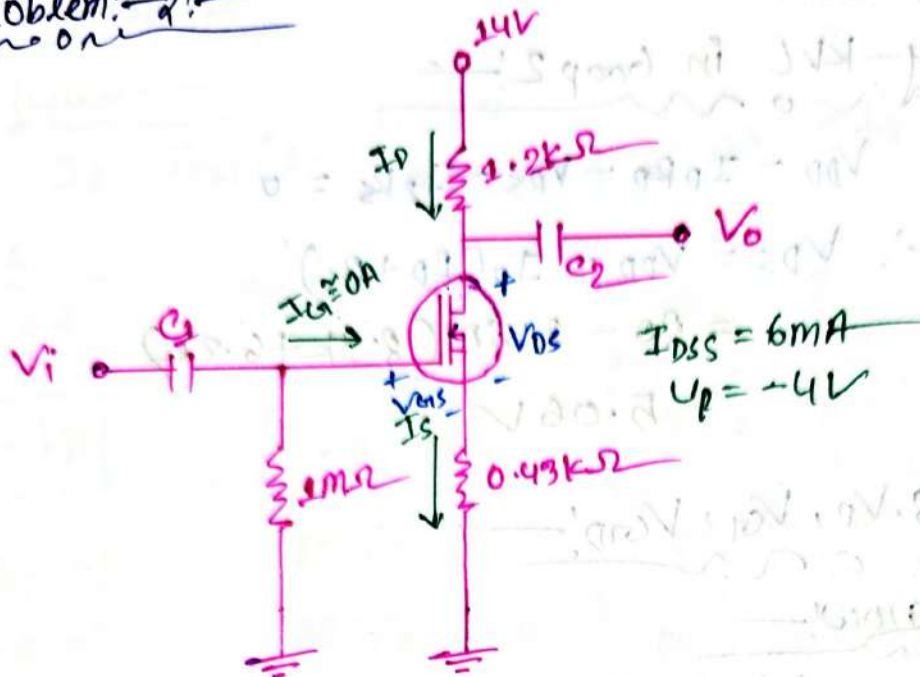
$$\therefore V_{GD} = V_D - V_{G1}$$

$$= 9.38 - 0$$

$$= 9.38 \text{V}$$

Ans:-

# problem: 2:-  
 none



⇒ Determine  $I_{DQ}$ ,  $V_{GSQ}$ ,  $V_{DS}$ ,  $V_D$

⇒ we know—  
 none

$$V_{GS} = -I_D R_S$$

$$I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_P}\right)^2$$

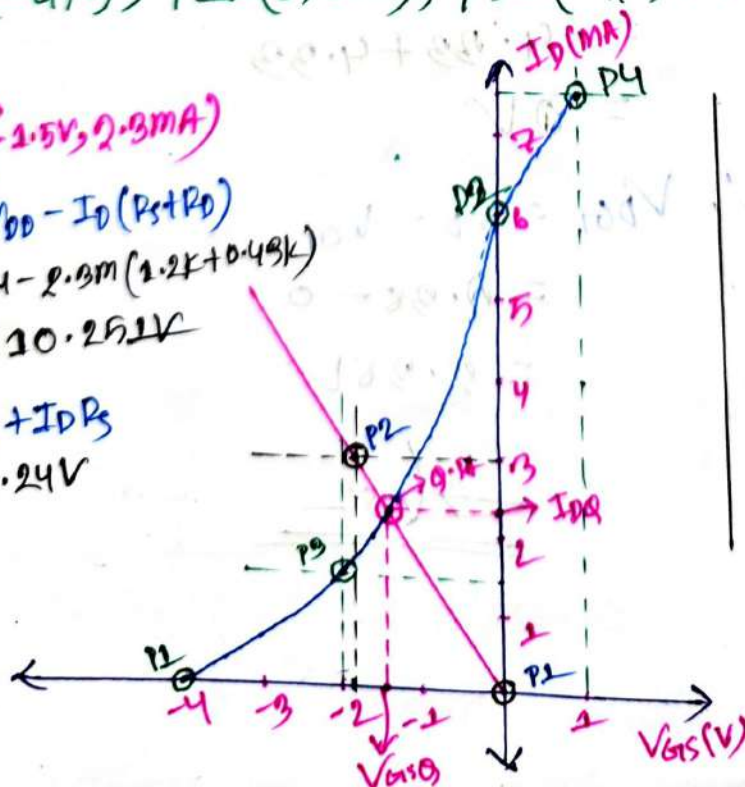
$$P1 \equiv (V_P, 0); P2 \equiv (0, I_{DSS}); P3 \equiv \left(\frac{V_P}{2}, \frac{I_{DSS}}{4}\right); P4 \equiv \left(-\frac{V_P}{4}, \frac{25I_{DSS}}{20}\right)$$

$$\therefore P1 \equiv (-4, 0); P2 \equiv (0, 6\text{mA}); P3 \equiv (-2\text{V}, 1.5\text{mA}); P4 \equiv (-1\text{V}, 7.5\text{mA})$$

$$\therefore Q \equiv (-1.5\text{V}, 2.3\text{mA})$$

$$\begin{aligned} V_{DS} &= V_{DD} - I_D (R_D + R_S) \\ &= 14 - 2.3\text{m}(1.2\text{k} + 0.49\text{k}) \\ &= 10.25\text{V} \end{aligned}$$

$$\begin{aligned} V_D &= V_{DS} + I_D R_S \\ &= 11.24\text{V} \end{aligned}$$



For load line:—  
 none

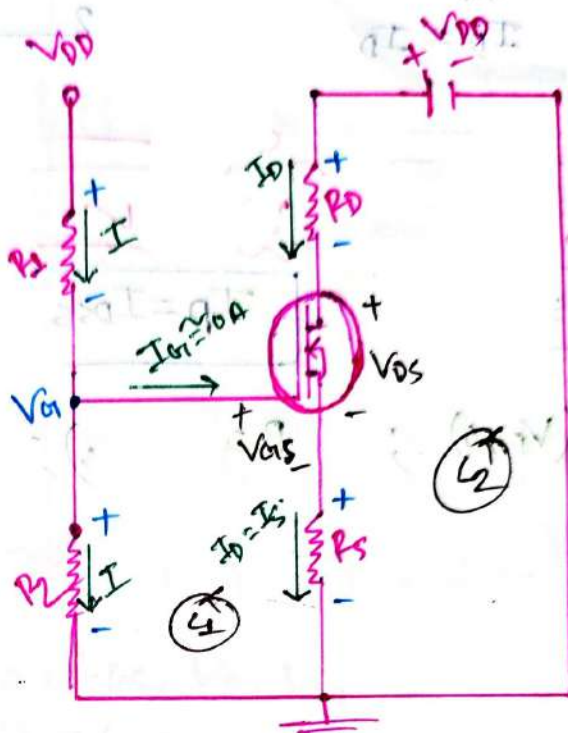
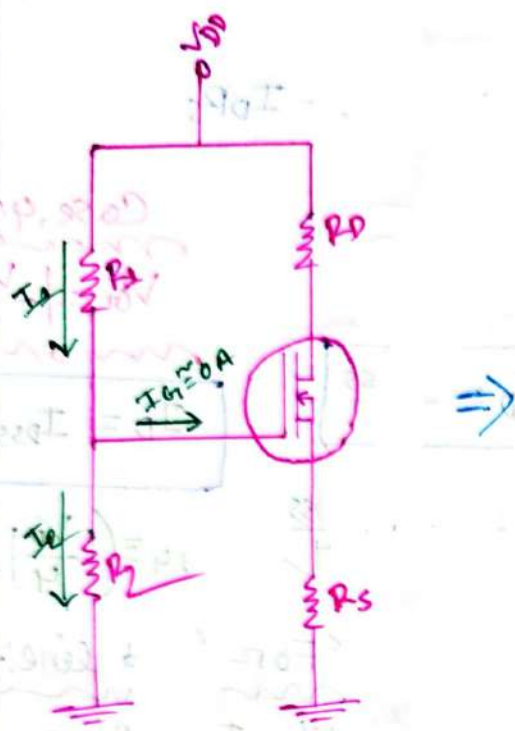
$$P1 \equiv (0, 0)$$

$$P2 \equiv \left(-\frac{I_{DSS} R_S}{2}, \frac{I_{DSS}}{2}\right)$$

$$\therefore P2 \equiv (-1.24\text{V}, 3\text{mA})$$



## # Voltage-Divider Bias of D-MOSFET



⇒ Voltage Drop Across  $R_1$ :

$$V_{G1} = IR_1$$

$$\Rightarrow V_{G1} = \frac{V_{DD} \times R_2}{R_1 + R_2}$$

Since,  $I_{G1} \approx 0A$

$$\therefore I_1 = I_2 = I = \frac{V_{DD} - 0}{R_1 + R_2}$$

$$\therefore I = \frac{V_{DD}}{R_1 + R_2}$$

For  $V_{GS}$ :

⇒ Apply KVL in Loop (L1) —

$$-V_{G1} + V_{GS} + I_D R_S = 0$$

$$\therefore V_{GS} = -I_D R_S + V_{G1}$$

$$\Rightarrow V_{GS} = V_{G1} - I_D R_S$$

For  $V_S$ :

$$V_S = I_D R_S$$

For  $V_D$ :

$$\therefore V_{DS} = V_D - V_S$$

$$\therefore V_D = V_{DS} + V_S$$

$$\text{OR, } V_D = I_D R_D$$

For  $V_{DS}$ :

⇒ Apply KVL in Loop (L2) —

$$V_{DD} - V_D - V_{DS} - V_S = 0$$

$$\therefore V_{DS} = V_{DD} - I_D R_D - I_D R_S$$

$$V_{DS} = V_{DD} - I_D (R_D + R_S)$$

For  $V_{DG1}$ :

$$V_{DG1} = V_D - V_{G1}$$

For  $V_{G1}$ :

$$V_{G1} = V_{GS} + V_S$$



Graphical Approach:-⇒ we know:-

$$I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_P}\right)^2; \quad V_{GS} = V_{G1} - I_D R_S$$

Case 1:- $I_D = 0A$ :-

$$V_{GS} = V_P$$

$$\therefore P_1 \equiv (V_P, 0);$$

Case 2:- $V_{GS} = 0V$ :-

$$I_D = I_{DSS}$$

$$P_2 \equiv (0, I_{DSS});$$

Case 3:- $V_{GS} = \frac{V_P}{2}$ :-

$$I_D = \frac{I_{DSS}}{4}$$

$$P_3 \equiv \left(-\frac{V_P}{2}, \frac{I_{DSS}}{4}\right);$$

Case 4:- $V_{GS} = \left|\frac{V_P}{4}\right|$ :-

$$I_D = I_{DSS} \times \left(\frac{5}{4}\right)^2$$

$$P_4 \equiv \left(\left|\frac{V_P}{4}\right|, \frac{25 I_{DSS}}{16}\right)$$

For load line:-Let  $I_D = 0A$ :-

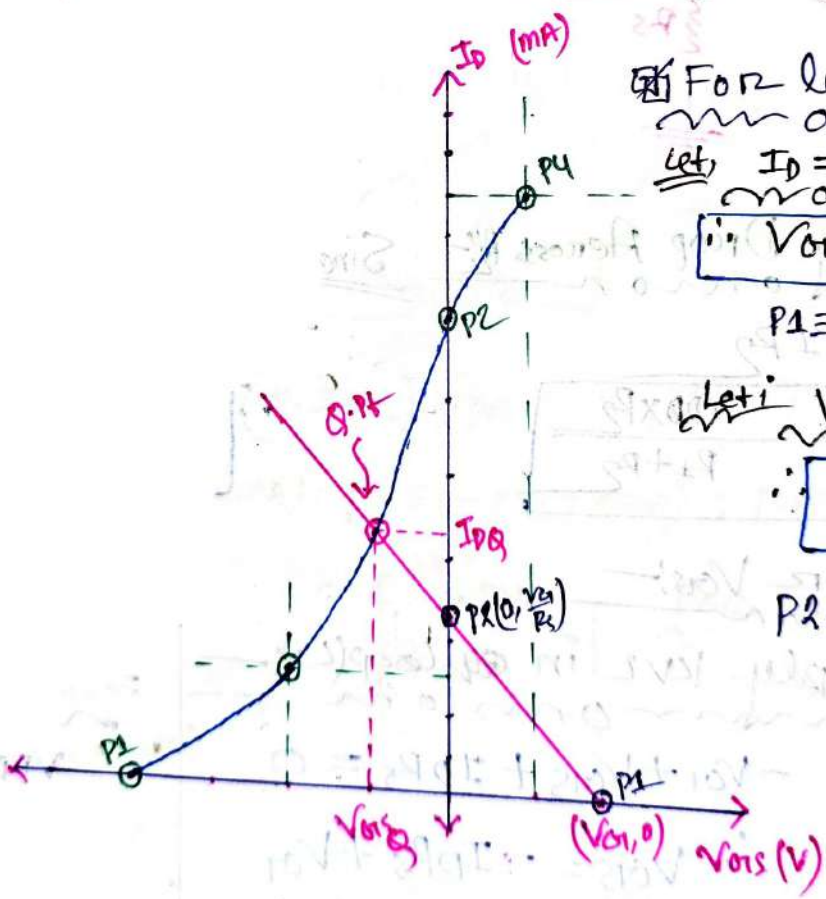
$$\therefore V_{GS} = V_{G1}$$

$$P_1 \equiv (V_{G1}, 0A)$$

Let  $V_{GS} = 0V$ :-

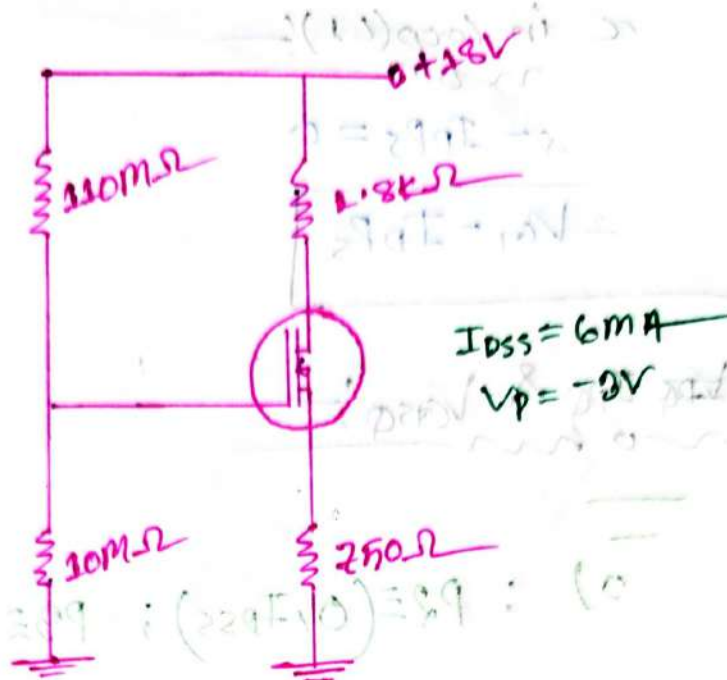
$$\therefore I_D = \frac{V_{G1}}{R_S}$$

$$P_2 \equiv \left(0V, \frac{V_{G1}}{R_S} A\right)$$



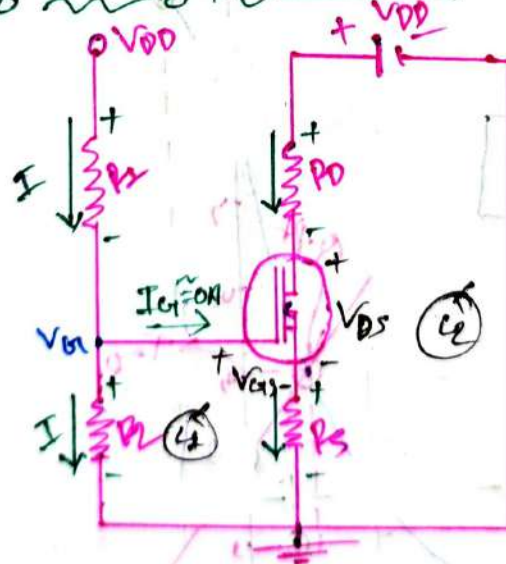
$$Q \equiv (V_{GSQ}, I_{DQ})$$

# problem: 3:-



⇒ Determine  $V_D$ ,  $I_{DQ}$ ,  $V_{GSQ}$ ,  $V_{DS}$ ,  $V_S$ ,  $V_{G1}$

⇒ Int redraw the Network:-



here,

$$R_1 = 110M\Omega$$

$$R_2 = 10M\Omega$$

$$R_D = 1.8K\Omega$$

$$R_S = 0.25K\Omega$$

$$I_{DSS} = 6mA$$

$$V_P = -3V$$

$$V_{DD} = 18V$$

We know:-

$$V_{G1} = \frac{V_{DD} R_2}{R_1 + R_2} = \frac{18 \times 10M\Omega}{(110M + 10M)\Omega}$$

$$V_{G1} = 1.5V$$

Ans:-



For  $V_{GS}$ :—

⇒ Apply KVC in loop (LL):—

$$V_{G1} - V_{GS} - I_D R_S = 0$$

$$\Rightarrow V_{GS} = V_{G1} - I_D R_S$$

For  $I_{DQ}$  &  $V_{GSQ}$ :—

We know:—

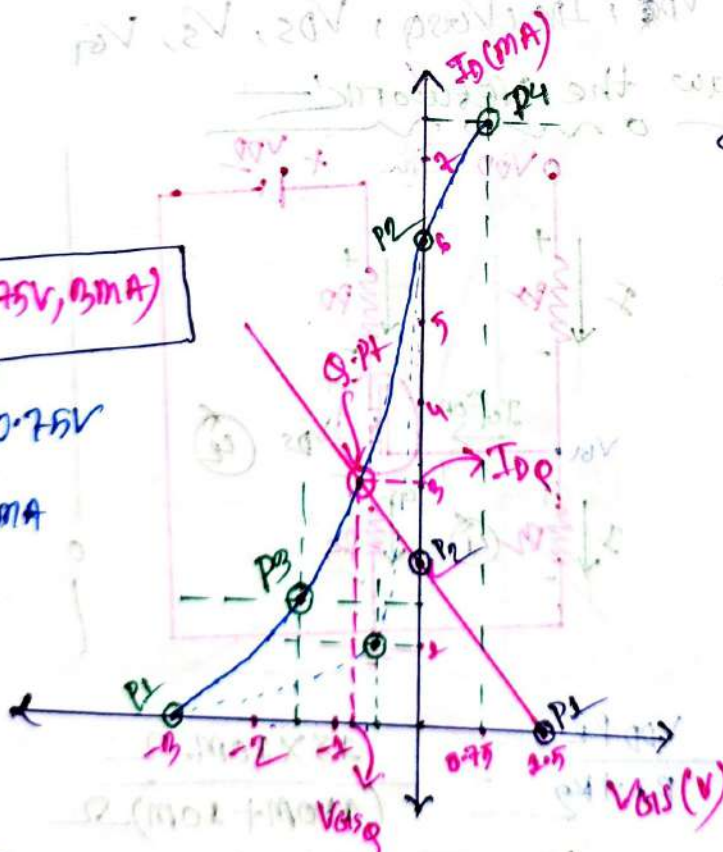
$$P_1 \equiv (V_P, 0) ; P_2 \equiv (0, I_{DSS}) ; P_3 \equiv \left( \frac{V_P}{2}, \frac{I_{DSS}}{4} \right) ; P_4 \equiv \left( -\frac{V_P}{4}, \frac{2I_{DSS}}{3} \right)$$

$$\therefore P_1 \equiv (-3, 0) ; P_2 \equiv (0, 6 \text{ mA}) ; P_3 \equiv (-1.5 \text{ V}, 1.5 \text{ mA}) ; P_4 \equiv (0.75 \text{ V}, 2.5 \text{ mA})$$

$$\therefore Q \equiv (-0.75 \text{ V}, 3 \text{ mA})$$

$$\therefore V_{GSQ} = -0.75 \text{ V}$$

$$I_{DQ} = 3 \text{ mA}$$



For load line:—

$$P_1 \equiv (V_{G1}, 0) \equiv (1.5, 0)$$

$$P_2 \equiv \left( 0, \frac{V_{G1}}{R_S} \right) \equiv (0, 2 \text{ mA})$$

$$V_D = I_D \times R_D = 3 \text{ mA} \times 1.8 \text{ k}\Omega = 5.4 \text{ V}$$

$$V_{G1} = 1.5 \text{ V}$$

$$V_S = I_D \times R_S = 3 \text{ mA} \times 0.75 \text{ k}\Omega = 2.25 \text{ V}$$

$$V_{DS} = V_D - V_S = 5.4 - 2.25 = 3.15 \text{ V}$$

Ans:—

Subject \_\_\_\_\_

Date :     /     /

#E-MOSFET  
Biasing

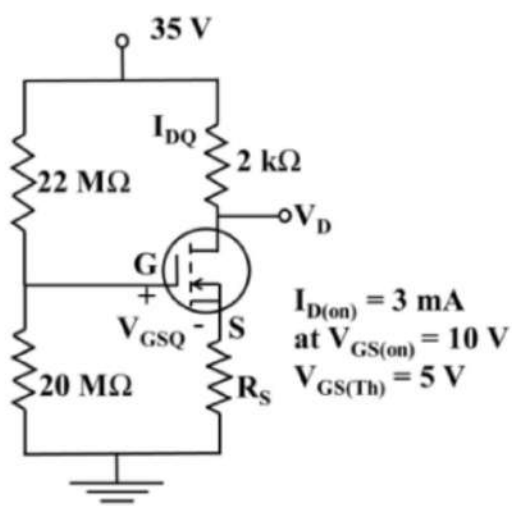


TOPIC NAME : \_\_\_\_\_

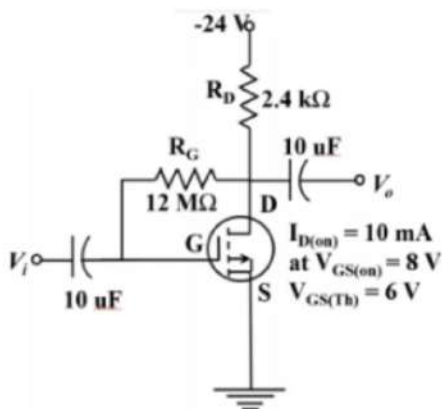
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TIME : \_\_\_\_\_ DATE : / /

4. Analyze Q points for the circuit in the mathematical and graphical approach for the following circuit. Here, ID = 12-34567-8, Rs = [Last three digits of your ID, here, 567 + 500] Ohm. [Consider your own ID].

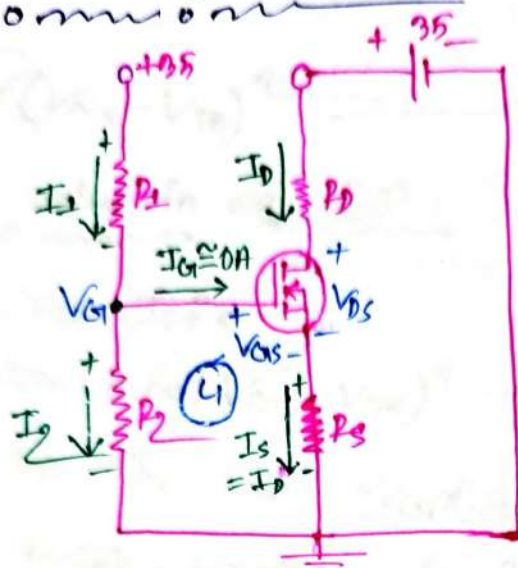


5. Analyze the Q point position of the following circuit if  $R_D = 1$  and  $2.4 \text{ k}\Omega$ .



Ans. to the Q. no. 4:-

1st we draw the circuit:-



here,

$$R_1 = 22M\Omega$$

$$R_2 = 20M\Omega$$

$$V_{DD} = 35V$$

$$I_1 \approx I_2 \quad [ \because I_{G1} \approx 0A ]$$

$$R_S = 604 \quad [ \because I_D = 23.50104A ]$$

$$R_D = 2k\Omega$$

$$I_D \approx I_S$$

we know:-

$$V_{G1} = \frac{V_{DD} R_2}{R_1 + R_2}$$

$$= \frac{35 \times 20M\Omega}{(20 + 22)M\Omega}$$

$$= 16.667V$$

Now Apply KVL in loop 1:-

$$V_{G1} - V_{GS} - I_D R_S = 0$$

$$\therefore V_{GS} = V_{G1} - I_D R_S \quad \text{--- (1)}$$

Again

we know:-

$$K = \frac{I_{D(on)}}{(V_{GS(on)} - V_{Th(on)})^2}$$

$$= \frac{3mA}{(10 - 5)^2}$$

$$= 120 \mu A/V^2$$

here

$$I_{D(on)} = 3mA$$

$$V_{GS(on)} = 10V$$

$$V_{Th(on)} = 5V$$



### MATHEMATICAL Approach:-

We know:-

$$I_D = K(V_{GS} - V_{TH})^2 \quad \text{--- (1)}$$

∴ put the value in eq (1) ⇒

$$V_{GS} = V_{G1} - I_D R_S$$

$$= V_{G1} - K R_S (V_{GS} - V_{TH})^2$$

$$= V_{G1} - K R_S (V_{GS}^2 - 2V_{GS} \times V_{TH} + V_{TH}^2)$$

$$= 16.667 - 120 \mu \times 694 (V_{GS}^2 - 10V_{GS} + 25)$$

$$= 16.667 - 83.28mV_{GS}^2 + 0.8328V_{GS} + (-2.082)$$

$$\therefore 83.28mV_{GS}^2 + 0.1672V_{GS} - 14.585 = 0$$

$$\therefore V_{GS} = -14.27, 12.26V$$

↓  
This is not possible for n-channel E-MOSFET.

$$\therefore V_{GSQ} = 12.26V$$

∴ put the value in eq (1) ⇒

$$\therefore I_{DQ} = K(V_{GSQ} - V_{TH})^2$$
$$= 120 \mu (12.26 - 5)^2$$

$$I_{DQ} = 6.324mA$$

OR put the value in eq (1) ⇒

$$\therefore I_{DQ} = \frac{V_{G1} - V_{GSQ}}{R_S} = \frac{16.667 - 12.26}{694} = 6.35mA$$

$$I_{DQ} = 6.35mA$$

$\therefore$  The Q.P  $\equiv (12.26V, 6.35mA)$

### Graphical Approach:-

We know:-

$$I_D = k(V_{GS} - V_{TH})^2$$

Case 1:-      Case 2:-      Case 3:-      Case 4:-

$V_{GS} \leq V_{TH}$        $V_{GS} = 2V_{TH}$        $V_{GS} = 3V_{TH}$        $V_{GS} = 4V_{TH}$   
 $\therefore I_D = 0A$ ,       $\therefore I_D = kV_{TH}^2$ ,       $\therefore I_D = 4kV_{TH}^2$ ,       $\therefore I_D = 9kV_{TH}^2$

$\therefore P1 \equiv (V_{TH}, 0)$ ,       $P2 \equiv (2V_{TH}, kV_{TH}^2)$ ,       $P3 \equiv (3V_{TH}, 4kV_{TH}^2)$ ,       $P4 \equiv (4V_{TH}, 9kV_{TH}^2)$

Now, put the values in those point:-

$P1 \equiv (5, 0)$

$P2 \equiv (10V, 3mA)$

$P3 \equiv (15V, 12mA)$

$P4 \equiv (20V, 27mA)$

Now For load line:-

Case 1:-      Case 2:-

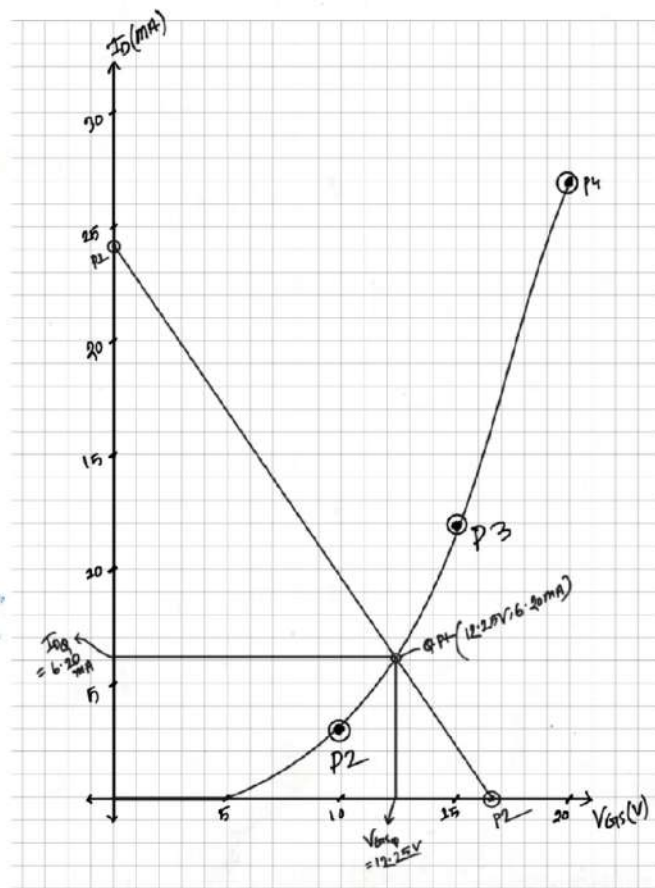
$V_{GS} = 0V$        $V_{GS} = V_{G1}$

$\therefore I_D = \frac{V_{G1}}{R_S}$        $I_D = 0A$

$P1 \equiv (0V, \frac{V_{G1}}{R_S})$ ,       $P2 \equiv (V_{G1}, 0)$

$\therefore P1 \equiv (0, 24.016mA)$

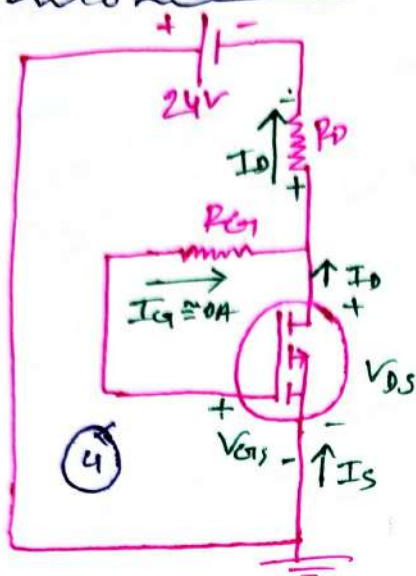
$P2 \equiv (16.667V, 0)$



The Q.P  $\equiv (12.25V, 6.20mA)$

Ans! to! the Q! no! 5!:-

as we draw the circuit:-



here

$$R_D = 1k\Omega \text{ OR } 1.4k\Omega$$

$$I_S = I_D \quad [\because I_G \cong 0A]$$

is given:-

$$I_{D(on)} = 10mA$$

$$\left. \begin{aligned} V_{GS(on)} &= -8V \\ V_{GS(th)} &= -6V \end{aligned} \right\} [\because P\text{-CH}]$$

Apply KVL in Loop L1:-

$$24 - I_D R_D + V_{GS} = 0$$

$$\therefore V_{GS} = I_D R_D - 24 \quad \text{--- (1)}$$

For 1k  $R_D$ :-

we know:-

$$k = \frac{I_{D(on)}}{(V_{GS(on)} - V_{TH(on)})^2} = \frac{10mA}{(-8 + 6)^2} = 2.5mA/V^2$$

we know:-

$$I_D = k(V_{GS} - V_{TH})^2$$

Case 1:-

$$V_{GS} \leq V_T$$

$$\therefore I_D = 0A$$

Case 2:-

$$V_{GS} = V_{GS(on)}$$

$$\therefore I_D = I_{D(on)}$$

Case 3:-

$$V_{GS} = -7V$$

$$I_D = 2.5mA$$

Case 4:-

$$V_{GS} = -9V$$

$$I_D = 22.5mA$$

$$\therefore P1 \equiv (V_T, 0A) \equiv (-6, 0)$$

$$P2 \equiv (V_{GS(on)}, I_{D(on)}) \equiv (-8, 10mA)$$

$$P3 \equiv (-7, 2.5mA)$$

$$P4 \equiv (-9, 22.5mA)$$



$R_D = 1 \text{ k}\Omega$

we know:-

$$V_{GS} = I_D R_D - 24$$

For load line:-

Case: 1:-

$$V_{GS} = 0 \text{ V}$$

$$\therefore I_D = \frac{24}{R_D}$$

$$P1 = (0, \frac{24}{R_D})$$

$$\therefore P1 = (0, 24 \text{ mA})$$

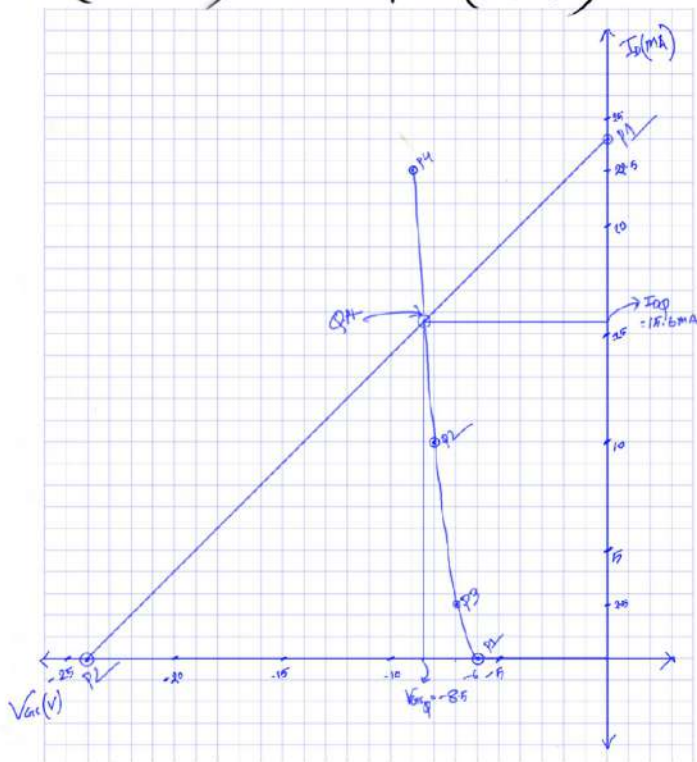
Case 2:-

$$I_D = 0 \text{ A}$$

$$\therefore V_{GS} = -24 \text{ V}$$

$$P2 = (-24, 0)$$

$$P2 = (-24, 0)$$



$$\therefore Q = (-8.5 \text{ V}, 15.6 \text{ mA})$$

$\Rightarrow$  Drain Current ( $I_D$ ) decreases from  $15.6 \text{ mA}$  to  $6.75 \text{ mA}$  when  $R_D$  increases.

Voltage ( $V_{GS}$ ) become less negative (closer to zero) as  $I_D$  increases, changing from  $-8.5 \text{ V}$  to  $-7.65 \text{ V}$ .

$R_D = 2.4 \text{ k}\Omega$

we know:-

$$V_{GS} = I_D R_D - 24$$

For load line:-

Case: 1

$$V_{GS} = 0 \text{ V}$$

$$\therefore I_D = \frac{24}{R_D}$$

$$P1 = (0, \frac{24}{R_D})$$

$$P1 = (0, 10 \text{ mA})$$

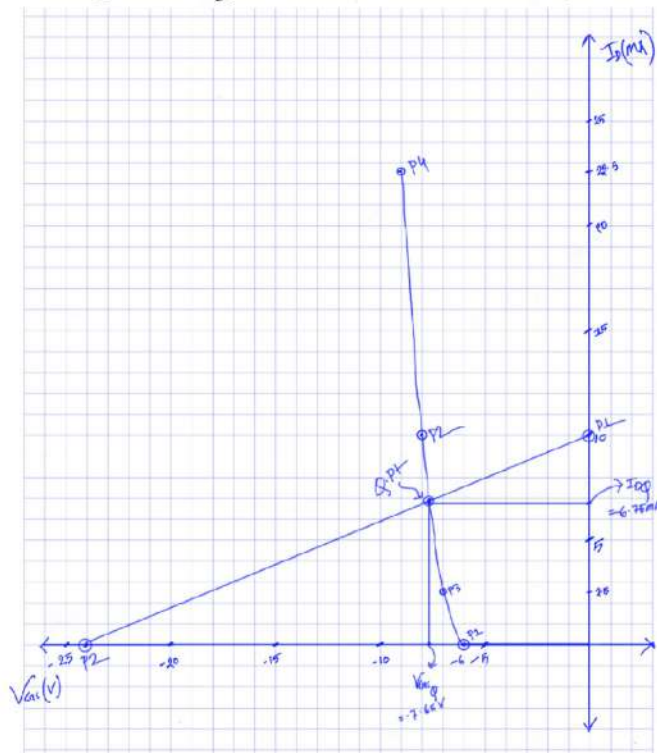
Case: 2

$$I_D = 0 \text{ A}$$

$$\therefore V_{GS} = -24 \text{ V}$$

$$P2 = (-24, 0)$$

$$P2 = (-24, 0)$$



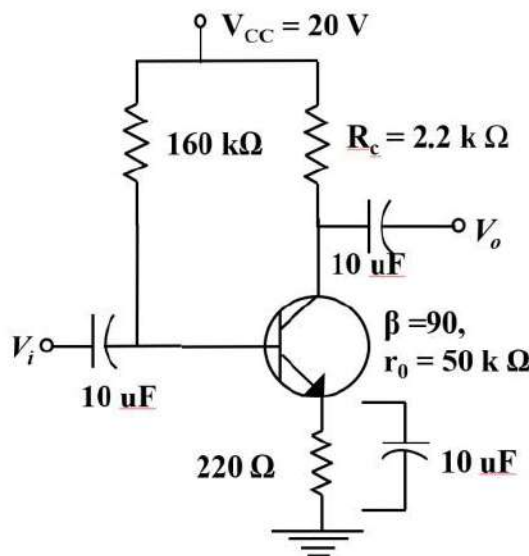
$$Q = (-7.65, 6.75 \text{ mA})$$



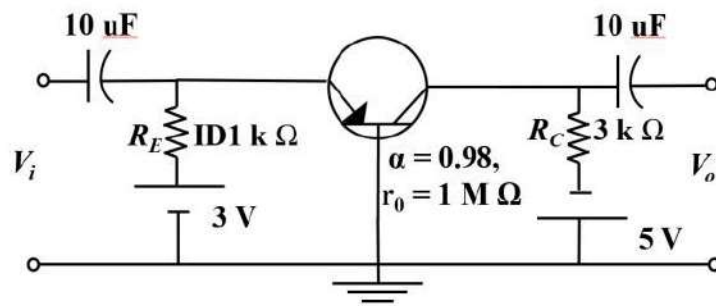
**American International University- Bangladesh (AIUB)**  
**Faculty of Engineering (EEE)**

## Sample Questions

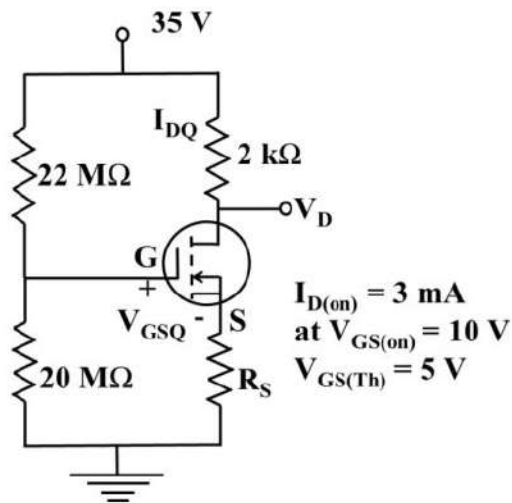
1. Conclude by analyzing (comparing) the effects in the ac parameters for the following circuit with and without bypass capacitor.



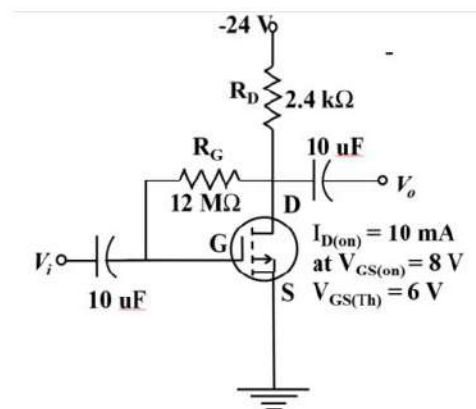
2. **Analyze** the transfer characteristics curve of JFET and D-MOSFET (any one channel is ok) using the Shockley's equation and shorthand method. Consider the pinch-off voltage is [use appropriate sign here] [three digits before the last digit of your ID] mV and  $I_{DSS}$  is [three digits before the last digit of your ID] mA.
3. **Analyze** the effect of  $ID1$  in the voltage gain, input impedance and output impedance of the following circuit. Here,  $ID = 12-34765-8$ ,  $ID1 = 12\text{ k}$  and [[Last three digits of your ID, here, 765] + 500 Ohm]. [Consider your own ID].



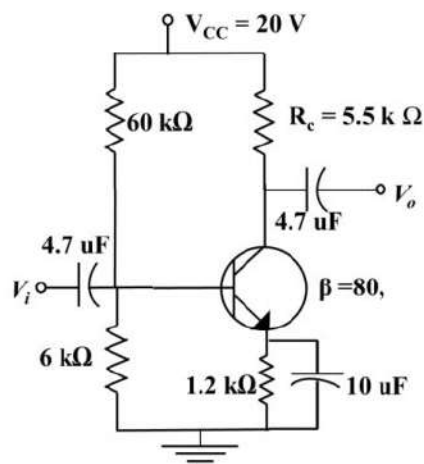
4. **Analyze** Q points for the circuit in the mathematical and graphical approach for the following circuit. Here,  $I_D = 12-34567-8$ ,  $R_S = [\text{Last three digits of your ID, here, } 567 + 500] \text{ Ohm}$ . [Consider your own ID].



5. Analyze the Q point position of the following circuit if  $R_D = 1$  and  $2.4 \text{ kOhm}$ .



6. Analyze the voltage gain of the following circuit. Design a circuit having same gain using a MOSFET.

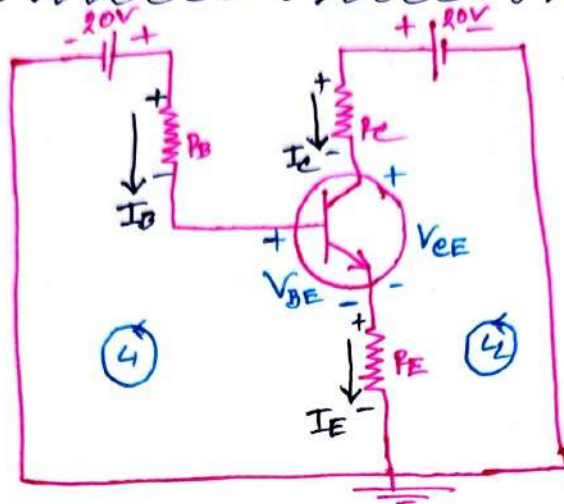




Ans to the q: no: 1

With capacitor

⇒ 1st redraw the circuit for DC Analyzing.



hence,

$$R_B = 160k$$

$$R_C = 2.2k\Omega$$

$$R_E = 220\Omega$$

$$R_0 = 50k\Omega$$

$$\beta = 100$$

For  $I_B$ :

Apply KVL in Loop L1:

$$20 - I_B R_B - V_{BE} - I_E R_E = 0$$

$$\Rightarrow 20 - I_B R_B - V_{BE} - (\beta + 1) I_B R_E = 0$$

$$\therefore I_B = \frac{20 - 0.7}{R_B + (\beta + 1) R_E}$$

$$= \frac{20 - 0.7}{160k + 101 \times 220}$$

$$= 107.22 \mu A$$

We know,

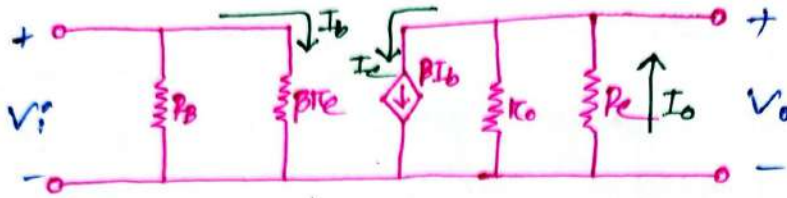
$$\begin{aligned} I_E &= (\beta + 1) I_B \\ &= 101 \times 107.22 \mu A \\ &= 10.75 mA \end{aligned}$$

$\therefore$  we know,

$$r_e = \frac{26mV}{I_E} = \frac{26mV}{10.75mA} = 2.667 \Omega$$

With capacitor:-

Now draw the re equivalent model:-



For Zi:-

$$Z_i = R_B \parallel r_{be}$$

$$= \left( \frac{1}{160k} + \frac{1}{90 \times 2.667} \right)^{-1}$$

$$= 232.67 \Omega$$

$$\therefore Z_i = 232.67 \Omega$$

For Zo:-

$$Z_o = R_C \parallel R_L$$

$$= \left( \frac{1}{80k} + \frac{1}{2.2k} \right)^{-1}$$

$$= 2.107 k\Omega$$

$$\therefore Z_o = 2.107 k\Omega$$

Now:-

For Av:-

$$A_v = \frac{V_o}{V_i} \quad \text{--- (1)}$$

Now,

$$V_o = -I_o \times (R_C \parallel R_L)$$

$$= -I_c (R_C \parallel R_L)$$

$$= -\beta I_b (R_C \parallel R_L)$$

$$\therefore (I_o = I_c)$$

$$\therefore (I_c = \beta I_b)$$

And:-

$$V_i = I_b \times r_{be}$$

$\therefore$  put the value of  $V_o, V_i$  in eq (1)  $\Rightarrow$

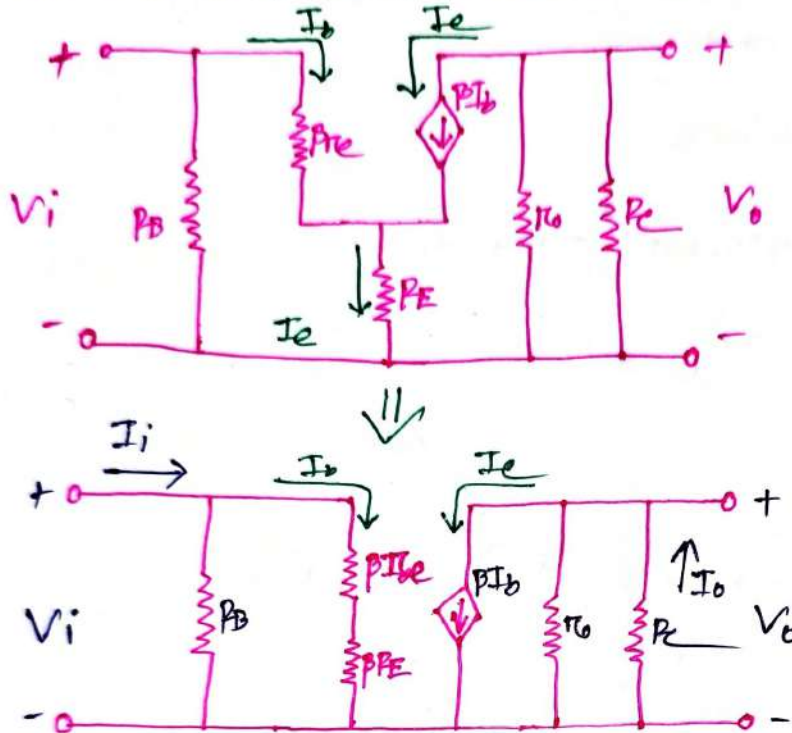
$$\therefore A_v = \frac{-\beta I_b (R_C \parallel R_L)}{I_b r_{be}}$$

$$= \frac{-(R_C \parallel R_L)}{r_{be}} = \frac{-2.107k}{2.667} = \frac{-2107 \Omega}{2.667 \Omega}$$

$$\therefore A_v = -790.02$$

Without Capacitor

⇒ Now draw the  $r_e$  equivalent model;



Since,

$$\begin{aligned} I_e &= I_b + I_c \\ &= I_b + \beta I_b \\ &= I_b(\beta + 1) \end{aligned}$$

Voltage drop Across  $R_E$

$$\begin{aligned} V_{RE} &= I_e R_E \\ &= I_b(\beta + 1) R_E \end{aligned}$$

Passing current through the  $R_E(\beta + 1)$  Resistance.

For  $Z_i$ :

$$Z_i = R_B \parallel \beta(r_e + R_E)$$

$$\therefore = \left( \frac{1}{160k} + \frac{1}{90(2.667 + 220)} \right)^{-1}$$

$$Z_i = 17.800k\Omega$$

For  $Z_o$ :

$$Z_o = r_o \parallel R_C$$

$$= \left( \frac{1}{50k} + \frac{1}{2.2k} \right)^{-1}$$

$$Z_o = 2.107k\Omega$$

Now,

For  $A_v$ :

$$A_v = \frac{V_o}{V_i} \quad \text{--- (1)}$$

$$\therefore V_o = -I_o(R_C \parallel r_o)$$

$$= -I_e(r_o \parallel R_C)$$

$$= -\beta I_b(r_o \parallel R_C)$$

$$\therefore V_i = I_b \beta(r_e + R_E)$$

Put the values in eq (1)

$$\therefore A_v = \frac{-\beta I_b(r_o \parallel R_C)}{I_b \beta(r_e + R_E)} = \frac{-r_o \parallel R_C}{r_e + R_E}$$

$$\therefore A_v = \frac{-2107\Omega}{(2.667 + 220)\Omega} = -0.4625$$

⇒ The bypass capacitor greatly lowers input impedance and boosts voltage gain  $A_v$  but does not affect the output impedance.



Ans: to the Q. no: 2

For n-channel JFET:-

NY ID: 23-50194-1

From Question we have:-

$$V_p = -104 \text{ mV} \text{ (": n-ch)}$$

$$I_{DSS} = 104 \text{ mA}$$

Now:-

We know:-

$$I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_p}\right)^2 \quad \text{--- (1)}$$

∴ Case 1:-

$$V_{GS} = 0 \text{ V}$$

$$∴ I_D = I_{DSS}$$

$$∴ P1 \equiv (0, I_{DSS})$$

Case 2:-

$$V_{GS} = V_p$$

$$∴ I_D = 0 \text{ A}$$

$$∴ P2 \equiv (V_p, 0)$$

Case 3:-

$$V_{GS} = \frac{V_p}{2}$$

$$∴ I_D = \frac{I_{DSS}}{4}$$

$$∴ P3 \equiv \left(\frac{V_p}{2}, \frac{I_{DSS}}{4}\right)$$

Case 4:-

$$V_{GS} = -\frac{V_p}{4}$$

$$∴ I_D = \frac{I_{DSS} \times 9}{16}$$

$$∴ P4 \equiv \left(-\frac{V_p}{4}, \frac{9I_{DSS}}{16}\right)$$

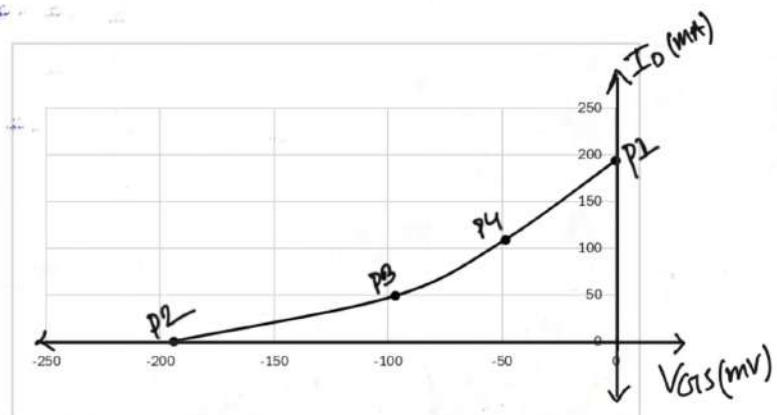
Now put the values in those point:-

$$∴ P1 \equiv (0, 104 \text{ mA})$$

$$P2 \equiv (-104 \text{ mV}, 0)$$

$$P3 \equiv (-52 \text{ mV}, 26 \text{ mA})$$

$$P4 \equiv (-26 \text{ mV}, 59.5 \text{ mA})$$



For n channel MOSFET:-

My ID:- 23-50104-1

From Question we have:-

$$V_p = -104 \text{ mV}$$

$$I_{DSS} = 104 \text{ mA}$$

We know:-

$$I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_p}\right)^2 \quad \text{--- (1)}$$

Case 1:-

$$V_{GS} = 0 \text{ V}$$

$$\therefore I_D = I_{DSS}$$

Case 2:-

$$V_{GS} = V_p$$

$$\therefore I_D = 0 \text{ A}$$

Case 3:-

$$V_{GS} = \frac{V_p}{2}$$

$$\therefore I_D = \left(\frac{I_{DSS}}{4}\right)$$

Case 4

$$V_{GS} = \frac{V_p}{4}$$

$$\therefore I_D = \frac{9 I_{DSS}}{16}$$

Case 5:-

$$V_{GS} > 0 \text{ V}$$

$$V_{GS} < \left|\frac{V_p}{2}\right|$$

$$\therefore V_{GS} = \left|\frac{V_p}{4}\right|$$

$$\therefore I_D = \frac{25 I_{DSS}}{16}$$

Now put the values in those points:-

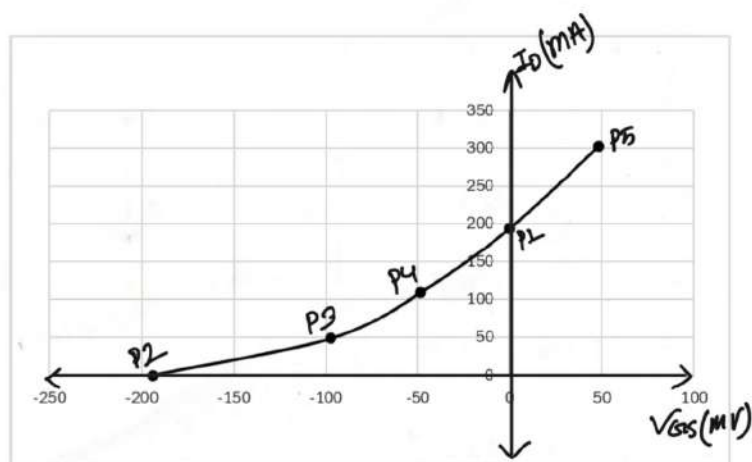
$$\therefore P1 \equiv (0, 104 \text{ mA})$$

$$P2 \equiv (-104 \text{ mV}, 0)$$

$$P3 \equiv (-52 \text{ mV}, 48.5 \text{ mA})$$

$$P4 \equiv (-26 \text{ mV}, 107.25 \text{ mA})$$

$$P5 \equiv (26 \text{ mV}, 303.125 \text{ mA})$$

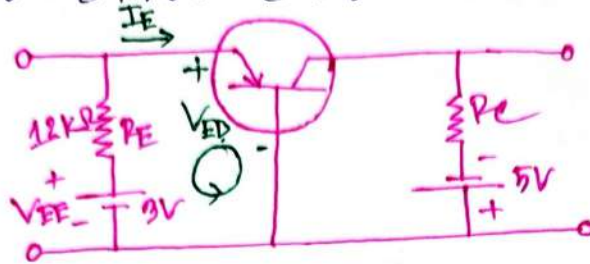


6

Ans: to the Q: no: 3

For 12k IDL:-

1st redraw the circuit for DC:-



Apply KVL in Loop (L1):-

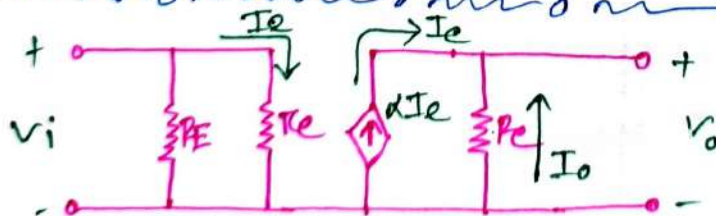
$$V_{EE} - I_E P_E - V_{EB} = 0$$

$$\begin{aligned} \therefore I_E &= \frac{V_{EE} - V_{EB}}{P_E} \\ &= \frac{3 - 0.7}{12k} \\ &= 191.667 \mu A \end{aligned}$$

$\therefore$  we know:-

$$r_e = \frac{26mV}{I_E} = \frac{26mV}{191.667 \mu A} = 135.65 \Omega$$

Now draw the re equivalent model:-



For Zi:-

$$\begin{aligned} Z_i &= P_E \parallel r_e \\ &= \left( \frac{1}{12k} + \frac{1}{135.65} \right)^{-1} \end{aligned}$$

$$\boxed{Z_i = 134.133 \Omega}$$

For Zo:-

$$Z_o = P_c$$

$$\boxed{Z_o = 3k \Omega}$$

here,

$$P_E = 12k \Omega$$

$$\alpha = 0.98$$

$$r_o = 1M \Omega$$

$$P_c = 3k \Omega$$

$$V_{EE} = 3V$$

Since  $r_o$  is quite high, so we can neglect  $r_o$  from the circuit.



For  $A_v$ :-

we know:-  $A_v = \frac{V_o}{V_i}$  — (1)

∴ Output voltage  $V_o$ :-

$$V_o = -I_o R_e$$

$$= -(-I_e R_e) \quad \because [I_o = -I_e]$$

$$= \alpha I_e R_e \quad \because [I_e = \alpha I_e]$$

Input voltage  $V_i$ :-

$$V_i = I_e R_e$$

Put the values in eq (1):-

$$\therefore A_v = \frac{\alpha I_e R_e}{I_e R_e}$$

$$= \frac{R_e}{R_e} \quad \left[ \begin{array}{l} \text{since } R_e \gg \alpha \\ \therefore R_{e\alpha} \approx R_e \end{array} \right]$$

$$\therefore A_v = \frac{3000 \Omega}{135.65 \Omega}$$

$$A_v = 22.11$$

Now:- For  $I_{D1} = 104 + 500 = 604 \mu A$  (since  $M_H I_D = 23 - 50 + 104 = 1$ )

From previous solution:-

we know:-

$$I_E = \frac{V_{EE} - V_{ED}}{R_E}$$

$$= \frac{3 - 0.2}{604}$$

$$= 3.31 \mu A$$

$$\therefore r_e = \frac{26 mV}{3.31 \mu A} = 7.854 \Omega$$

Input Impedance:-

$$Z_i = R_E \parallel r_e$$

$$= \left( \frac{1}{604} + \frac{1}{7.854} \right)^{-1}$$

$$Z_i = 7.766 \Omega$$

Output Impedance:-

$$Z_o = R_e$$

$$\therefore Z_o = 3 k\Omega$$

Voltage gain ( $A_v$ ):-

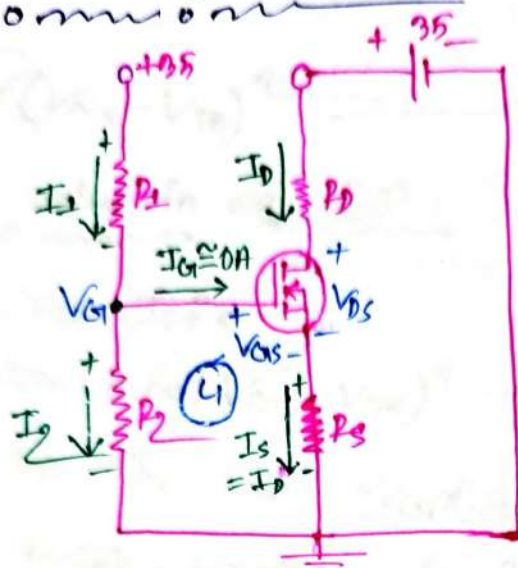
$$A_v = \frac{R_e}{r_e} = \frac{3000}{7.854} = 381.97$$

$$\therefore A_v = 381.97$$

⇒ Higher  $R_E$  increases the input impedance but decreases the voltage gain. But the lower  $R_E$  decreases the input impedance & significantly boosts the voltage gain. The output impedance remains constant in both case

Ans. to the Q. no. 4:-

1st we draw the circuit:-



here,

$$R_1 = 22M\Omega$$

$$R_2 = 20M\Omega$$

$$V_{DD} = 35V$$

$$I_1 \approx I_2 \quad [ \because I_{G1} \approx 0A ]$$

$$R_S = 604 \quad [ \because I_D = 23.50104A ]$$

$$R_D = 2k\Omega$$

$$I_D \approx I_S$$

we know:-

$$V_{G1} = \frac{V_{DD} R_2}{R_1 + R_2}$$

$$= \frac{35 \times 20M\Omega}{(20+22)M\Omega}$$

$$= 16.667V$$

Now Apply KVL in loop 1:-

$$V_{G1} - V_{GS} - I_D R_S = 0$$

$$\therefore V_{GS} = V_{G1} - I_D R_S \quad \text{--- (1)}$$

Again we know:-

$$K = \frac{I_D (on)}{(V_{GS(on)} - V_{Th(on)})^2}$$

$$= \frac{3mA}{(10-5)^2}$$

$$= 120 \mu A/V^2$$

here

$$I_{D(on)} = 3mA$$

$$V_{GS(on)} = 10V$$

$$V_{Th(on)} = 5V$$



### MATHEMATICAL Approach:-

We know:-

$$I_D = K(V_{GS} - V_{TH})^2 \quad \text{--- (1)}$$

∴ put the value in eq (1) ⇒

$$V_{GS} = V_{G1} - I_D R_S$$

$$= V_{G1} - K R_S (V_{GS} - V_{TH})^2$$

$$= V_{G1} - K R_S (V_{GS}^2 - 2V_{GS} \times V_{TH} + V_{TH}^2)$$

$$= 16.667 - 120 \mu \times 694 (V_{GS}^2 - 10V_{GS} + 25)$$

$$= 16.667 - 83.28 mV_{GS}^2 + 0.8328 V_{GS} + (-2.082)$$

$$\therefore 83.28 mV_{GS}^2 + 0.1672 V_{GS} - 14.585 = 0$$

$$\therefore V_{GS} = -14.27, 12.26V$$

↓  
This is not possible for n-channel E-MOSFET.

$$\therefore V_{GSQ} = 12.26V$$

∴ put the value in eq (1) ⇒

$$\therefore I_{DQ} = K(V_{GSQ} - V_{TH})^2$$
$$= 120 \mu (12.26 - 5)^2$$

$$I_{DQ} = 6.324 mA$$

OR put the value in eq (1) ⇒

$$\therefore I_{DQ} = \frac{V_{G1} - V_{GSQ}}{R_S} = \frac{16.667 - 12.26}{694} = 6.35 mA$$

$$I_{DQ} = 6.35 mA$$



$\therefore$  The Q.P  $\equiv (12.26V, 6.35mA)$

### Graphical Approach:-

We know:-

$$I_D = k(V_{GS} - V_{TH})^2$$

Case 1:-      Case 2:-      Case 3:-      Case 4:-

$V_{GS} \leq V_{TH}$        $V_{GS} = 2V_{TH}$        $V_{GS} = 3V_{TH}$        $V_{GS} = 4V_{TH}$   
 $\therefore I_D = 0A$ ,       $\therefore I_D = kV_{TH}^2$ ,       $\therefore I_D = 4kV_{TH}^2$ ,       $\therefore I_D = 9kV_{TH}^2$

$\therefore P_1 \equiv (V_{TH}, 0)$ ,       $P_2 \equiv (2V_{TH}, kV_{TH}^2)$ ,       $P_3 \equiv (3V_{TH}, 4kV_{TH}^2)$ ,       $P_4 \equiv (4V_{TH}, 9kV_{TH}^2)$

Now, put the values in those point:-

$P_1 \equiv (5, 0)$

$P_2 \equiv (10V, 3mA)$

$P_3 \equiv (15V, 12mA)$

$P_4 \equiv (20V, 27mA)$

Now For load line:-

Case 1:-      Case 2:-

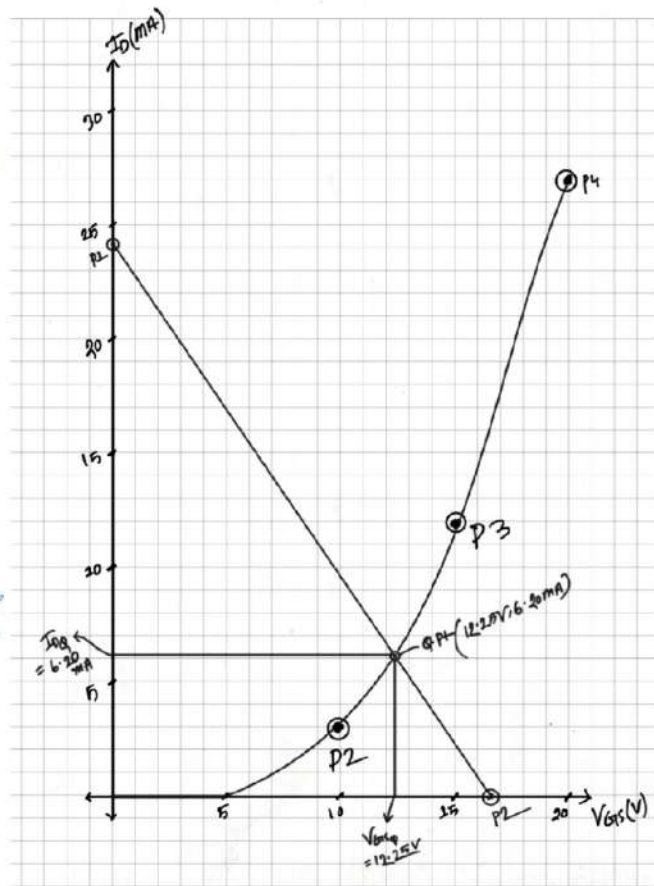
$V_{GS} = 0V$        $V_{GS} = V_{G1}$

$\therefore I_D = \frac{V_{G1}}{R_S}$        $I_D = 0A$

$P_1 \equiv (0V, \frac{V_{G1}}{R_S})$ ,       $P_2 \equiv (V_{G1}, 0)$

$\therefore P_1 \equiv (0, 24.016mA)$

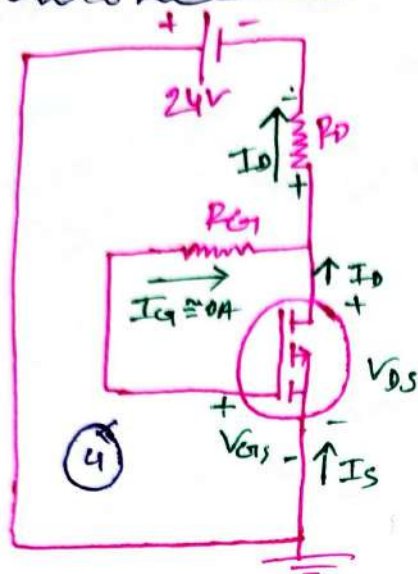
$P_2 \equiv (16.667V, 0)$



The Q.P  $\equiv (12.25V, 6.20mA)$

Ans! to! the Q! no! 5!—

as we draw the circuit!—



here

$$R_D = 1k\Omega \text{ OR } 1.4k\Omega$$

$$I_S = I_D \quad [\because I_G \approx 0A]$$

is given!—

$$I_{D(on)} = 10mA$$

$$\left. \begin{aligned} V_{GS(on)} &= -8V \\ V_{GS(th)} &= -6V \end{aligned} \right\} [\because P\text{-}ch]$$

Apply KVL in Loop L1!—

$$24 - I_D R_D + V_{GS} = 0$$

$$\therefore V_{GS} = I_D R_D - 24 \quad \text{--- (1)}$$

we know!—

$$\begin{aligned} k &= \frac{I_{D(on)}}{(V_{GS(on)} - V_{TH(on)})^2} \\ &= \frac{10mA}{(-8 + 6)^2} = 2.5mA/V^2 \end{aligned}$$

we know!—

$$I_D = k(V_{GS} - V_{TH})^2$$

Case 1!—

$$V_{GS} \leq V_T$$

$$\therefore I_D = 0A$$

Case 2!—

$$V_{GS} = V_{GS(on)}$$

$$\therefore I_D = I_{D(on)}$$

Case 3!—

$$V_{GS} = -7V$$

$$I_D = 2.5mA$$

Case 4!—

$$V_{GS} = -9V$$

$$I_D = 22.5mA$$

$$\therefore P1 \equiv (V_T, 0A) \equiv (-6, 0)$$

$$P2 \equiv (V_{GS(on)}, I_{D(on)}) \equiv (-8, 10mA)$$

$$P3 \equiv (-7, 2.5mA)$$

$$P4 \equiv (-9, 22.5mA)$$



$R_D = 1\text{K}\Omega$

we know:-

$$V_{GS} = I_D R_D - 24$$

For load line:-

Case: 1:-

$$V_{GS} = 0\text{V}$$

$$\therefore I_D = \frac{24}{R_D}$$

$$P1 = (0, \frac{24}{R_D})$$

$$\therefore P1 = (0, 24\text{mA})$$

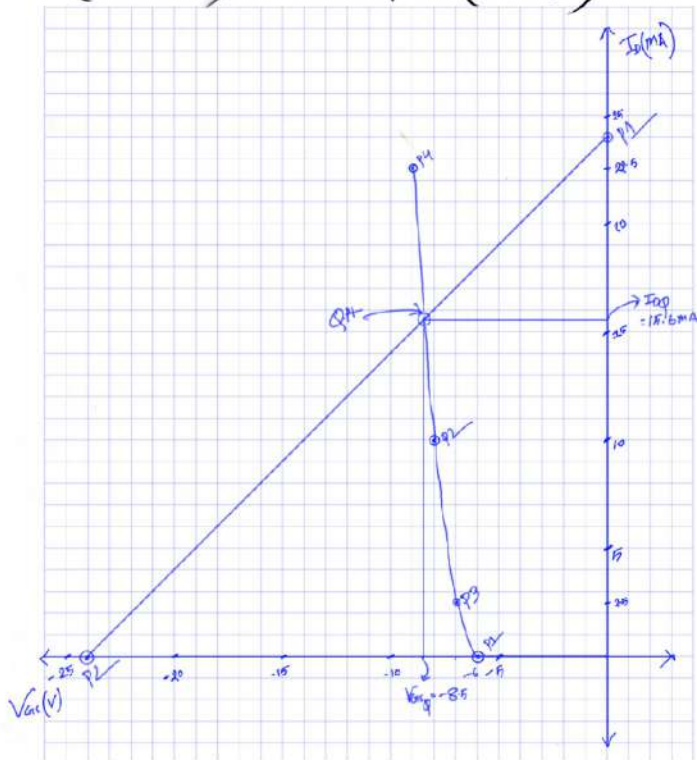
Case 2:-

$$I_D = 0\text{A}$$

$$\therefore V_{GS} = -24\text{V}$$

$$P2 = (-24, 0)$$

$$P2 = (-24, 0)$$



$$\therefore Q = (-8.5\text{V}, 15.6\text{mA})$$

$\Rightarrow$  Drain Current ( $I_D$ ) decreases from  $15.6\text{mA}$  to  $6.75\text{mA}$  when  $R_D$  increases.

Voltage ( $V_{GS}$ ) become less negative (closer to zero) as  $I_D$  increases, changing from  $-8.5\text{V}$  to  $-7.65\text{V}$ .

$R_D = 2.4\text{K}\Omega$

we know:-

$$V_{GS} = I_D R_D - 24$$

For load line:-

Case: 1

$$V_{GS} = 0\text{V}$$

$$\therefore I_D = \frac{24}{R_D}$$

$$P1 = (0, \frac{24}{R_D})$$

$$P1 = (0, 10\text{mA})$$

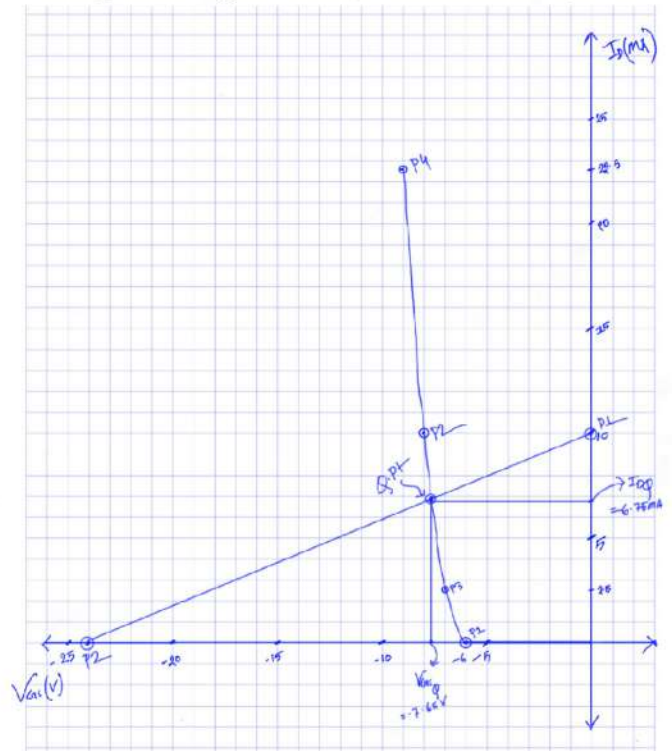
Case: 2

$$I_D = 0\text{A}$$

$$\therefore V_{GS} = -24\text{V}$$

$$P2 = (-24, 0)$$

$$P2 = (-24, 0)$$



$$Q = (-7.65\text{V}, 6.75\text{mA})$$



①

Ans. to the Q. no. 6:-

1st we draw the circuit for DC:-

Now,

For  $R_{Th}$ :-

$$R_{Th} = R_1 || R_2$$

$$= \left( \frac{1}{60k} + \frac{1}{6k} \right)^{-1}$$

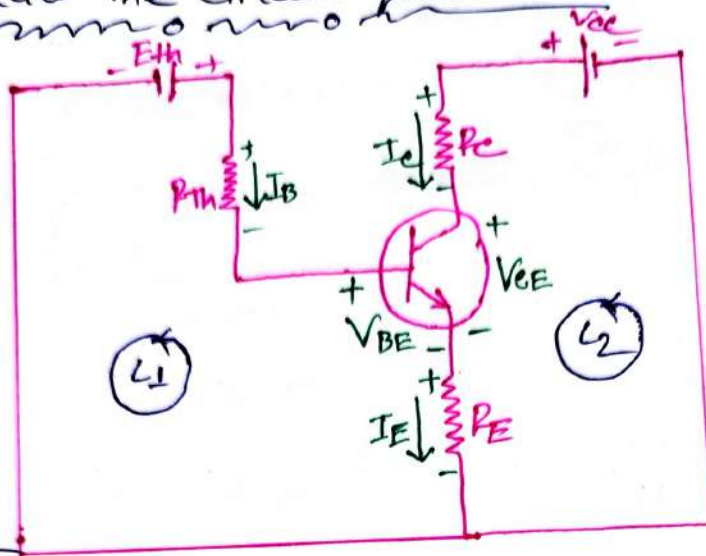
$$= 5.45k\Omega$$

For  $E_{Th}$ :-

$$E_{Th} = \frac{R_2 V_{CC}}{R_1 + R_2}$$

$$= \frac{6k \times 20}{60k + 6k}$$

$$= 1.818V$$



hence

$$R_1 = 60k\Omega$$

$$R_2 = 6k\Omega$$

$$R_C = 5.5k\Omega$$

$$R_E = 1.2k\Omega$$

$$V_{CC} = 20V$$

$$\beta = 80$$

$\therefore$  Apply KVC in Loop L1:-

$$E_{Th} - R_{Th} I_B - V_{BE} - R_E I_E = 0$$

$$I_B = \frac{E_{Th} - V_{BE}}{R_{Th} + (\beta + 1) R_E} \quad [ \because I_E = (\beta + 1) I_B ]$$

$$= \frac{1.818 - 0.7}{5.45k + 81 \times 1.2k}$$

$$= 10.89 \mu A$$

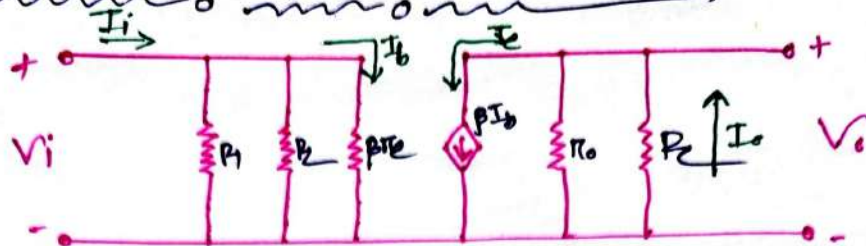
$$\therefore I_E = (\beta + 1) I_B$$

$$= 81 \times 10.89 \mu A$$

$$= 0.882 mA$$

$$\therefore r_e = \frac{26mV}{0.882mA} = 29.47 \Omega$$

Now:- Draw the re equivalent model:-



For Zi:-

$$Z_i = R_1 \parallel R_2 \parallel R_{pi}$$

$$= \left( \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_{pi}} \right)^{-1}$$

$$= \left( \frac{1}{60k} + \frac{1}{6k} + \frac{1}{80 \times 20.47} \right)^{-1}$$

$$Z_i = 1.646k\Omega$$

For Zo:-

$$Z_o = r_{co} \parallel R_c$$

$$\therefore Z_o = R_c \quad [\because r_{co} = \infty]$$

$$Z_o = 5.5k\Omega$$

Now:- For Voltage gain ( $A_v$ ):-

we know:-

$$A_v = \frac{V_o}{V_i} \quad \text{--- (1)}$$

For  $V_o$ :-

$$V_o = -I_o \times (r_{co} \parallel R_c)$$

$$= -I_o R_c \quad [\because r_{co} = \infty]$$

$$= -I_e R_c \quad [\because I_o = I_e]$$

$$= -\beta I_b R_c \quad [\because I_e = \beta I_b]$$

For  $V_i$ :-

$$V_i = I_b \times R_{pi}$$

$\therefore$  put the values in eq (1):-

$$\therefore A_v = \frac{-\beta I_b R_c}{I_b R_{pi}} = - \frac{R_c}{R_{pi}}$$

$$\therefore A_v = - \frac{5500\Omega}{29.47\Omega} = -186.63$$

$$\therefore A_v = -186.63$$

② Let

Now, I'm Designing a D-MOSFET circuit for the same voltage gain.

here,

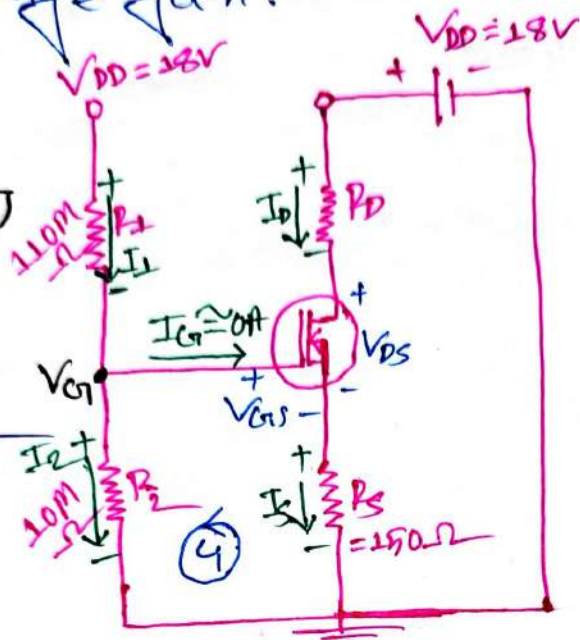
$$I_1 = I_2 \quad (\because I_G \approx 0A)$$

$$I_D = I_S$$

$$\therefore V_{GS} = \frac{R_2 V_{DD}}{R_1 + R_2}$$

$$= \frac{10M \times 18V}{(120)M}$$

$$= 1.5V$$



Let

$$I_{DSS} = 8mA$$

$$V_p = -6V$$

Apply KVL in loop (L1):

$$-V_{GS} + V_{GDS} + I_D R_S = 0$$

$$\therefore V_{GDS} = V_{GS} - I_D R_S \quad \text{--- (1)}$$

We know:

$$I_D = I_{DSS} \left(1 - \frac{V_{GDS}}{V_p}\right)^2$$

For Transfer ch. Curve:

Case 1:

$$V_{GDS} = V_p$$

$$\therefore I_D = 0A$$

$$\therefore P_1 = (V_p, 0)$$

$$P_1 = (-6V, 0)$$

Case 2:

$$V_{GDS} = 0V$$

$$\therefore I_D = I_{DSS}$$

$$P_2 = (0, I_{DSS})$$

$$P_2 = (0, 8mA)$$

Case 3:

$$V_{GDS} = \frac{V_p}{2}$$

$$\therefore I_D = \frac{I_{DSS}}{4}$$

$$P_3 = \left(\frac{V_p}{2}, \frac{I_{DSS}}{4}\right)$$

$$P_3 = (-3V, 2mA)$$

Case 4:

$$V_{GDS} = \left|\frac{V_p}{4}\right|$$

$$\therefore I_D = \frac{25 I_{DSS}}{20}$$

$$P_4 = \left(\left|\frac{V_p}{4}\right|, \frac{25 I_{DSS}}{20}\right)$$

$$P_4 = (1.5V, 10mA)$$



Now For load line:-

Case 1:-

$$I_D = 0A$$

$$\therefore V_{GS} = V_{G1}$$

$$\therefore P1 \equiv (V_{G1}, 0)$$

$$\therefore P1 \equiv (1.5V, 0)$$

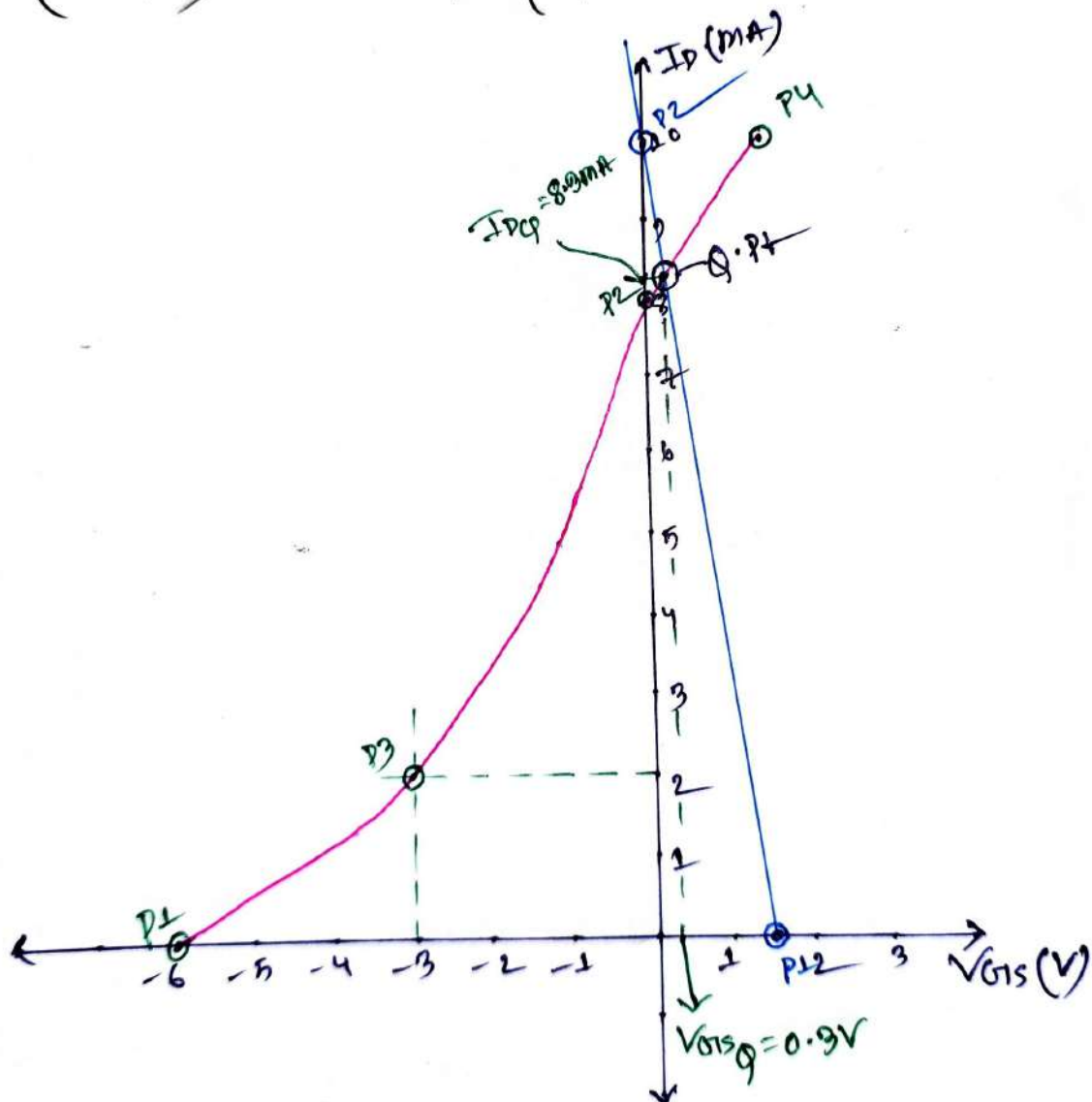
Case 2:-

$$V_{GS} = 0V$$

$$\therefore I_D = \frac{V_{G1}}{R_S}$$

$$P2 \equiv (0, \frac{V_{G1}}{R_S})$$

$$P2 \equiv (0, 10mA)$$



$$\therefore Q \equiv (0.9V, 8.3mA)$$

Now, we know:-

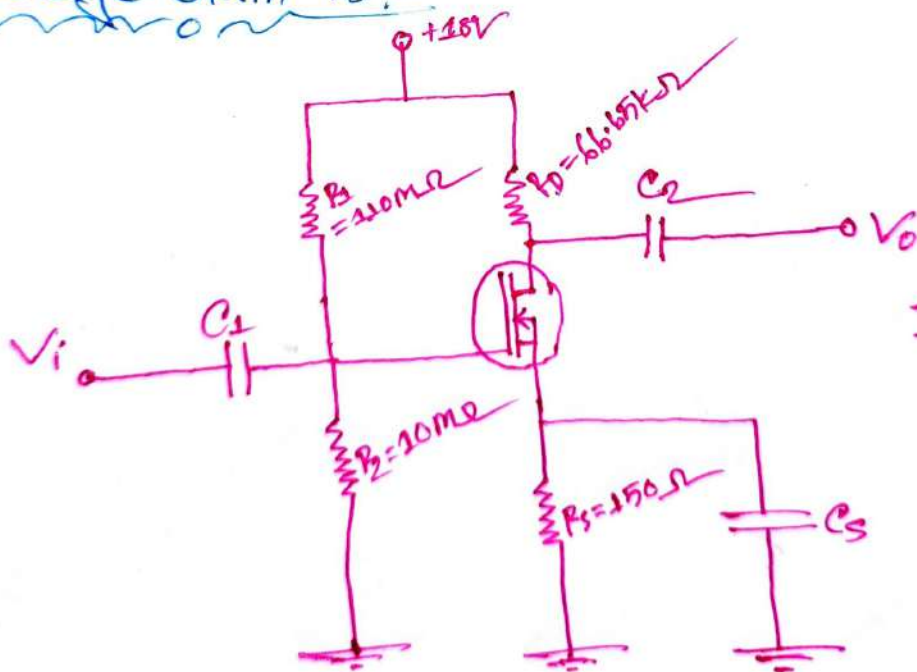
$$g_{m0} = \frac{2I_{DSS}}{|V_p|} = \frac{2 \times 8 \text{ mA}}{|-6|} = 2.667 \text{ mS}$$

$$\begin{aligned} \therefore g_m &= g_{m0} \left( 1 - \frac{V_{GSQ}}{V_p} \right) \\ &= 2.667 \text{ mS} \left( 1 - \frac{0.3}{-6} \right) \\ &= 2.8 \text{ mS} \end{aligned}$$

$$\therefore A_v = -g_m R_D$$

$$\therefore R_D = \frac{A_v}{-g_m} = \frac{-186.63}{-2.8 \text{ mS}} = 66.65 \text{ k}\Omega$$

$\therefore$  The correct D-MOSFET circuit for the same Voltage Gain is:-



$$I_{DSS} = 8 \text{ mA}$$

$$V_p = -6 \text{ V}$$